ERS403 (MD4) Endocrine and Reproductive System II: Complete Pathology Study Guide

This guide combines **everything from the teacher lecture decks and SGD handouts, Pathoma lecture notes and Pathoma slides** for the pathology of ERS II, with extra explanations so that the *why* behind each fact is clear. Delete anything you do not need.

How to use this guide

1. **Read the explanation first, memorise the table second.** Every chapter explains the mechanism (WHY boxes), then gives a table to memorise (HY boxes), then pictures.
2. **Use the pictures actively.** Each image caption says which slide it came from. Cover the caption, name the lesion, and say what you expect to see (the READ boxes tell you how to read each type of picture).
3. **Finish each chapter with the "Test yourself" answers** (teacher MCQs and SGD questions are answered in the text).
4. **Spot the exam traps.** Where the teacher's slide or handout contains an error or an ambiguity, it is flagged in the text (for example the mismatched histology table in the ovarian tumour SGD, and the vaginal lymph drainage note).

What the coloured boxes mean

- **Memory hook:** a mnemonic or one-line anchor.
- **How to read this picture:** what to look for on a slide.
- **Why it happens:** the mechanism behind a fact.
- **High yield:** the facts and answers most likely to be tested.

Module map

Chapter	Topic	Sources
1	Endocrine pancreas and diabetes mellitus	Diabetes deck, Pathoma 15.6
2	Male reproductive tract (penis, testis, prostate)	Testicular and Penile deck, Prostate deck, SGD Prostate, Pathoma Ch 14
3	Female reproductive tract (vulva, vagina, cervix, uterus)	Female Reproductive Tract deck, SGD Uterine, Pathoma Ch 13
4	Developmental disorders of the reproductive tract	Developmental Disorders deck
5	Ovary: PCOS and ovarian tumours	Ovary deck, SGD Ovarian Tumors, Pathoma 13.5
6	Gestational disorders	Gestational deck, Pathoma Ch 13 gestational
7	Breast	Breast deck, Pathoma Ch 16
8	Pituitary	Pituitary decks, Pathoma 15.1-15.2
9	Thyroid	Thyroid deck, Pathoma 15.3
10	Parathyroid and calcium	Parathyroid deck, Pathoma 15.4
11	Adrenal	Adrenal deck, SGD Adrenal Medullary, Pathoma 15.5

12	ERS II review picture quiz	ERS II review deck
13	Master quick-recall tables	Everything above

Two ideas that unlock most of the module

1. Every endocrine problem is one of: too much hormone, too little hormone, or a mass. Ask which axis (hypothalamus, pituitary, target gland) is at fault, and whether the feedback loop explains the lab pattern (for example high ACTH with low cortisol means the adrenal is the problem; low ACTH with low cortisol means the pituitary is). **2. The reproductive tract is embryology plus hormones plus HPV.** Gonad and duct development explains the developmental disorders; estrogen without progesterone explains endometrial hyperplasia and cancer; HPV explains cervical, vulval, vaginal and penile squamous disease.

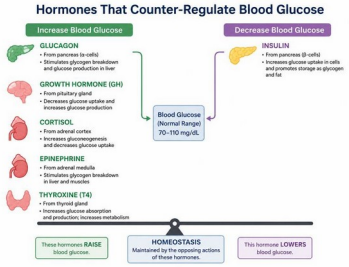
High yield: Image sources. Pictures come from the teacher decks (slide numbers shown) and from Pathoma's slide deck (captions starting "Pathoma 17" or "Pathoma 18"). Pictures that appear in several decks are shown once.

1. Endocrine Pancreas and Diabetes Mellitus

Sources: Teacher deck "Diabetes" (67 slides) + Pathoma Ch 15 "Endocrine Pancreas" (lecture notes pp. 313-315) + Pathoma slides Ch 18.

1.1 The big idea in plain language

The pancreas has two jobs. The **exocrine** part (most of the gland) makes digestive enzymes and sends them to the gut through ducts. The **endocrine** part is made of tiny clusters of cells called **islets of Langerhans**, which release hormones directly into the blood. Diabetes mellitus (DM) is what happens when the main islet hormone, **insulin**, is missing (type 1) or the body stops listening to it (type 2). Nearly every sign and complication of diabetes can be worked out from one sentence: **insulin is the "fed state" hormone, so without its action the body behaves as if it is starving even though blood glucose is sky high.**

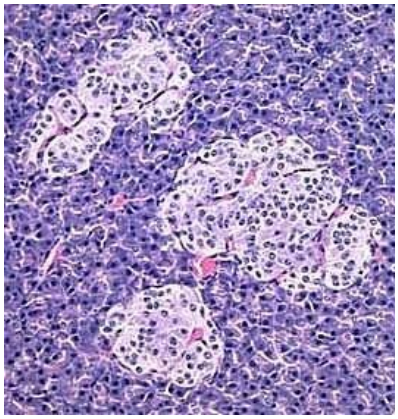


slide 3

1.2 Cells of the islet and what they make

Cell	Hormone	Share of islet	What it does
Beta (B) cell	Insulin and amylin (made together; C-peptide is released with insulin)	65-80%, sit in the centre of the islet	Insulin lowers blood glucose; amylin slows gastric emptying and promotes fullness
Alpha cell	Glucagon	15-20%, at the edge	Raises blood glucose (gluconeogenesis, glycogenolysis, lipolysis)
Delta cell	Somatostatin (D1 cells make VIP)	3-10%	"Brake" on other hormones and on gut secretion
PP (gamma) cell	Pancreatic polypeptide	3-5%	Regulates pancreatic secretion and food intake
Epsilon cell	Ghrelin	<1%	Hunger, energy balance, stimulates GH
Enterochromaffin cells	Serotonin	rare	-
Gastrin cells	Gastrin	not normally in pancreas	Gastrinomas are the common tumour here

How to read this picture: The teacher slides 4, 6, 7, 9, 10 and 11 below show islet anatomy. On an H&E slide the islet is a pale, round island of cells surrounded by darker, grainy exocrine acini (the digestive-enzyme cells). Beta cells are in the middle, alpha cells at the rim.



slide 4 *- Exocrine and Endocrine Pancreas*

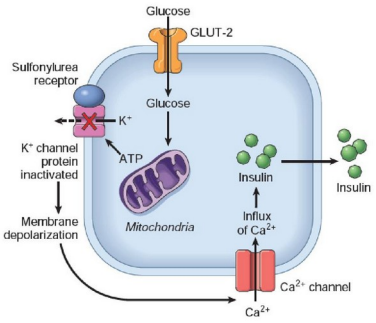
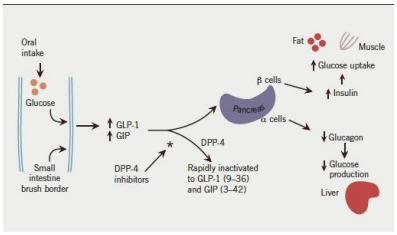
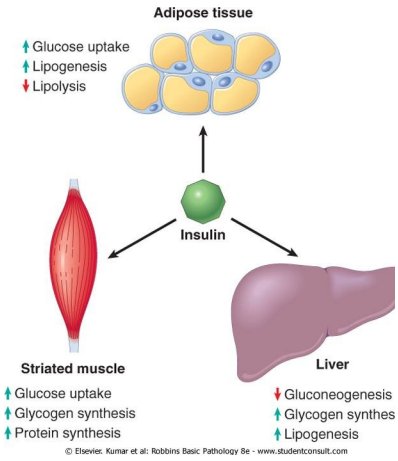


Figure 24-28 Insulin synthesis and secretion. The influx of glucose into β cells through the GLUT-2 receptors initiates a cascade of signaling events that culminates in Ca^{2+}-induced release of stored insulin (see text for details).

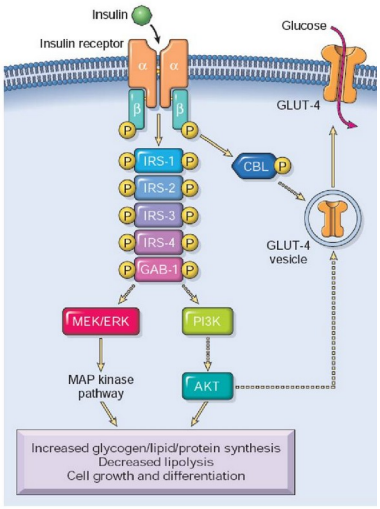
slide 6



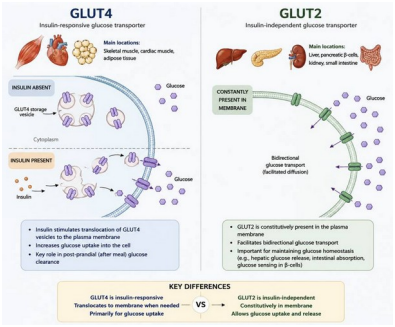
slide 7 *- Glucose-dependent insulinotropic polypeptide (GIP) | Glucagon-like peptide-1 (GLP-1) | Reduces gastric emptying |*



slide 9 *- Insulin Anabolic "Steroid" | only these tissue….!*



slide 10



slide 11

1.3 Insulin versus glucagon (Pathoma)

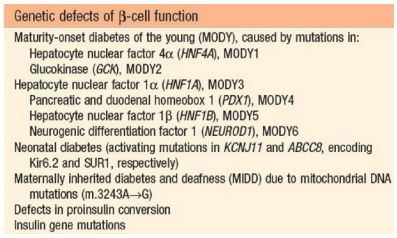
- **Insulin** = the major **anabolic** hormone ("build and store"). It moves **GLUT4** transporters to the surface of muscle and fat cells so glucose can enter, and drives glycogen, fat and protein synthesis. Teacher slide 8 calls it the anabolic "steroid".
- **Glucagon** = the major **catabolic** hormone ("break down and release"). It causes gluconeogenesis from amino acids, lipolysis and glycogenolysis.
- Insulin-dependent tissues (need insulin to take in glucose): striated muscle, adipose tissue, liver. Insulin-**independent** tissues (take in glucose passively): blood vessels, nerves, kidney, lens and retina, brain. This split explains the complications: tissues that cannot shut their glucose door are flooded when blood sugar stays high (see 1.9).
- **Incretins** (teacher slide 7): the gut hormones **GIP** and **GLP-1** raise insulin release after a meal and slow gastric emptying. They are broken down by the enzyme **DPP-4**, which is why DPP-4 inhibitors and GLP-1 agonists are diabetes drugs.

Memory hook: Insulin = "I store it", glucagon = "G-unleash it".

1.4 Classification (teacher slides 12-16)

DM is **not one disease**.

Type	Share	Mechanism	Key facts
Type 1 (IDDM, "juvenile")	5-10%	Absolute insulin deficiency from autoimmune destruction of beta cells	Children/young adults; ketoacidosis-prone; HLA-DR3/DR4
Type 2 (NIDDM, "adult onset")	90-95%	Peripheral insulin resistance plus inadequate beta-cell response	Obesity, strong genetics; usually silent for years
MODY (maturity-onset diabetes of the young)	about 5%	Autosomal dominant, single-gene defects in beta-cell function (MODY 1-6; types 2 and 3 most common)	Young, non-obese, no antibodies, no ketosis. MODY2 = glucokinase enzyme defect (the beta cell's glucose "sensor"). Genetic testing. Treatment depends on the subtype
LADA (latent autoimmune diabetes in adults)	6-10% in UK data	Slow autoimmune beta-cell loss	Looks like type 2 at first, but antibodies are present and insulin is needed within a few years. Called "type 1.5"
Gestational	-	Pregnancy hormones cause insulin resistance	See 1.12
Secondary DM	-	Another disease: excess hormones that raise glucose (Cushing, pheochromocytoma, acromegaly, steroid therapy), beta-cell destruction (pancreatitis, tumour, hemochromatosis), infections (congenital rubella, CMV), Down syndrome	Treat the cause



Genetic defects of β-cell function

Maturity-onset diabetes of the young (MODY), caused by mutations in:
- Hepatocyte nuclear factor 4α (*HNF4A*), MODY1
- Glucokinase (*GCK*), MODY2

Hepatocyte nuclear factor 1α (*HNF1A*), MODY3
- Pancreatic and duodenal homeobox 1 (*PDX1*), MODY4
- Hepatocyte nuclear factor 1β (*HNF1B*), MODY5
- Neurogenic differentiation factor 1 (*NEUROD1*), MODY6

Neonatal diabetes (activating mutations in *KCNJ11* and *ABCC8*, encoding Kir6.2 and SUR1, respectively)

Maternally inherited diabetes and deafness (MIDD) due to mitochondrial DNA mutations (m.3243A→G)

Defects in proinsulin conversion

Insulin gene mutations

slide 15

Table 24-7 Type 1 Versus Type 2 Diabetes Mellitus

Type 1 Diabetes Mellitus	Type 2 Diabetes Mellitus
Clinical	
Onset: usually childhood and adolescence	Onset: usually adult; increasing incidence in childhood and adolescence
Normal weight or weight loss preceding diagnosis	Vast majority are obese (80%)
Progressive decrease in insulin levels	Increased blood insulin (early); normal or moderate decrease in insulin (late)
Circulating islet autoantibodies (anti-insulin, anti-GAD, anti-ICA512)	No islet autoantibodies
Diabetic ketoacidosis in absence of insulin therapy	Nonketotic hyperosmolar coma more common
Genetics	
Major linkage to MHC class II genes; also linked to polymorphisms in *CTLA4* and *PTPN22*, and insulin gene VNTRs	No HLA linkage; linkage to candidate diabetogenic and obesity-related genes (*TCF7L2*, *PPARG*, *FTO*, etc.)
Pathogenesis	
Dysfunction in T cell selection and regulation leading to breakdown in self-tolerance to islet autoantigens	Insulin resistance in peripheral tissues, failure of compensation by β-cells
	Multiple obesity-associated factors (circulating nonesterified fatty acids, inflammatory mediators, adipocytokines) linked to pathogenesis of insulin resistance
Pathology	
Insulitis (inflammatory infiltrate of T cells and macrophages) β-cell depletion, islet atrophy	No insulitis; amyloid deposition in islets Mild β-cell depletion

HLA, Human leukocyte antigen; MHC, major histocompatibility complex; VNTRs, variable number of tandem repeats.

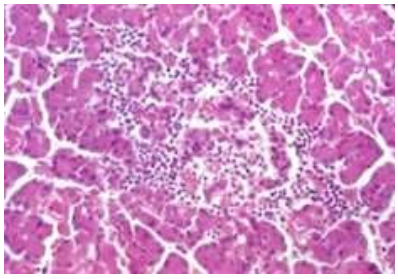
slide 17

1.5 Type 1 DM

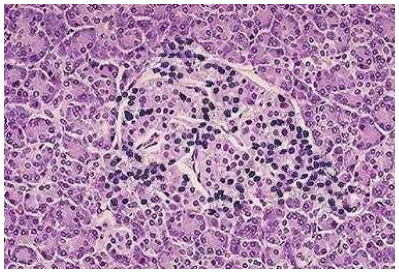
Pathogenesis (teacher slide 18, Pathoma):

5. Genetic susceptibility (**HLA-DR3 / DR4**) plus an environmental trigger (a viral infection is the classic suspect).
6. T lymphocytes attack the beta cells (a **type IV hypersensitivity**). Under the microscope this is **insulitis**: lymphocytes crowding into the islet.
7. Autoantibodies appear, sometimes **years before** symptoms: islet cell antibody (**ICA**), insulin autoantibody (**IAA**), glutamic acid decarboxylase (**GAD65**).
8. Once most beta cells are gone, insulin is absent. The disease runs in families with other autoimmune diseases (Graves, Hashimoto, SLE, rheumatoid arthritis, post-strep glomerulonephritis, rheumatic heart disease).

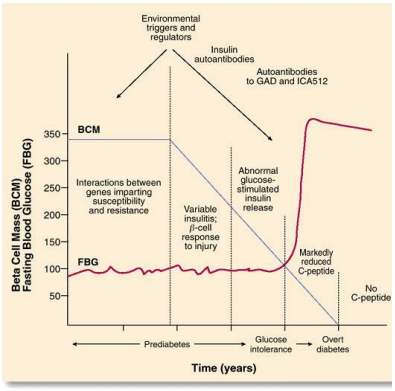
Presentation (Pathoma): child with high blood glucose, **weight loss despite polyphagia** (very hungry because cells are starving, thin because glucagon is unopposed and muscle and fat are broken down), **polyuria** (glucose spills into urine and drags water), **polydipsia** (thirst from fluid loss), glycosuria. Treatment: **lifelong insulin**.



slide 18 *- Pathogenesis of Type I DM | Genetic / HLA-DR3/4 | Environment / Viral infe..? | Insulin*



slide 19 *- Insulitis – Type I | Lymphocytes.*



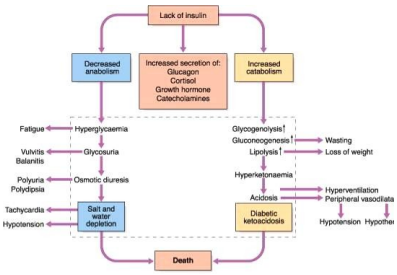
slide 20 *- DM1: Course*

How to read this picture: Slide 19 is "Insulitis - Type I". Look for a normal islet outlined by a dense blue cluster of small round lymphocytes. The islet is infiltrated and the pink beta cells are disappearing. Slide 20 shows the time course ("DM1: Course"): beta-cell mass falls silently for years until about 80-90% is lost, then symptoms start abruptly.

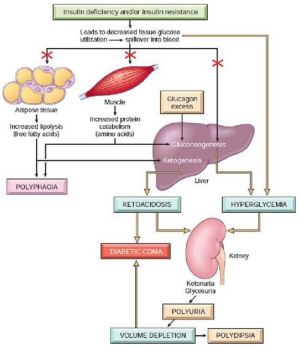
Diabetic ketoacidosis (DKA) - the feared complication of type 1

- **Why it happens:** no insulin means fat cells break down fat unchecked (lipolysis). Free fatty acids go to the liver and are turned into **ketone bodies** (acidic). Stress such as infection raises epinephrine and glucagon and pushes this further.
- **Findings:** glucose usually **>300 mg/dL**, **anion-gap metabolic acidosis** (the gap is made of unmeasured ketoacids), **hyperkalemia** (but the body is actually depleted of potassium, see below), **Kussmaul breathing** (deep, fast breathing to blow off CO2), dehydration from osmotic diuresis, abdominal pain, nausea and vomiting, altered mental status, **fruity (acetone) breath**.

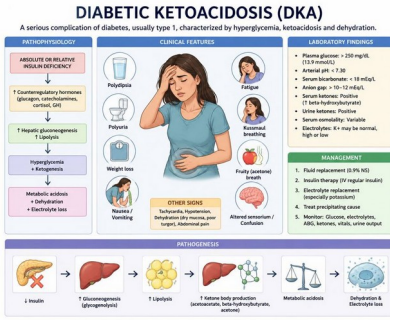
- **Why serum K is high even though total body K is low:** without insulin, the Na/K pump is not stimulated so K stays out of cells, and acidosis swaps H+ into cells for K+ out of cells. The blood K looks high, yet most K is being lost in the urine.
- **Treatment (Pathoma):** (1) **fluids** for dehydration, (2) **insulin**, (3) **potassium replacement**, because once insulin works K moves back into cells and serum K drops (and acidosis correction also lowers it). Giving insulin without watching K can cause dangerous hypokalemia.



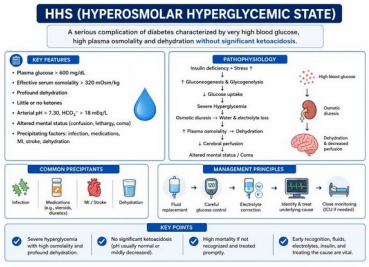
slide 29



slide 30



slide 31

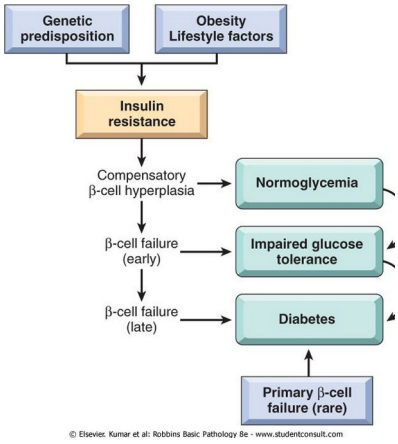


slide 32

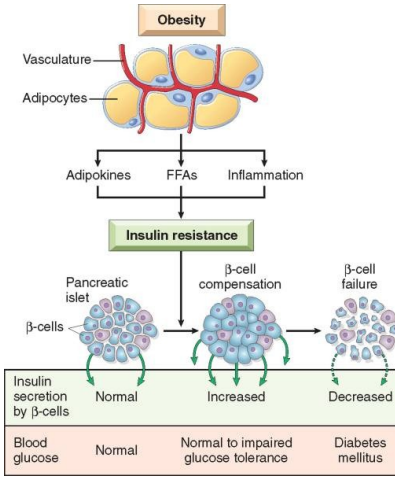
Memory hook: DKA = "Dehydrated, Kussmaul, Acetone breath". Give fluids, insulin, then K.

1.6 Type 2 DM

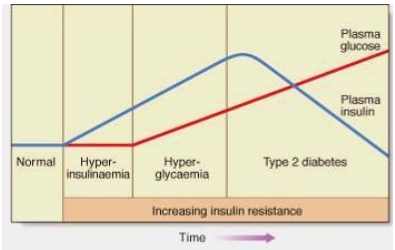
- **Core problem: end-organ insulin resistance**, followed by beta-cell exhaustion. Obesity is the main risk factor (excess fat reduces insulin-receptor signalling). The genetic component is stronger than in type 1.
- **Teacher slide 23 - mechanisms:** liver and skeletal-muscle insulin resistance; beta-cell hypersecretion that eventually fails; raised glucagon; **lipotoxicity** (long-term toxic effect of free fatty acids); increased renal glucose reabsorption (the target of SGLT2 inhibitors); appetite dysregulation.
- **Time course (Pathoma, slides 21-24):** early on **insulin is high** (the beta cells over-work); later the beta cells **burn out** and insulin becomes low.
- **Histology: amyloid deposition in islets** (hyaline change). The amyloid is made of **amylin**, which is co-secreted with insulin. Early islets look normal, late islets lose beta cells and are replaced by pink amyloid.



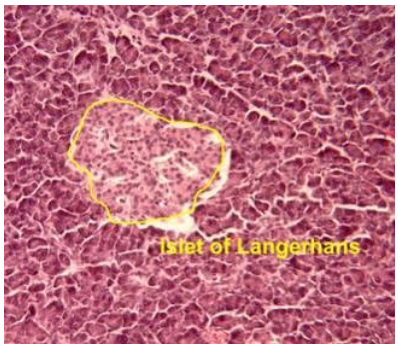
slide 21 *- Pathogenesis of Type II DM | Relative / Insulin Def. | ß cell / Exhaustion / (IDDM)*



slide 22 *- Insulin Resistance: | Diabetes*



slide 24 *- Progression of Type II | Years ..*



slide 25 *- DM2 Islets: Normal early □ amyloid late: | Normal (left) / / Loss of ß cells (right) replaced by Amylo*

How to read this picture: Slide 25 compares a normal islet (left) with a late type 2 islet (right). The abnormal islet has fewer cells and the space is filled with homogeneous, glassy pink material. That is amyloid (Congo red would stain it apple-green under polarised light).

- **Presentation:** polyuria, polydipsia, polyphagia and weight loss, but often **clinically silent**; found on screening or when a complication appears.
- **Diagnosis (any one confirms; teacher slide 26 + Pathoma):**

Test	Diagnostic value
Random glucose with symptoms	>200 mg/dL
Fasting glucose	>126 mg/dL
2-hour glucose after 75 g oral glucose load	>200 mg/dL
Hemoglobin A1c (glycated hemoglobin)	6.5% or higher

Normal glucose is held in a narrow range of about 70-120 mg/dL (slide 26 notes). **C-peptide** is the lab measure of the body's own insulin output (it is released in equal amounts with insulin).

- **Treatment:** weight loss and exercise first line, then oral drugs, then insulin if needed. Diet control, exercise, insulin and oral drugs per teacher slide 26.

Hyperosmolar hyperglycemic state (HHS), also called hyperosmolar non-ketotic coma

- A type 2 complication: glucose >500 mg/dL causes massive diuresis, dehydration, hypotension and coma (from the high osmolarity in the brain).
- **No ketoacidosis** because there is still a little insulin, enough to stop lipolysis and ketone production.

Memory hook: Type 1 -> ketoacidosis (no insulin at all). Type 2 -> hyperosmolar coma (a little insulin, but not enough for the glucose).

1.7 Acute versus chronic complications (teacher slides 27-28, 33)

Acute (metabolic)	Chronic (vascular)
Hypoglycemia (type 1 and 2; dizziness, confusion, sweating, palpitations, tachycardia)	**Microangiopathy** (small vessels): retinopathy, nephropathy, neuropathy, dermopathy
Diabetic ketoacidosis (type 1)	**Macroangiopathy**: accelerated atherosclerosis (MI, stroke, peripheral vascular disease and leg gangrene)
Hyperosmolar hyperglycemic state (type 2)	Infections
Lactic acidosis	

1.8 Why complications happen (Pathoma "two major mechanisms" and teacher slide 34)

High glucose damages the body in two different ways. Learn both.

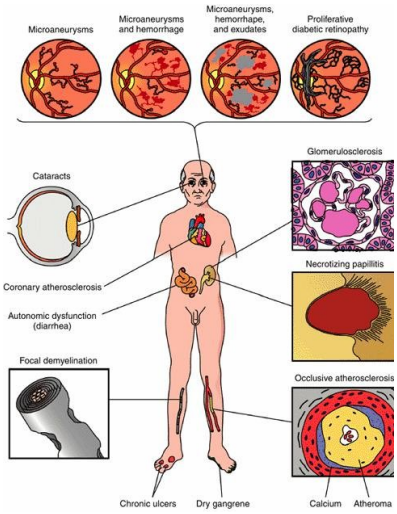
1. Non-enzymatic glycosylation (NEG). Glucose sticks to proteins without an enzyme, forming **advanced glycosylation end-products (AGEs)**. This happens in the blood vessel basement membrane and traps plasma proteins, LDL and cholesterol in the wall.

- NEG of **large and medium arteries** -> **atherosclerosis** (peripheral vascular disease -> amputation, MI, stroke).
- NEG of **small arterioles** -> **hyaline arteriolosclerosis** (the vessel wall turns pink and glassy, narrow and leaky) -> **diabetic nephropathy** (the kidney's efferent arteriole is affected, causing glomerular hypertension, proteinuria and nephrotic-range protein loss) and retinal damage.
- NEG of **hemoglobin** -> **HbA1c**, the 2-3 month marker of glucose control.

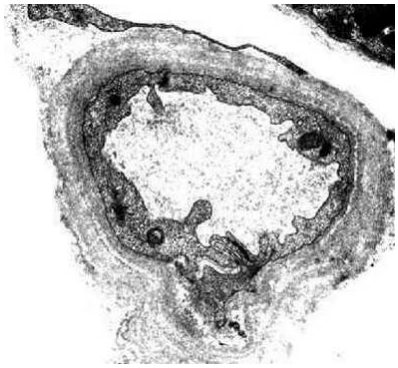
2. Osmotic damage via the polyol (sorbitol) pathway. Insulin-independent cells (Schwann cells, retinal pericytes, lens) take in glucose freely. The enzyme **aldose reductase** turns glucose into **sorbitol**, which cannot leave the cell, so water follows it in and the cell swells and is damaged.

- **Schwann cells** -> **peripheral neuropathy**.
- **Retinal pericytes** die -> weak capillaries -> **microaneurysms** that leak or rupture -> vision loss.
- **Lens** -> swelling and clouding -> **cataract** (teacher slide 65).
- A third route (slide 34): excess glucose also makes **diacylglycerol (DAG)**, which activates **protein kinase C** and promotes abnormal angiogenesis and basement-membrane matrix.

Memory hook: "Sticky and swollen": sticky (glycosylation) hurts big and small vessels, swollen (sorbitol) hurts nerve, retina, lens.



slide 35 *- DM: Complications:*



slide 37 *- Diabetic Microangiopathy | Normal | Diabetic | Glucose / Glycosylation / BM damage lea*

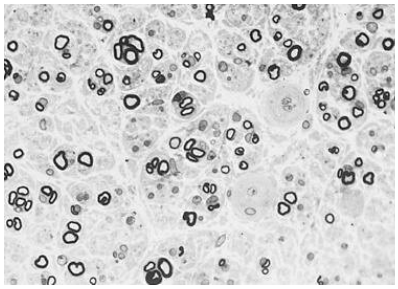
How to read this picture: Slide 37 contrasts a normal capillary with a diabetic one. In the diabetic vessel the basement membrane is thick and layered, the lumen is narrower and proteins have leaked out, which explains oedema, protein loss and ischaemia.

1.9 Microangiopathy in detail (teacher slides 36-57)

Chronic hyperglycaemia -> glycosylation of basement-membrane proteins -> thick, narrow, fragile, **leaky** vessels with deposited proteins, matrix and LDL -> ischaemia and defective inflammation. Leaks cause oedema and **proteinuria** in the kidney and **microaneurysms** in the retina.

Diabetic neuropathy (slide 38-40)

- Sensory first, then motor (myelin loss). **Peripheral neuropathy** is **bilateral, symmetric, progressive, irreversible**: paraesthesia, pain, loss of sensation, muscle atrophy ("glove and stocking" pattern).
- Visceral (autonomic) neuropathy: constipation or diarrhoea, **orthostatic hypotension**, gastroparesis, bladder and sexual dysfunction.
- Cranial nerve involvement: diplopia, Bell palsy (unilateral facial paralysis, CN VII).
- **Neuropathic ulcer**: loss of pain sensation + trauma + foot deformity, made worse by ischaemia, callus formation and oedema. The foot is injured without the patient noticing.



slide 39 *- DM-Neuropathy – Myelin stain | Myelin loss in nerve | Normal*

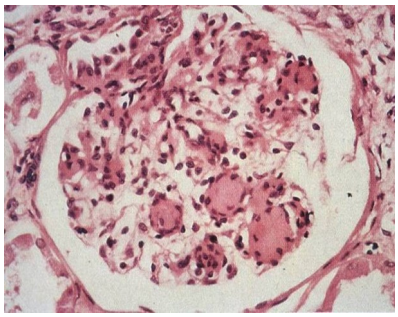


slide 40 *- Neuropathic ulcer | Etiology: / peripheral sensory neuropathy, Trauma & deformit*

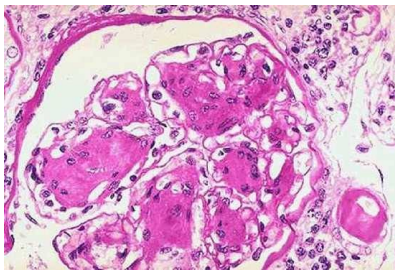
How to read this picture: Slide 39 compares nerve with a myelin stain. In the normal nerve the black-stained myelin rings are densely packed; in the diabetic nerve the stain is patchy and axons have lost their rings.

Diabetic nephropathy (slides 41-48)

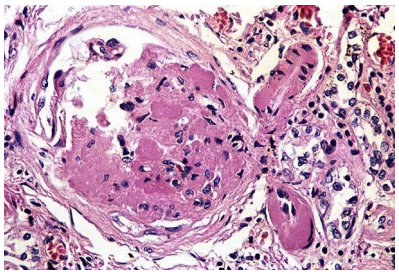
- Leading cause of death and illness in long-standing diabetes. It combines microangiopathy, atherosclerosis and infections.
- **Diffuse or nodular glomerulosclerosis** (the **Kimmelstiel-Wilson lesion**): pink, round, acellular nodules in the centre of glomerular tufts made of AGE-laden matrix. Nodular = pathognomonic.
- **Hyaline arteriolosclerosis** of both afferent **and** efferent arterioles (in hypertension it is mainly the afferent, so this is a diabetes clue).
- **Thickened basement membrane** (slide 46: proximal and distal convoluted tubules).
- **Necrotizing papillitis** (papillary necrosis) from infarction and infection, and **pyelonephritis**.
- Clinical course: microalbuminuria -> overt **nephrotic syndrome** -> **end-stage renal failure**.



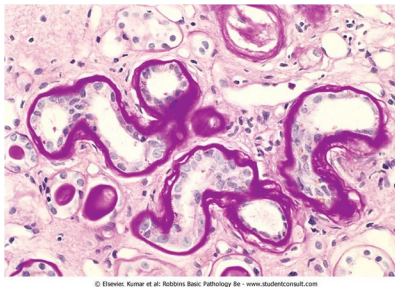
slide 43 *- Nodular Glomerulosclerosis – KW*



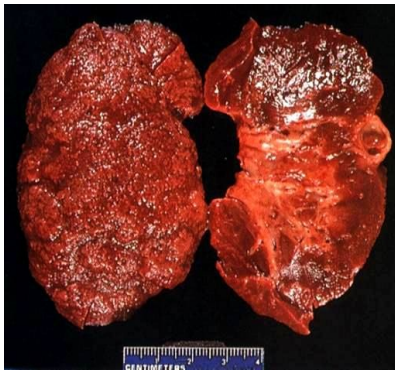
slide 44 *- Diabetic Glomerulosclerosis | B | A | A: Nodular glomerulosclerosis. / B: Hyaline Arteri*



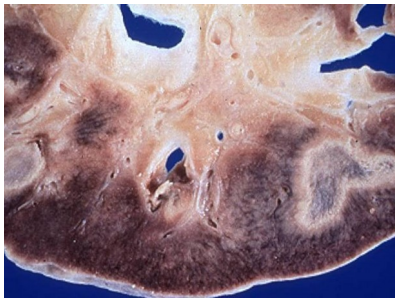
slide 45 *- DM Kidney: advanced (KW Lesion) | B | A*



slide 46 *- DM Kidney: thickening of BM (PCT) | PCT | DCT | PCT: Proximal Convoluted Tubule, DCT: Distal Con*



slide 47 *- Gross : Diabetic kidney*



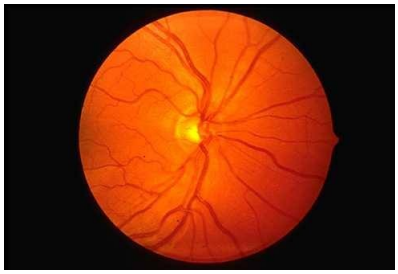
slide 48 *- DM with Infarction: | Papillary necrosis*

How to read this picture: On slides 43-45 look for glomeruli whose centre contains one or several round, glassy pink nodules pushing the capillary loops to the edge (Kimmelstiel-Wilson). Slide 44 labels A = nodular glomerulosclerosis and B = hyaline arteriolosclerosis (a pink, thick-walled arteriole). Slide 47 is the gross kidney: finely granular (scarred) surface. Slide 48 shows papillary necrosis: pale, wedge-shaped dead tissue at the tip of the pyramids.

Diabetic retinopathy (slides 49-57)

Stage	What you see	Why
Non-proliferative	Venous dilation, **microaneurysms** (small red dots), **dot-and-blot haemorrhages**, **hard exudates** (yellow, deep; chronic oedema/lipid leakage), **cotton-wool spots** (soft white microinfarcts from ischaemia that obscure vessels), macular oedema	Pericyte loss and leaky capillaries
Proliferative	**Neovascularisation** (new fragile vessels grow into the vitreous), large pre-retinal/vitreous haemorrhage with sudden vision loss, fibrovascular membranes that cause **traction retinal detachment** and permanent blindness	Ischaemic retina releases growth factors (VEGF)

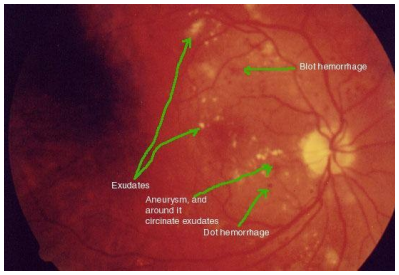
- Most common cause of visual loss in non-proliferative disease is **macular oedema**. Best test: **fluorescein angiography**. Treatment of proliferative disease: **panretinal photocoagulation** (laser), plus anti-VEGF drugs.



slide 52 *- Normal Retina*



slide 53 *- Diabetic Retinopathy & Cataract*



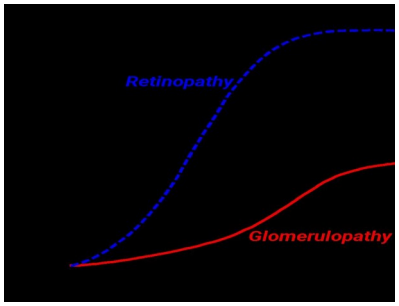
slide 54 *- Diabetic Retinopathy*



slide 55 *- Diabetic Retinopathy | Fluorescein angiogram of the eye of a diabetic patient. Note*



slide 56 *- Diabetic Retinopathy | Pre retinal Hemorrhage - detachment*

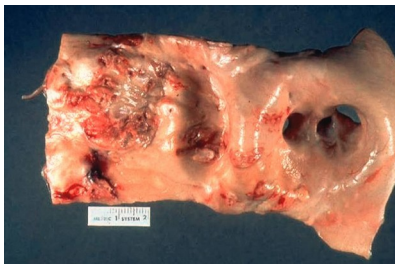


slide 57 *- Frequency of Clinically Evident Retinopathy and Glomerulopathy in Type I Diabetes*

How to read this picture: Compare the normal retina (slide 52) with the diabetic ones. Dots and blots are red spots (dots = microaneurysms, blots = larger haemorrhages). Hard exudates are sharply bordered yellow deposits. Cotton-wool spots are fluffy, pale white patches with blurry edges. On the fluorescein angiogram (slide 55) microaneurysms are tiny bright dots. Slide 57 plots how retinopathy and glomerulopathy become more frequent with years of type 1 diabetes.

1.10 Macroangiopathy / atherosclerosis (slides 58-59, 64)

Contributing factors: dyslipidaemia (**low HDL**), non-enzymatic glycosylation, **increased platelet adhesiveness**, **increased thromboxane A2**, decreased prostacyclin, endothelial damage. Results: MI, stroke (CVA), **gangrene of the leg** from peripheral vascular disease, renal artery disease. **Medial calcification** of digital arteries (slide 64: calcified vessels in an amputated toe, "Monckeberg-type") is also common.



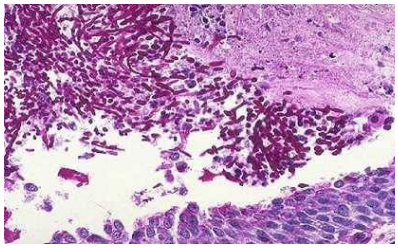
slide 59 *- Atherosclerosis:*



slide 64 *- Blood vessel calcification: | In digital arteries in DM / Amputated Toe | Calcified BV*

1.11 Infections in diabetics (slides 60-61)

Multifactorial: impaired inflammation (vessel thickening), reduced white cell function, glycosylated antibodies and mediators, tissue ischaemia and infarction, and poor metabolism. High glucose alone is **not** the cause. Typical: **candidiasis**, foot infections, pyelonephritis, **mucormycosis**.



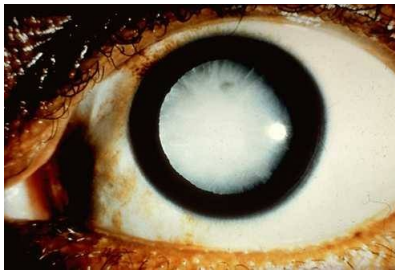
slide 61 *- Fungal infections: Candidiasis*

1.12 Diabetes and pregnancy, other skin/eye signs (slides 62-66)

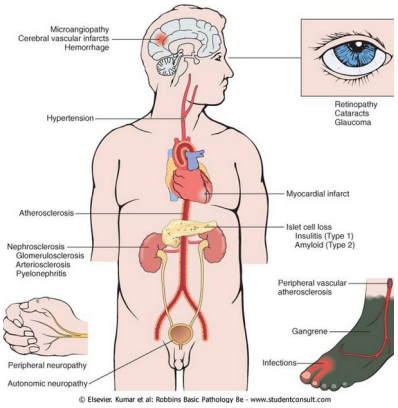
- **Pre-existing diabetes** in pregnancy: stillbirth and **congenital malformations** (hyperglycaemia is teratogenic in the first trimester).
- **Gestational diabetes**: **macrosomia** (large baby, because fetal hyperinsulinaemia is a growth signal) with **polycythaemia** (slide 63); risk of neonatal hypoglycaemia after birth.
- **Acanthosis nigricans** (slide 66): velvety, dark skin in the neck folds and armpits; a skin sign of **insulin resistance**.
- **Cataract** (slide 65) from sorbitol, see 1.8.



slide 63 *- Macrosomia | With Polycythemia*



slide 65 *- Cataract – Sorbitol.. Polyol..osmotic.. | Lens epithelium (Insulin independent) is exposed to Hypergly*



slide 67 *- Complications Summary:*

1.13 Pancreatic endocrine tumours (Pathoma)

Tumour	Hormone	Presentation	Remember
Insulinoma	Insulin	Episodes of confusion / altered mental status from severe hypoglycaemia that is **relieved by glucose**	Low glucose, **high insulin, high C-peptide** (C-peptide is low if the patient is injecting insulin)
Gastrinoma	Gastrin	Peptic ulcers that do not respond to treatment, extending into the jejunum (**Zollinger-Ellison syndrome**)	Gastrin makes parietal cells secrete acid
Somatostatinoma	Somatostatin	Achlorhydria (no stomach acid), gallstones, steatorrhoea	Somatostatin lowers acid and gallbladder contraction
VIPoma	Vasoactive intestinal peptide	**Watery diarrhoea, hypokalaemia, achlorhydria** ("WDHA")	VIP drives gut water and electrolyte secretion

- **MEN 1** (multiple endocrine neoplasia type 1) = **3 Ps**: **P**ituitary adenoma, **P**arathyroid hyperplasia, **P**ancreatic endocrine tumour (see the adrenal/MEN section).
- Proteins made together with insulin: **amylin** (forms amyloid in type 2 DM) and **C-peptide** (a blood test to tell whether a patient has been giving themselves insulin).

1.14 Summary table: type 1 versus type 2

Feature	Type 1	Type 2
Core defect	Autoimmune destruction of beta cells	Insulin resistance, then beta-cell failure
Insulin level	Very low or absent	High early, low late
Age / body type	Children, thin	Adults, obese
Genetic link	HLA-DR3/DR4	Stronger family history, no HLA
Antibodies	ICA, IAA, GAD65	None
Islet histology	Insulitis (lymphocytes)	Amyloid (amylin)
Acute crisis	DKA	HHS (hyperosmolar coma)
Treatment	Insulin	Diet/exercise, oral drugs, then insulin

2. Male Reproductive Tract: Penis, Testis, Prostate

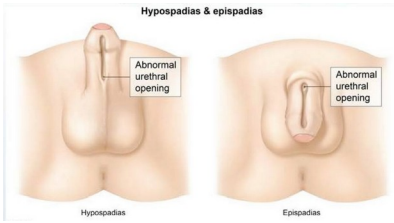
Sources: Teacher decks "Testicular and Penile Disorders" (81 slides) and "Disorders of the Prostate Gland" (37 slides), SGD handout "Male Reproductive Disorders: Prostate", Pathoma Ch 14 (Male Genital System, lecture notes pp. 287-297) and Pathoma slides Ch 17.

2.1 Penis

Developmental problems

Condition	What it is	Why / key facts
Hypospadias	Urethra opens on the **underside (ventral)** of the penis	Failure of the urethral folds to fuse. Hypo = below. Common. Linked with undescended testes, UTI and infertility
Epispadias	Urethra opens on the **top (dorsal)** surface	Abnormal position of the genital tubercle; a partial form of failure of abdominal/pelvic fusion. Penis is bent upward. **Associated with bladder exstrophy** (bladder open onto the lower abdominal wall). Also linked with undescended testes, UTI, infertility
Phimosis	Foreskin opening is too small to retract over the glans	Hygiene problems, balanoposthitis (infection of glans and foreskin). Treatment: circumcision
Paraphimosis	Retracted foreskin gets trapped behind the glans and strangulates it	A medical emergency: swelling prevents it coming back
Peyronie disease	Progressive fibrosis of the tunica albuginea causes **penile curvature** and pain	Associated with **Dupuytren contracture** of the hand (both are fibromatoses)
Priapism	Painful erection lasting **>4 hours**, unrelated to arousal (Pathoma: medical emergency, ischaemia and thrombosis can cause gangrene)	Causes: **sickle cell disease**, myeloproliferative disorders, drugs (**sildenafil, trazodone**)

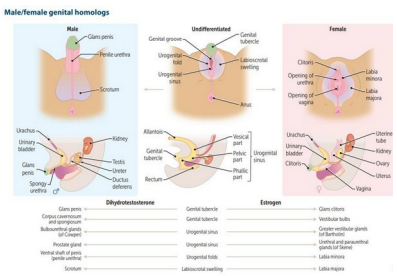
Memory hook: Hypospadias = urethra **below** (hypo = under). **Epi**spadias = **epi** is on top (like epidermis), with bladder exstrophy.



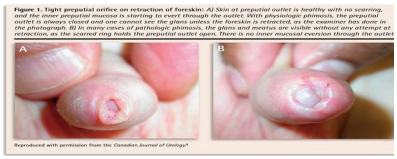
slide 5



slide 6 *- Bladder extrophy*



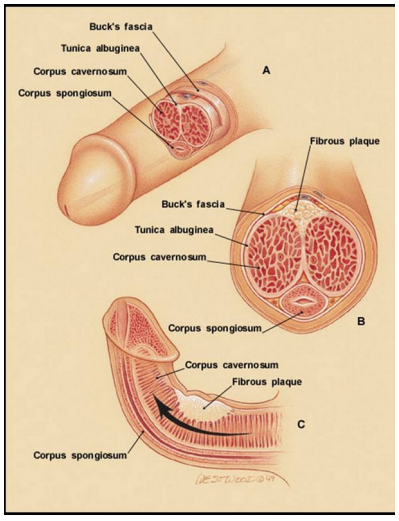
slide 7



slide 10 *- Paraphimosis | Phimosis*



slide 10 *- Paraphimosis | Phimosis*



slide 13



slide 14 *- PEYRONIE DISEASE*

How to read this picture: Slides 10 (phimosis vs paraphimosis) and 14 (Peyronie) are clinical photographs. In phimosis you see the foreskin ballooning or unable to pull back; in paraphimosis a swollen, constricted collar of foreskin sits behind the glans. In Peyronie disease the penis bends toward the side of the fibrous plaque.

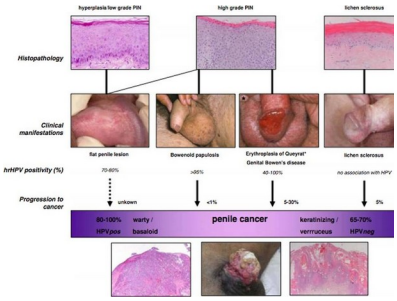
Test yourself (teacher slides 8, 11). Newborn with incomplete dorsal urethra that extends to an open bladder on the abdominal wall = **epispadias** (bladder exstrophy). A 35-year-old whose prepuce cannot be fully retracted and nothing else wrong = **phimosis**.

Infections and skin disease of the penis (and vulva)

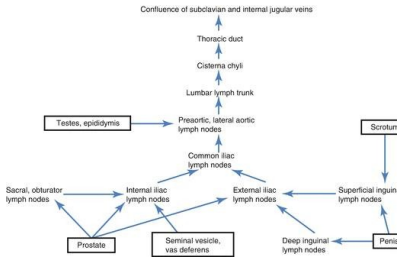
- **Condyloma acuminatum** (genital warts): caused by **HPV 6 and 11** (low-risk types). Cauliflower-like, exophytic, papillary growth. Microscopy: **koilocytes** (cells with a wrinkled, enlarged, hyperchromatic nucleus and a clear perinuclear halo), acanthosis (thick epidermis), hyperkeratosis and parakeratosis. Most common sexually transmitted viral disease; in anorectal disease linked with high-grade anal dysplasia and anal squamous carcinoma, especially in HIV-positive patients.
- **Lymphogranuloma venereum (LGV)**: *Chlamydia trachomatis* **serotypes L1-L3**. Necrotising granulomatous inflammation of inguinal lymphatics and nodes; perianal disease may heal by fibrosis causing rectal stricture. (See also the microbiology lectures.)
- **Lichen sclerosus** (on the penis called balanitis xerotica obliterans, BXO): smooth white plaques; thinned epidermis, hyperkeratosis, sclerosis of the upper dermis and a band-like lymphocytic infiltrate. In

women (postmenopausal vulva) raises the risk of vulvar squamous carcinoma; on the penis it can precede squamous carcinoma.

- **Lichen simplex chronicus**: squamous cell hyperplasia (acanthosis, hyperkeratosis); shows as leukoplakia (white plaque) from chronic rubbing.



slide 23



***slide 25** - SKIN | GLANS | CORPUS / CAVERNOSUM*

How to read this picture: Condyloma (slides 17-18): look for finger-like, papillary projections with a thick squamous lining. In the upper layers find cells with a clear halo around a small dark, irregular nucleus = koilocytes.

Penile and scrotal carcinoma

- **Squamous cell carcinoma (SCC) of the penis:** age 40-70, slow growing, painless, spreads first to **local (inguinal) lymph nodes**. Plaque -> ulcerated papule or fungating mass. Risk factors: **HPV 16, 18** (and 31, 33; about two-thirds of cases) and **lack of circumcision** (retained smegma and chronic inflammation under the foreskin). Phimosis and lichen-type conditions also raise risk (slide 26 answer: balanitis xerotica obliterans).
- **Precursor lesions (carcinoma in situ)** - exam favourites:

Lesion	Where / what it looks like	Patient
Bowen disease	CIS of the penile **shaft or scrotum**, **leukoplakia** (white plaque)	Older, around the 5th decade
Erythroplasia of Queyrat	CIS on the **glans**, red velvety **erythroplakia**	Older
Bowenoid papulosis	**Multiple reddish-brown papules**; HPV 16; CIS that **rarely progresses to invasion**	Younger (about 40s)

- **Scrotal SCC:** older men (60-80 years); historically chimney sweeps (soot) and exposure to industrial chemicals (arsenic); invades the scrotum then regional nodes (the first described occupational cancer: Percivall Pott).

Test yourself (slide 28): 43-year-old uncircumcised man, 0.9 cm rough tan raised lesion on the glans with difficulty retracting the foreskin = **Bowen disease** (SCC in situ). Slide 27: pearly penile papules at the corona in a circumcised, sexually inactive 17-year-old = **normal variant**.

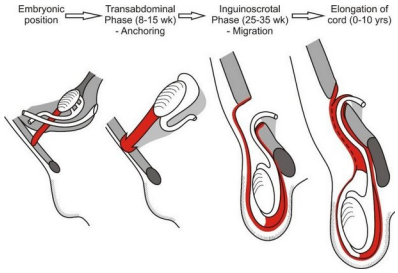
2.2 Testis (non-neoplastic)

Embryology reminder: testes form in the abdomen and descend through the inguinal canal into the scrotum, guided by hormones: **anti-Mullerian hormone** from **Sertoli cells**, **insulin-like hormone 3 (INSL3)** from **Leydig cells**, and androgen-driven release of calcitonin-gene-related peptide from the genitofemoral

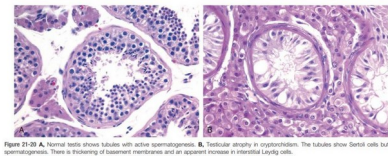
nerve (slide 33). The cool scrotum matters because sperm production needs a temperature below body temperature.

Cryptorchidism (undescended testis)

- Failure of the testis to descend to the scrotal sac, usually **unilateral**. **Most common congenital male reproductive abnormality** (about 1% of male births; more with prematurity). Most resolve spontaneously; if not, **orchiopexy before about 2 years** (surgery to place the testis in the scrotum).
- Why it is damaging: **Sertoli cells and germ cells are temperature-sensitive**, so at body temperature they fail: FSH rises, inhibin falls, sperm count falls, **atrophy and infertility**. **Leydig cells tolerate heat**, so testosterone stays normal in unilateral cases (low in bilateral). Histology can show prominent Leydig cells (Pathoma).
- **Complications:** infertility and a **5-10 times higher risk of germ cell tumour (seminoma)**, even after surgical correction (the risk is in the testis, not only the position). Associated with hypospadias.



slide 33 *- Anti-Müllerian hormone - Sertoli | Insulin-like hormone 3 (INSL3) - Leydig cells | Androgen-ind*



slide 34

Test yourself (slide 35): adult with a solid left inguinal mass and an empty left hemiscrotum: the most appropriate approach is to remove the mass (a cancer risk), not to move it into the scrotum.

Testicular torsion vs epididymitis (slides 45-48; Pathoma)

- **Torsion:** the spermatic cord twists, obstructing venous outflow then arterial flow -> **haemorrhagic infarction within about 6 hours**. Cause: congenital failure of the testis to attach firmly to the scrotal lining (the "bell-clapper" anomaly, via the processus vaginalis). Teen or young man; **sudden severe pain**, nausea and vomiting, high-riding horizontal testis, **absent cremasteric reflex** (the reflex travels in cord nerves). **Prehn sign negative**. Doppler ultrasound: **no or reduced blood flow**. Surgical emergency.
- **Epididymitis:** inflammation of the epididymis, gradual onset, fever, urinary symptoms, tender swollen epididymis, **Prehn sign positive** (pain relieved on lifting the scrotum), Doppler shows increased blood flow. **Age <35:** *Neisseria gonorrhoeae* and *Chlamydia trachomatis*. **Age >35 (and in older men):** *E. coli* and *Pseudomonas* (spread of a UTI/prostate infection). Chronic epididymitis: **TB**.

Feature	Torsion	Epididymitis
Age	Adolescents, young adults	Any age
Onset	Sudden	Gradual
Cremasteric reflex	Absent	Present
Prehn sign	Negative	Positive
Doppler	Decreased/absent flow	Increased flow
Treatment	Emergency surgery	Antibiotics

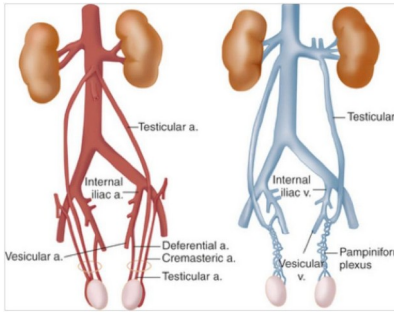
Test yourself (slide 48): 19-year-old, sudden severe scrotal pain, tender enlarged left testis, decreased Doppler flow = **torsion**.

Orchitis (testicular inflammation; Pathoma table)

Cause	Notes	Sterility?
Chlamydia and/or *Neisseria gonorrhoeae*	Sexually active young adults; libido preserved because Leydig cells are spared	Yes
E. coli and *Pseudomonas*	Older adults; spread of UTI organisms	-
Mumps	Teenagers and adults (not before puberty, so not <10 years). Very painful, swollen	Yes (can cause sterility)
Autoimmune orchitis	**Granulomas in the seminiferous tubules**	-

Varicocele, hydrocele, spermatocele

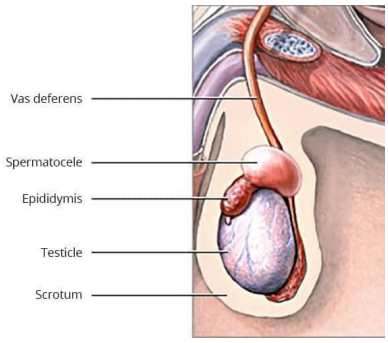
Condition	Definition	Key points
Varicocele	Dilated veins of the **pampiniform plexus** (incompetent valves)	Feels like a "**bag of worms**"; shows when standing, vanishes when lying down. Mostly **left side** because the left testicular vein drains into the **left renal vein** at a right angle, while the right drains straight into the IVC. A new left varicocele should make you think of **renal cell carcinoma** invading the renal vein. A frequent cause of male infertility (oligospermia)
Hydrocele	Fluid in the **tunica vaginalis**	Soft, pear-shaped, painless, **transilluminates**. In children: patent processus vaginalis (peritoneal fluid communicates). In adults: lymphatic blockage or inflammation (epididymitis, tumour)
Spermatocele	Cystic dilation of the epididymis / rete testis containing sperm	Painless cyst above the testis



slide 37



slide 38 - *Varicocele*



slide 43

Test yourself (slides 39, 44): infertile man with a left cord that feels like a bag of worms and oligospermia = **varicocele**. Painless scrotal swelling that transilluminates and is cystic on ultrasound = **hydrocele**.

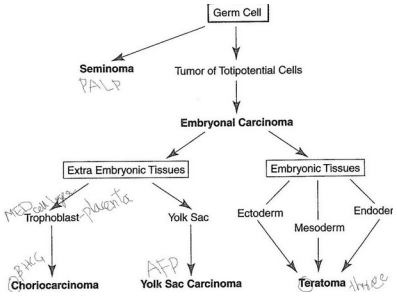
2.3 Testicular tumours

Big picture

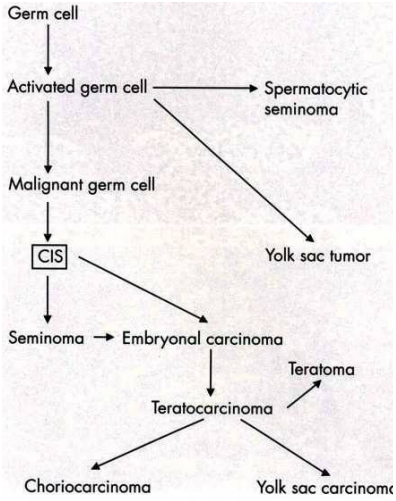
- A **firm, painless testicular mass that does not transilluminate** is cancer until proven otherwise. Ultrasound shows a hypoechoic solid intratesticular mass.
- **Management: radical (inguinal) orchiectomy**, never a transscrotal biopsy (risk of seeding the scrotum and altering lymph drainage). Then stage with chest X-ray and abdominal/chest CT.
- **Epidemiology:** most common cancer in men **15-34**; >95% are **germ cell tumours (GCT)**. Risk factors: **cryptorchidism (5-10 times)**, **testicular dysgenesis (Klinefelter syndrome, testicular feminisation)**, Caucasian more than African-American, family history. Can also occur in extragonadal midline sites (mediastinum, retroperitoneum).
- **Why no epithelial tumours (carcinoma) in the testis?** The testis is not lined by epithelium; it is surrounded by the tunica vaginalis (mesothelium/mesenchyme).
- **Spread:** lymph drains to the **retroperitoneal (para-aortic) nodes**, not the inguinal nodes (the testis developed near the kidney). Right testis -> **inter-aortocaval** nodes just below the renal vessels; left testis -> **para-aortic** nodes (lateral to the aorta). Scrotal skin tumours spread to the inguinal nodes. Back pain may come from retroperitoneal metastases. Staging: Stage I limited to testis/epididymis/cord; II retroperitoneal nodes; III beyond the retroperitoneum (supradiaphragmatic nodes or organs).
- **Classification:**
  - **Germ cell:** seminomatous (seminoma, spermatocytic seminoma) and **non-seminomatous** (embryonal carcinoma, yolk sac tumour, choriocarcinoma, teratoma, and **mixed**).
  - **Sex cord-stromal:** Leydig cell tumour, Sertoli cell tumour.
  - **Lymphoma.**
- **Tumour markers: beta-hCG** (choriocarcinoma, sometimes embryonal/seminoma slightly), **AFP** (yolk sac, embryonal), **placental alkaline phosphatase (PLAP)** (seminoma). Markers are used for diagnosis, staging, treatment response and recurrence.



slide 53



slide 55



slide 56

Seminoma versus non-seminoma

Feature	Seminoma	Non-seminomatous GCT
Share	about 55% of GCT, most common	about 45%

	testicular tumour	
Radiosensitivity	**Highly radiosensitive** and chemosensitive	Not radiosensitive; chemosensitive
Metastasis	**Late**	**Early**, to retroperitoneal nodes
Prognosis	**Excellent** (>90% 5-year survival in early disease)	More aggressive, variable

Test yourself (slide 58): the testicular tumour that responds best to radiation = **seminoma**.

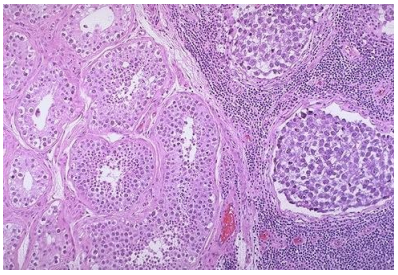
Features of each tumour

Tumour	Key points	Histology	Marker
Seminoma	Age 15-35 (peak 30s). Homogeneous, gray-tan, **bulky rubbery lobulated mass with no haemorrhage or necrosis**. Spreads to retroperitoneal nodes, may skip mediastinum and reach left supraclavicular (Virchow) node	**Large polygonal cells with clear (glycogen-rich) cytoplasm, central round nuclei, "fried-egg" look**, in lobules separated by fibrous septa with **lymphocytes** (and granulomas)	**PLAP**; hCG may be mildly raised
Spermatocytic seminoma	Older men, excellent prognosis, does not metastasize	-	-
Embryonal carcinoma	Age 20-30. **Most undifferentiated** GCT. Bulky mass with **haemorrhage and necrosis**; invades, metastasizes to lung/liver (about 30% metastatic at diagnosis)	**Sheets of large primitive cells with overlapping hyperchromatic nuclei** forming glands/papillae	**AFP and/or hCG**
Yolk sac (endodermal sinus) tumour	**Most common testicular tumour in infants and children**, rare in adults	Interlacing strands of epithelial cells in loose connective tissue forming glomerulus-like structures: **Schiller-Duval bodies**	**AFP**
Choriocarcinoma	Age 15-25. Often a **small primary with extensive haemorrhage and necrosis**; very early haematogenous spread to lung, liver, brain. May cause gynaecomastia	Both **syncytiotrophoblasts (multinucleated, dark cytoplasm)** and **cytotrophoblasts (mononuclear, pale)** are needed for the diagnosis; no villi	**beta-hCG** (correlates with disease). The alpha subunit of hCG resembles FSH, LH, TSH, so can cause **hyperthyroidism and gynaecomastia**
Teratoma	Any age; benign in children, variable in adults (in females mostly benign). Cystic mass containing **cartilage, teeth, hair, muscle, nerve**	Mixture of ectoderm, endoderm and mesoderm in haphazard arrangement; **mature or immature**; may transform into sarcoma etc.	AFP or hCG may be raised
Mixed GCT	**About 60% of GCT are mixed.** Prognosis follows the **worst component**. Teratocarcinoma = teratoma + embryonal carcinoma	-	-

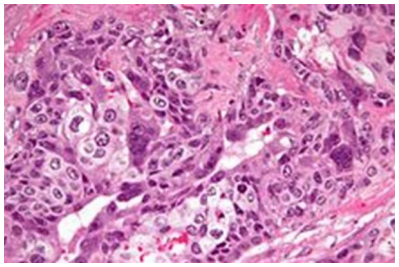
Memory hook: Seminoma = "**S**heets of **S**ame-looking cells with **S**uperb prognosis". Choriocarcinoma = hCG, hyperthyroid and gynaecomastia. Yolk sac = AFP + Schiller-Duval (**D**uval looks like **D**isc/glomerulus).



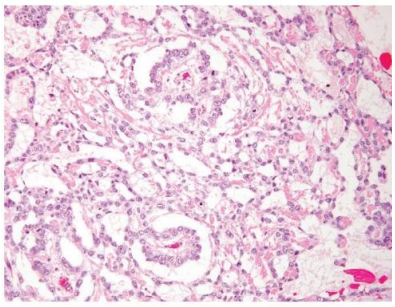
slide 60 *- Seminoma / Nodular tumor is bulging, suggesting that the tumor is firm and rubbery*



slide 61 *- Normal testis appears at the left, and seminoma is present at the right. Note the differen*



slide 67 *- Choriocarcinoma | Syncytiotrophoblasts are multinucleated and have a dark stai*



slide 80 *- ?*

How to read this picture: Slide 60 is the gross seminoma: a uniform, pale, bulging, rubbery tumour replacing the testis, with no bleeding. Slide 61 histology compares normal testis (left) with seminoma nests separated by lymphocyte-rich stroma (right). For choriocarcinoma (slides 66-67) the gross is a small haemorrhagic nodule and the micrograph shows two cell types mixed together (large dark syncytiotrophoblasts and pale cytotrophoblasts). Slide 69 should show a Schiller-Duval body: a central capillary surrounded by tumour cells inside a space, resembling a tiny glomerulus. Slides 71-72 are gross teratomas with hair and teeth.

Sex cord-stromal tumours and lymphoma

Tumour	Key points
Leydig cell tumour	Painless unilateral well-circumscribed mass; median age 70 but any age; produces **androgens (sometimes oestrogens)**. Children: **precocious puberty (masculinisation)**; adults:

	gynaecomastia. **Reinke crystals** (rectangular eosinophilic intracytoplasmic rods) in about 30%. 90% benign
Sertoli cell tumour	Rare (<1%), >90% benign, usually clinically silent; subtypes: sclerosing, large cell calcifying, NOS
Testicular lymphoma	**Most common testicular mass in men >60**; often **bilateral**; **diffuse large B-cell lymphoma**; poor prognosis, relapse in the CNS or opposite testis

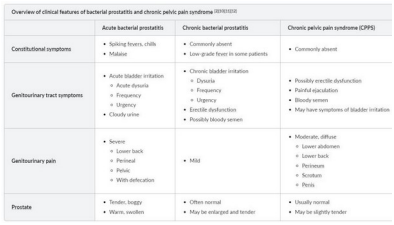
Test yourself (slides 77-79):

9. 30-year-old, 4 cm solid mass, normal AFP and hCG, tan lobulated mass without haemorrhage/necrosis, treated with radiation = **seminoma**.
10. 31-year-old, AFP 81, soft reddish-brown mass, primitive cells in cords and sheets = **embryonal carcinoma**.
11. 2-year-old boy with testicular enlargement and AFP 226 = **yolk sac tumour**.

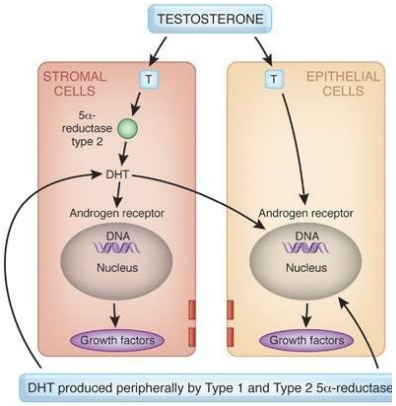
2.4 Prostate

Anatomy and histology (Pathoma; deck slide 11)

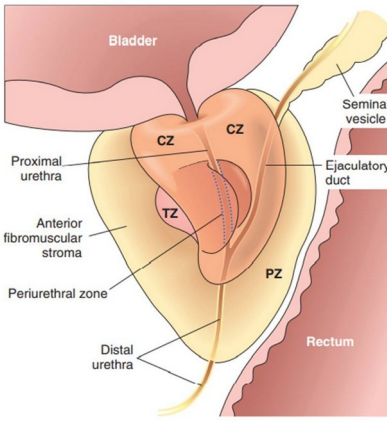
- The prostate has **glands and stroma** (connective tissue); stromal cells make **DHT** (dihydrotestosterone) from testosterone using **5-alpha-reductase**. DHT is the main driver of prostate growth.
- Glands are lined by **two cell layers**: **inner luminal** cells and **outer basal** cells.
- **Zones:** the **transition zone and periurethral zone** (around the urethra) give **BPH**; the **peripheral zone** (posterior, felt on rectal exam) gives **cancer**. The central zone surrounds the ejaculatory ducts.



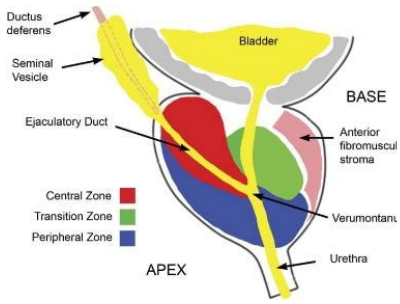
slide 6



slide 9



slide 11 *- Central zone (CZ) / Peripheral zone (PZ) / Transitional zone (TZ) / Periurethral zone.*



slide 12

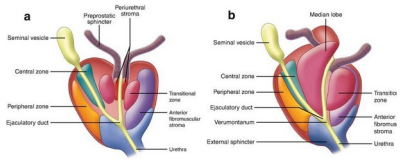
Prostatitis

	Acute prostatitis	Chronic prostatitis
Cause	Bacteria; **young adults: *Chlamydia* and *N. gonorrhoeae***; **older: *E. coli* and *Pseudomonas*** (UTI organisms) by direct spread from the urethra/bladder (or blood)	Chronic inflammation. **Bacterial** (recurrent UTI, may follow acute) or **non-bacterial / chronic pelvic pain syndrome** (no organism, diagnosis of exclusion)
Symptoms	**Dysuria, fever, chills**, perineal pain	Dysuria, burning, suprapubic/perineal/low-back pain, nocturia
Exam	**Tender, boggy prostate** (avoid vigorous massage: risk of bacteraemia)	Often non-tender
Prostatic fluid	High WBC, **positive culture**	High WBC, **culture negative** (in the non-bacterial form)
Microscopy	Neutrophils	Lymphocytes, plasma cells, macrophages
Treatment	Antibiotics	Prolonged antibiotics (bacterial); none specific (non-bacterial)

Chronic prostatitis is a common cause of **recurrent UTI in men** and can raise PSA; it is often found when a biopsy is done to exclude cancer. **Test yourself (slide 7):** 55-year-old, dysuria, E. coli culture, recurs 5 times in 6 months = **chronic bacterial prostatitis**.

Benign prostatic hyperplasia (BPH)

- **Definition:** nodular hyperplasia of **both glands and stroma** (collagen and smooth muscle) in the **transition/periurethral zones**. Extremely common (75% of men 80 or older), age-related, **not premalignant** (no increased cancer risk).
- **Pathogenesis: DHT** acting on androgen receptors of stromal and epithelial cells; sensitivity to androgens rises with age.
- **Gross:** enlarged gland with **well-demarcated nodules** compressing the urethra into a **slit-like lumen**.
- **Symptoms:** *Obstructive:* hesitancy, weak and thin stream, incomplete emptying, straining, post-void dribbling. *Irritative:* urgency, frequency, **nocturia**. Chronic obstruction: bladder wall hypertrophy -> bladder **diverticula**, urinary retention -> **UTIs, hydronephrosis, renal failure**, microscopic haematuria.
- **Labs:** PSA slightly raised (usually <10 ng/mL), because the enlarged gland leaks more PSA (and the prostate makes PSA, which liquefies semen).
- **Treatment:**
  - **Alpha-1 blockers** relax prostate and bladder-neck smooth muscle: **terazosin** (also lowers blood pressure, so useful in hypertensive patients) and **tamsulosin** (alpha-1A selective, spares blood-vessel smooth muscle, so better in normotensive patients).
  - **5-alpha-reductase inhibitors (finasteride)** block conversion of testosterone to DHT, **shrinking** the gland (takes months; also helps baldness; side effects: **decreased libido, sexual dysfunction, gynaecomastia**).
  - **TURP** (transurethral resection of the prostate).



slide 14

How to read this picture: Slides 15-17 (BPH): the gross picture shows well-defined nodules around the urethra, which is squeezed into a slit. Microscopically the nodule is made of large glands with papillary infoldings (crowded, cystically dilated glands with pink secretions, and plenty of fibromuscular stroma). The gland lining is still two-layered.

Test yourself (slide 19): drug that lowers DHT synthesis in stromal cells = blocks **5-alpha-reductase** (finasteride).

Prostate adenocarcinoma

- **Epidemiology: most common cancer in men** (usually after 50); **second leading cause of cancer death in men** (after lung). Risk factors: **age**, **race (African-American > Caucasian > Asian)**, family history, diet high in saturated fat. Genes: **TMPRSS2-ERG fusion** (the androgen-driven TMPRSS2 promoter switches on the ERG oncogene) and **BRCA2** mutations.
- **Location: posterior peripheral zone**, so it does not squeeze the urethra early, hence **clinically silent** until late. It is a hard, irregular, ill-defined, yellow nodule felt on **digital rectal exam**.
- **Screening/diagnosis:** DRE and **PSA** (PSA is **organ-specific but not cancer-specific**: BPH, prostatitis, instrumentation and age can all raise it). Normal PSA rises with age (about 2.5 ng/mL at 40-49; about 7.5 at 70-79); **PSA >10 is very worrying at any age**. In cancer the **free-PSA fraction falls** (cancer PSA is mostly bound to protein). Always confirm with **transrectal ultrasound-guided biopsy**.
- **Microscopy: small, crowded, closely packed glands with prominent nucleoli**; the glands are lined by **one cell layer only (basal cell layer absent)**, which is the key diagnostic clue. Perineural invasion is common. **Prostatic intraepithelial neoplasia (PIN)** is the precursor.
- **Grading: Gleason score** is based on **architecture alone, not on nuclear atypia**. Grade the two most common patterns (each 1 to 5) and add them (total 2-10). Higher score = worse prognosis.
- **Metastasis:** via lymphatics to the obturator and pelvic nodes; via the vertebral venous plexus (Batson plexus) to the **lumbar spine and pelvis** as **osteoblastic (bone-forming, "white" on imaging) lesions**, causing **lower back pain**; **alkaline phosphatase, PSA and prostatic acid phosphatase** rise. (Osteolytic lesions, e.g. from many other cancers, are "black" holes.)
- **Treatment:** localised disease: **prostatectomy**, external beam radiation or cryosurgery; PIN/low-risk disease in an elderly man: monitor PSA. Advanced disease: **androgen deprivation**: orchiectomy or **GnRH analogues (leuprolide)** (continuous use suppresses LH and FSH), plus **androgen-receptor blockers (flutamide)**.

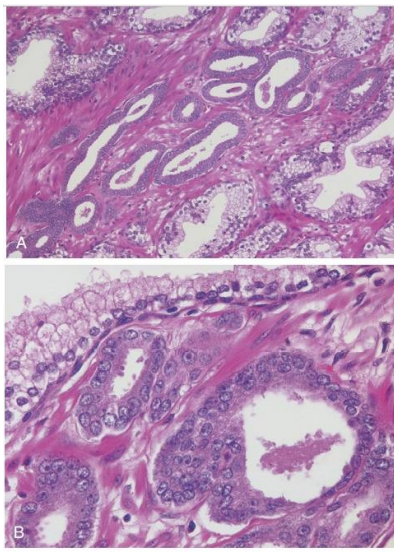
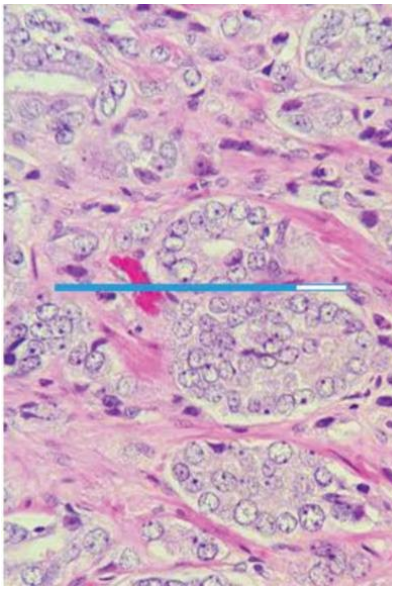
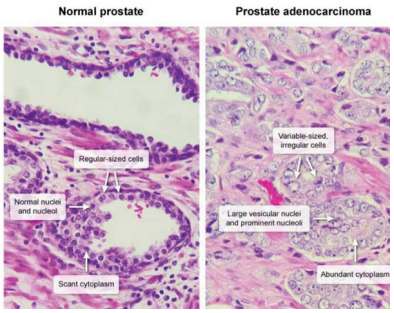


Figure 17–14 **A,** Adenocarcinoma of the prostate demonstrating sm glands crowded in between larger benign glands. **B,** Higher magnificatic shows several small malignant glands with enlarged nuclei, promine nucleoli, and dark cytoplasm, as compared with the larger, benign gla (top).

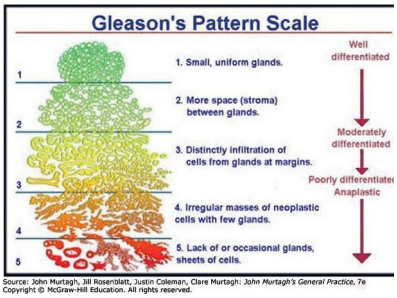
slide 23



slide 24



slide 25



slide 28

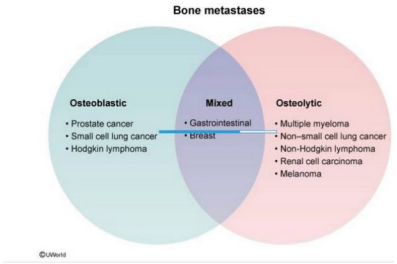
TRADITIONAL GLEASON SCORE	NEW GRADING SYSTEM GROUP 1
GLEASON 3+3=6 Only individual discrete well-formed glands	GRADE 1
GLEASON 3+4=7 Predominantly well-formed glands with a lesser component of poorly-formed/fused/cribiform glands.	GRADE 2
GLEASON 4+3=7 Predominantly poorly-formed/ fused/cribriform glands with a lesser component of well-formed glands.	GRADE 3
GLEASON 4+4=8 Only poorly-formed/fused/cribriform glands or -Predominantly well-formed glands with a lesser component lacking or -Predominantly lacking glands with a lesser component of well-formed glands.	GRADE 4
GLEASON 9-10 Lacks gland formation (or with necrosis) with or without poorly-formed/fused/cribriform gland.	GRADE 5

slide 29

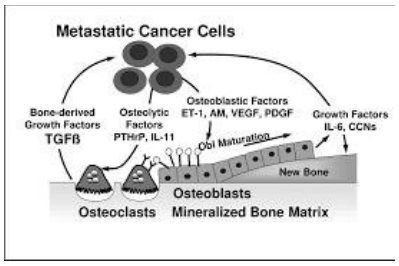
Table 4. American Joint Committee on Cancer Prognostic Stage Grouping[a]

WHEN T IS...	AND N IS...	AND M IS...	AND PSA IS...	AND GRADE GROUP IS...	THEN THE STAGE GROUP IS...
cT1a-c, cT2a	N0	M0	<10 ng/mL	1	I
pT2	N0	M0	<10 ng/mL	1	I
cT1a-c, cT2a	N0	M0	≥10, <20 ng/mL	1	IIA
pT2	N0	M0	≥10, <20 ng/mL	1	IIA
cT2b-c	N0	M0	<20 ng/mL	1	IIA
T1-2	N0	M0	<20 ng/mL	2	IIB
T1-2	N0	M0	<20 ng/mL	3	IIC
T1-2	N0	M0	<20 ng/mL	4	IIC
T1-2	N0	M0	≥20 ng/mL	1-4	IIIA
T3-4	N0	M0	Any	1-4	IIIB
Any T	N0	M0	Any	5	IIIC
Any T	N1	M0	Any	Any	IVA
Any T	Any	M1	Any	Any	IVB

slide 30



slide 31



slide 33

How to read this picture: In the biopsy (slides 26-27) look for a crowded cluster of small glands "back to back" lacking the normal fibrous stroma between them; each cell has a prominent central nucleolus. Compare to normal glands, which are large and folded with a two-layer lining. Slide 32 contrasts osteolytic with osteoblastic bone lesions: osteoblastic (prostate) are dense white patches on a plain film, osteolytic show dark holes.

Test yourself (slides 13, 26, 35, 36):

12. Zone where prostate cancer arises = **peripheral zone**.
13. 70-year-old with a firm nodule, small crowded glands with prominent nucleoli = **adenocarcinoma**.
14. 77-year-old, PSA 6, biopsy shows PIN = **monitor PSA** (do not do radical surgery).
15. 95-year-old with BPH and a focus of grade 1-2 adenocarcinoma on TURP chips = **no further therapy** (indolent disease; he is more likely to die of something else).

2.5 SGD: Prostate (answers to the guiding questions)

BPH case (68-year-old with hesitancy, weak stream, nocturia, smooth rubbery enlarged prostate).

- **Zone involved:** the **transition (and periurethral) zone** that surrounds the urethra.
- **Why obstruction:** the nodules lie around the urethra and **squeeze it into a slit**; the bladder must push harder, so the stream is weak and emptying incomplete.
- **Hormone: DHT**, made from **testosterone** by **5-alpha-reductase** in the stromal cells; DHT binds the androgen receptor in stromal and epithelial cells, promoting growth. It rises relative to testosterone with age, and the tissue becomes more sensitive.
- **Histology:** nodular proliferation of both glands (with papillary infolding) and **stroma (smooth muscle and collagen)**; two cell layers preserved.
- **Symptoms:** hesitancy and weak stream from outlet obstruction; **nocturia** because the bladder never empties completely and the hypertrophied detrusor becomes unstable.
- **Complications:** retention, UTI, bladder stones, diverticula, hydronephrosis, renal failure.
- **Mild PSA rise:** an enlarged gland with more epithelium and distorted architecture leaks more PSA into the blood.
- **Finasteride** (5-alpha-reductase inhibitor) reduces DHT and shrinks the **gland over months**; **tamsulosin** (alpha-1A blocker) relaxes smooth muscle within days without shrinking anything.

Prostatitis case (45-year-old with fever, dysuria, tender boggy prostate).

- Organisms: **E. coli** and other Gram-negative rods; younger men also *Chlamydia* and *Gonorrhoeae*. Reach the prostate by reflux of infected urine, direct extension from urethra/bladder, or blood.
- **Neutrophils** dominate in acute disease; chronic prostatitis shows **lymphocytes, plasma cells and macrophages** with fibrosis.
- **Avoid vigorous massage in acute prostatitis** because it may push bacteria into the blood (bacteraemia/sepsis).
- Prostatitis causes pain, fever and dysuria (and boggy, tender gland) whereas BPH causes painless obstructive symptoms with a smooth rubbery gland and cancer is usually silent with a hard nodule.
- Untreated acute prostatitis can lead to **abscess**, sepsis, chronic prostatitis and urinary retention.

Adenocarcinoma case (72-year-old, PSA 18, hard irregular nodule, biopsy glands lacking basal cells).

- **Zone:** the **peripheral zone**, posterior. Early cancer does not compress the urethra, so **urinary symptoms are absent early**; it presents later than BPH.
- **Genes: TMPRSS2-ERG** fusion (the androgen-responsive TMPRSS2 promoter drives the ERG oncogene) and **BRCA2** mutation (aggressive disease, hereditary breast/ovarian-cancer gene).

- **Absent basal cells** = the diagnostic clue to malignancy (benign glands always have a basal layer; immunostains for basal markers such as p63 or high-molecular-weight keratin are negative in cancer). **Prominent nucleoli** are the nuclear feature of malignant prostatic cells.
- **PSA is not specific**: BPH, prostatitis, ejaculation, rectal exam, instrumentation and age all raise it. Still useful because it is **organ-specific, cheap and tracks response and recurrence**.
- **Metastasis: bone** (axial skeleton, lumbar spine and pelvis). The lesions are **osteoblastic** because prostate cells stimulate osteoblasts (they secrete bone-forming factors such as endothelin-1 and BMPs). **Alkaline phosphatase** rises with bone formation.

Integrative questions.

- Difficulty initiating urination + a hard nodular prostate + raised alkaline phosphatase = **prostatic adenocarcinoma with osteoblastic bone metastases** (the nodule is hard, nodular, the enzyme comes from new bone formation).
- **Finasteride and low libido:** it blocks the conversion of testosterone to DHT; lower androgen signalling reduces libido and can impair erection.
- **Why peripheral-zone cancer presents later than BPH:** BPH arises right around the urethra and obstructs early; peripheral-zone cancer sits far from the urethra and grows silently.

2.6 Quick compare: prostate diseases

	Acute prostatitis	Chronic prostatitis	BPH	Adenocarcinoma
Zone	Whole gland	Whole gland	Transition/ periurethral	**Peripheral (posterior)**
Exam	Tender, boggy	Variable	Smooth, rubbery, symmetric	**Hard, irregular nodule**
PSA	May rise	May rise	Mild (<10)	Often >10
Symptoms	Fever, dysuria	Pelvic pain, recurrent UTI	Obstructive and irritative	Silent early; back pain with metastases
Treatment	Antibiotics	Prolonged antibiotics	Alpha blocker, 5-ARI, TURP	Surgery/radiation; androgen deprivation

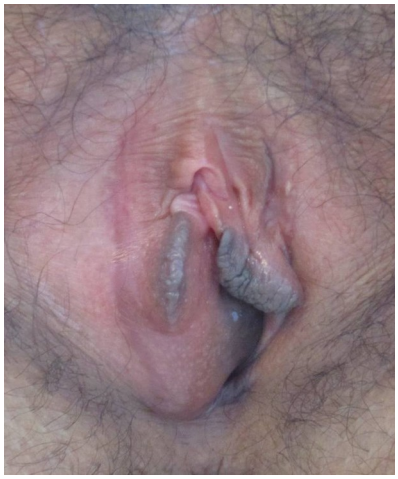
3. Female Reproductive Tract: Vulva, Vagina, Cervix, Uterus

Sources: Teacher deck "Female Reproductive Tract 1" (79 slides), SGD handout "Uterine Disorders", Pathoma Ch 13 (Female Genital System, lecture notes pp. 267-278) and Pathoma slides Ch 17.

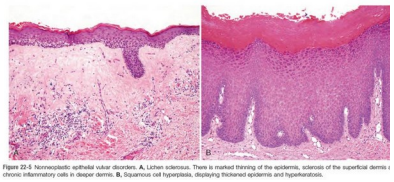
3.1 Vulva

Anatomy and histology: the vulva is the skin and mucosa outside the hymen (labia, mons pubis, vestibule), lined by **keratinised squamous epithelium**.

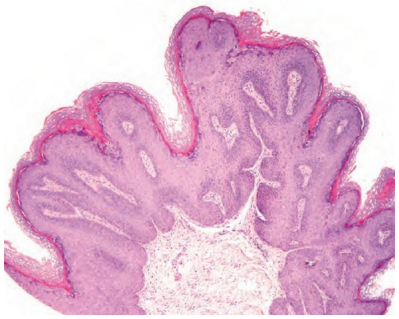
Condition	Key points
Bartholin gland cyst	Reproductive age. **Duct blocked** by inflammation or trauma, secretions pile up. Unilateral, may be painless or give mild dyspareunia (painful intercourse); can become a **Bartholin abscess**. Glands sit at 4 and 8 o'clock and drain to the **inferior vestibule**
Condyloma acuminatum	Genital warts, **HPV 6 and 11** (most common STD). Verrucous, cauliflower-like, may grow profusely in pregnancy. **Koilocytes** are the hallmark. Rarely progress to cancer. **Condyloma latum** (flat, moist, grey plaques) is **secondary syphilis**. Do a Pap smear to rule out coexisting CIS
Lichen sclerosus	Autoimmune, **thin epidermis with dermal sclerosis** (fibrosis) and a band-like lymphocytic infiltrate; smooth white "**parchment-like**" plaques/macules; **postmenopausal** women; itching. Benign but **raises risk of squamous cell carcinoma**
Lichen simplex chronicus	**Squamous hyperplasia** from chronic rubbing/scratching; thick, leathery leukoplakia; any age, often young. **No increased risk** of carcinoma
Papillary hidradenoma	Benign sweat-gland tumour, a sharply circumscribed nodule on the labia majora or interlabial folds; resembles an intraductal papilloma of the breast (ectopic breast-like tissue along the milk line)
Extramammary Paget disease	**Carcinoma in situ** of the vulvar epidermis: **malignant "Paget" cells within the epidermis** (pagetoid spread). Red, itchy, scaly eczema-like, sharply bordered plaque, usually labia majora. **Usually no underlying carcinoma** (unlike Paget disease of the nipple, which means an underlying breast carcinoma). **Paget cells are PAS positive, keratin positive, S100 negative; melanoma is PAS negative, keratin negative, S100 positive** (Pathoma's table lists S100 negative for Paget, positive for melanoma)
Candida vulvovaginitis	Extremely common, especially in diabetics and after antibiotics; erythema and thick white curd-like discharge



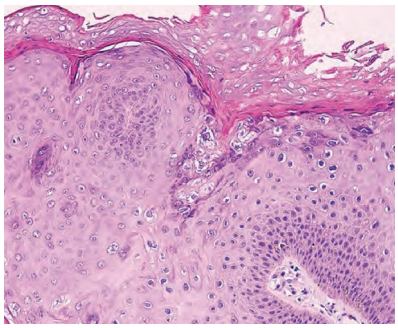
slide 4 *- BARTHOLIN CYST*



slide 6



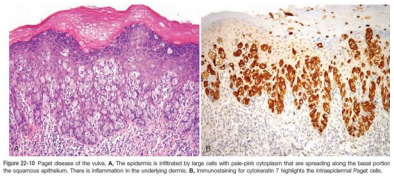
slide 10 *- Condyloma acuminatum | Exophytic, papillary architecture | koilocytic atypia – / n*



slide 10 *- Condyloma acuminatum | Exophytic, papillary architecture | koilocytic atypia – / n*



slide 14



slide 15

How to read this picture: Bartholin cyst (slide 4): a tense smooth swelling at the lower vulva/introitus on one side. Lichen sclerosus (slide 6): pale white atrophic skin, sometimes with a "figure-8" pattern around vulva and anus. Condyloma (slide 10): koilocytes have a raisin-like wrinkled nucleus with a halo. Extramammary Paget (slides 14-15): single large pale cells with abundant clear cytoplasm scattered through the epidermis, often in clusters.

Test yourself (slide 7): 64-year-old with itchy pale grey patches, thinning epithelium with dermal fibrosis and chronic inflammation = **lichen sclerosus**.

Vulvar carcinoma

- Relatively rare. Two pathways:

	HPV-related	**Non-HPV**
Age	Reproductive age to 60	**>70 (average 75)**
Cause	HPV **16, 18**	Long-standing **lichen sclerosus**
Precursor	**Vulvar intraepithelial neoplasia (VIN)**: koilocytic change, disordered maturation, nuclear atypia	Differentiated VIN
Histology	**Basaloid and warty** carcinomas	**Keratinising** SCC

- Presents as leukoplakia (white plaque), needs **biopsy**.

3.2 Vagina

- Lined by **non-keratinising squamous epithelium**. The lower third comes from the urogenital sinus; the upper part from the Mullerian (paramesonephric) ducts, which originally had a columnar lining.
- **Vaginal adenosis:** persistence of **columnar (Mullerian) epithelium in the upper vagina**. Cause: **diethylstilbestrol (DES)** taken by the mother in pregnancy (crosses the placenta). Benign but a **precursor to clear cell adenocarcinoma**. DES-exposed women also have early abortion, ectopic pregnancy and premature birth.
- **Clear cell adenocarcinoma:** malignant glands with clear cytoplasm; rare; DES exposure in utero. Presents with spotting or a vaginal nodule (slide 19).
- **Embryonal rhabdomyosarcoma (sarcoma botryoides):** malignant **immature skeletal muscle** ("rhabdo" = skeletal). Most common malignancy of infants and young children (<5, typically <4). A **soft, polypoid, grape-like mass** protruding from the vagina (or penis/bladder in boys), bleeding. Microscopy: spindle cells, **rare cross-striations**, a **cambium layer** (tumour hugging the layer under the epithelium). **Desmin and myogenin positive** (skeletal-muscle markers).
- **Vaginal squamous cell carcinoma: HPV 16, 18, 31, 33**; precursor **VAIN**. Lymph spread: **upper two-thirds -> pelvic (iliac) nodes; lower third -> inguinal nodes**. Careful: your Pathoma note (p. 271) states it the other way round (upper third to inguinal, lower two-thirds to iliac), which conflicts with the standard teaching (First Aid and most lecturers). Check which version your teacher uses; the rule most exams follow is *upper = pelvic, lower = inguinal* (like the vulva, which drains to the inguinal nodes).
- **Embryologic remnants: Gartner duct cyst** (mesonephric/Wolffian remnant, anterolateral wall of the upper vagina, cuboidal/columnar non-mucinous lining) and **paraovarian / paratubal cysts** (hydatid cysts of Morgagni, around the broad ligament; thin-walled, translucent, near the fimbria).
- **Imperforate hymen:** girl with pelvic pain, no menses and a fluid-filled distended vagina (haematocolpos).



slide 36

Test yourself (slides 19, 23, 26, 27): nodular upper vaginal mass with clear cell carcinoma -> **DES exposure**. 14-year-old with pain, no menses, vagina distended with fluid -> **imperforate hymen**. Small translucent fluid cysts near the fimbriated end of the tube -> **paratubal (paraovarian) cysts**. Tender cyst in the lateral vaginal wall lined by cuboidal epithelium -> **embryologic remnant (Gartner duct cyst)**.

3.3 Cervix

Anatomy

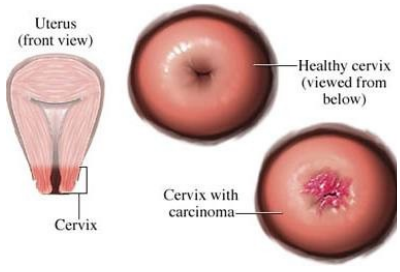
- **Exocervix (ectocervix):** visible at speculum exam; **non-keratinising squamous epithelium**. **Endocervix:** not visible; **single layer of columnar (mucus) cells**. The junction is the **transformation zone (squamocolumnar junction)**, where squamous metaplasia happens and **where HPV does its damage**.
- **Microglandular hyperplasia:** benign, non-neoplastic endocervical gland proliferation in reproductive-age women, linked to hormones (oral contraceptives, pregnancy); can cause contact bleeding (slide 31).
- **Cervicitis / pelvic inflammatory disease (PID):** see 3.4.

HPV and cervical dysplasia (Pathoma + slides 38-54)

- **HPV** infects the basal cells of the transformation zone. Most infections are cleared by the immune system. Persistent infection with **high-risk HPV (16, 18, 31, 33)** produces **E6 (degrades p53)** and **E7 (inactivates Rb)**, disabling the two main tumour-suppressor brakes. Low-risk types 6 and 11 cause genital warts but not cancer.
- **Vaccination:** covers 6 and 11 (prevents condyloma) and 16 and 18 (prevents CIN and cervical cancer).
- **Cervical intraepithelial neoplasia (CIN):** dysplasia with koilocytic change. Progression is slow (**10-20 years**) and **not inevitable**.

Grade	Thickness involved	Behaviour
CIN I (mild dysplasia)	less than 1/3	Often regresses, reversible
CIN II (moderate)	up to 2/3	Reversible less often
CIN III (severe)	almost the full thickness	High risk
Carcinoma in situ	Full thickness, **basement membrane intact**	Irreversible, the step before invasion
Invasive SCC	Breaks through the basement membrane	Cancer

Note that cytology reports LSIL (about CIN I) and HSIL (CIN II and III) (slide 42: CIN III = **HSIL**, caused by **HPV**; slide 54: dysplasia through the full thickness = **HSIL / CIN III**).



slide 39

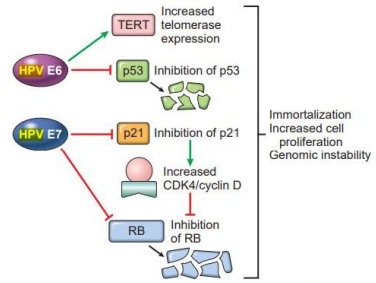
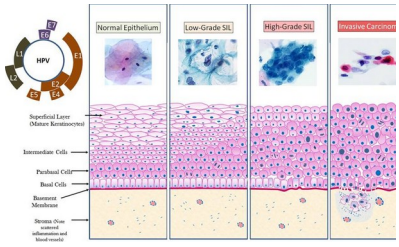
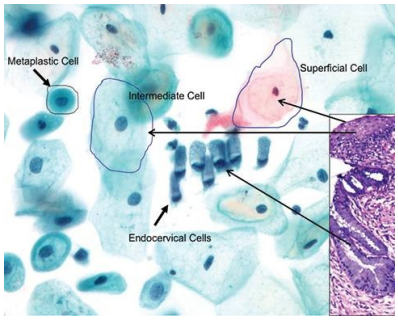


Figure 7.42 Transforming effects of human papillomavirus (*HPV*) E6 and E7 proteins. The net effect of HPV E6 and E7 proteins is to immortalize cells and remove the restraints on cell proliferation (see Fig. 7.26). *TERT*, Telomerase reverse transcriptase. (Modified from Münger K, Howley PM: Human papillomavirus immortalization and transformation functions, *Virus Res* 89:213–228, 2002.)

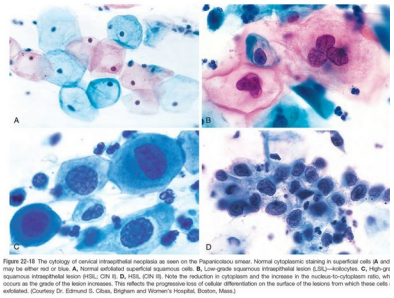


slide 43 *- Figure 1. Sketch demonstrating the progressive nature of the dysplastic changes leading to | Article Copyright © 2018 Authors, Source DOI: 10.1177/237428*

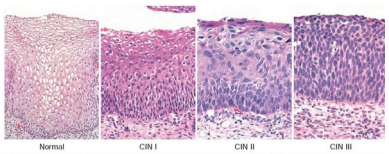
slide 41



slide 44



slide 45



slide 46

Table 22-1 Classification Systems for Squamous Cervical Precursor Lesions

Dysplasia/Carcinoma in Situ	Cervical Intraepithelial Neoplasia (CIN)	Squamous Intraepithelial Lesion (SIL), Current Classification
Mild dysplasia	CIN I	Low-grade SIL (LSIL)
Moderate dysplasia	CIN II	High-grade SIL (HSIL)
Severe dysplasia	CIN III	High-grade SIL (HSIL)
Carcinoma in situ	CIN III	High-grade SIL (HSIL)

CIN, Cervical intraepithelial neoplasia; SIL, squamous intraepithelial lesion.

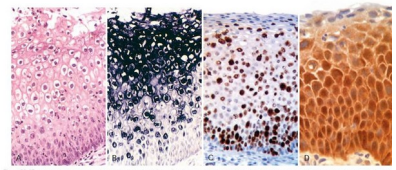
slide 47

Table 22-2 Natural History of Squamous Intraepithelial Lesions with Approximate 2-Year Follow-up

Lesion	Regress	Persist	Progress
LSIL	60%	30%	10% to HSIL
HSIL	30%	60%	10% to carcinoma*

HSIL, High-grade squamous intraepithelial lesion; LSIL, low-grade squamous intraepithelial lesion.
*Progression within 2 to 10 years.

slide 47



slide 48

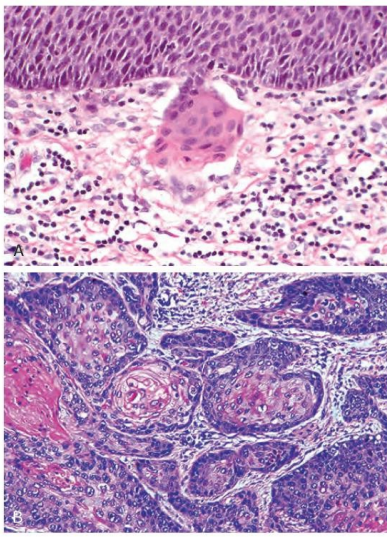


Figure 22-16 Squamous cell carcinoma of the cervix. **A,** Microinvasive squamous cell carcinoma with invasive nest breaking through the basement membrane of high-grade squamous intraepithelial lesion. **B,** Invasive squamous cell carcinoma.

slide 49

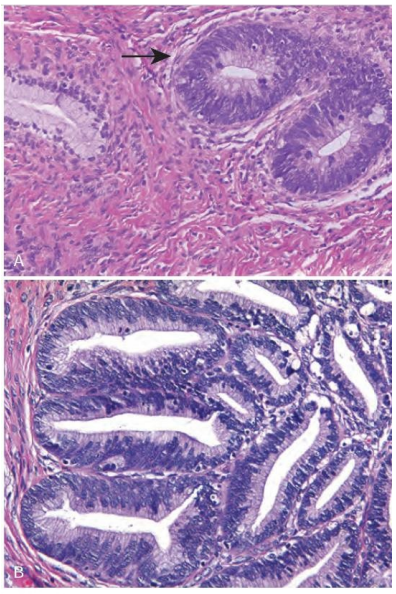


Figure 22-17 Adenocarcinoma of the cervix. **A,** Adenocarcinoma in situ *(arrow)* showing dark glands adjacent to normal pale endocervical glands. **B,** Invasive adenocarcinoma.

slide 50

How to read this picture: Slides 43-50 are pap-smear and biopsy pictures. On a normal Pap (slide 45) squamous cells are flat with a small round nucleus and abundant cytoplasm; endocervical columnar cells are blue-staining and sometimes show cilia. **Koilocytes** (abnormal) show a large, irregular, dark nucleus with a clear halo and a sharply outlined cytoplasm. On cervical biopsy, normal squamous epithelium matures from small basal cells to flat surface cells; in CIN the immature (basaloid) cells with high nuclear:cytoplasmic ratio occupy more and more of the thickness, until in CIN III/CIS they occupy it all.

Cervical carcinoma

- **Types: squamous cell carcinoma 80%**, adenocarcinoma 15%, others (clear cell, transitional).

- **Epidemiology:** third most common gynaecologic cancer in the US (endometrial > ovarian > cervical); peak in the 40s.
- **Risk factors:** HPV (main), early first intercourse, multiple partners, multiple pregnancies, oral contraceptives, **smoking**, and **immunosuppression** (cervical carcinoma is an AIDS-defining illness).
- **Presentation:** often asymptomatic; **postcoital bleeding**, dyspareunia, malodorous discharge, abnormal bleeding.
- **Spread and complications:** local invasion into the bladder/ureters causes **hydroureter, hydronephrosis and post-renal failure (the commonest cause of death)**; iliac and obturator lymph node involvement -> **leg lymphoedema**; lumbosacral plexus invasion -> back, hip and leg pain; vesicovaginal and rectovaginal fistulas.
- **Screening - the Pap smear** is "the most successful screening test ever developed": cervical cancer moved from the most common to among the least common gynaecologic cancers in the US. Brush cells from the transformation zone; start **at 21 and repeat every 3 years** (with HPV co-testing older than 30 in current practice). Abnormal -> **colposcopy** (magnified view) with biopsy; the **Schiller test** (Lugol iodine) stains normal glycogen-rich epithelium brown while abnormal epithelium stays pale. **Limitations:** good for squamous disease, poor for **adenocarcinoma** (which hides in the endocervical canal), and false negatives occur if the transformation zone is not sampled.

3.4 Pelvic inflammatory disease (PID)

- **Ascending infection** from the cervix to the endometrium, tubes and pelvis. Organisms: ***Neisseria gonorrhoeae*** (Gram-negative diplococci) and ***Chlamydia trachomatis*** (intracytoplasmic inclusions); puerperal infections: staphylococci, streptococci, coliforms, *Clostridium perfringens*.
- **By site:** cervicitis (cervical motion tenderness, the "**chandelier sign**"), **endometritis**, **salpingitis**, pelvic peritonitis and abscess. **Fitz-Hugh-Curtis syndrome:** perihepatitis with "**violin-string**" adhesions between the tube/peritoneum and the liver capsule, causing right upper quadrant pleuritic pain.
- **Presentation:** fever, chills, discharge, bilateral lower abdominal pain, cervical motion and adnexal tenderness.
- **Complications: tubal scarring -> infertility and ectopic pregnancy**, tubo-ovarian abscess, adhesions with intestinal obstruction. Tuberculous salpingitis occurs in developing countries (pelvic masses not responsive to antibiotics).

Test yourself (slide 37): 10 years of infertility with mild pelvic pain, normal exam, ultrasound and sperm count = **old PID with tubal scarring**.

3.5 Uterus: the menstrual cycle first (Pathoma)

Phase	Hormone	Endometrium
Proliferative	**Estrogen**	Grows
Secretory	**Progesterone** (from the corpus luteum)	Prepares for implantation
Menstrual	**Loss of progesterone**	Sheds

Estrogen without progesterone ("unopposed estrogen") is the cause of hyperplasia and most endometrial cancer, so keep that thought.

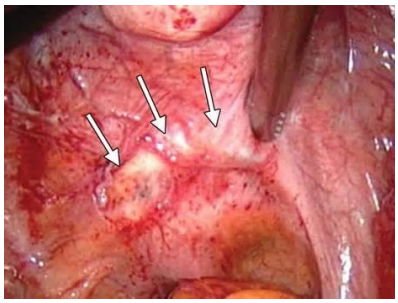
Non-neoplastic uterine disorders

Condition	Key points
Asherman syndrome	Secondary amenorrhoea from **loss of the basalis (stem-cell) layer and scarring** after aggressive dilation and curettage (D&C)
Anovulatory cycle	No ovulation -> no corpus luteum -> no progesterone; endometrium grows from estrogen but isn't maintained and falls off. **Common cause of dysfunctional (abnormal) uterine bleeding at menarche and menopause**
Acute endometritis	Bacterial; usually **retained products of conception** (after delivery/miscarriage) acting as a focus. Fever, bleeding, pelvic pain. Ascending infection associated with pregnancy/abortion
Chronic endometritis	**Plasma cells (essential for the diagnosis)** and lymphocytes in the endometrium. Causes: retained products, **IUD**, **TB**, **PID** (gonorrhoea/chlamydia). Bleeding, pain, infertility. Organisms include *Ureaplasma*, *Gardnerella*, group B strep
Endometrial polyp	Hyperplastic protrusion of endometrium (focal overgrowth). Postmenopausal women. **Tamoxifen** (anti-estrogen in breast, pro-estrogen in endometrium), hypertension, obesity, HRT. Abnormal bleeding
Endometriosis	Functional endometrial **glands and stroma outside the uterine lining** (they respond to the menstrual cycle). Most likely **retrograde menstruation** (plus metaplasia/lymphatic spread). Reproductive age
Adenomyosis	**Endometrial glands and stroma inside the myometrium** (basal-layer hyperplasia grows into the wall). **Uniformly enlarged, soft, globular, tender uterus**; dysmenorrhoea, menorrhagia. Definitive treatment: hysterectomy
Endometrial hyperplasia	Hyperplasia of endometrial glands relative to stroma from **unopposed estrogen**: obesity (fat makes estrone), **PCOS**, granulosa-cell tumours, estrogen replacement without progestin, follicle persistence in anovulatory cycles. Classic presentation: postmenopausal bleeding. **With atypia = high risk of carcinoma**; without atypia low risk. Classified by architecture (simple vs complex) and by atypia

Endometriosis in more detail

- **Common sites:** ovary (most common; a haemorrhagic **"chocolate cyst" = endometrioma**), uterine and ovarian ligaments (pelvic pain), **pouch of Douglas** (pain with defecation), bladder wall (dysuria and haematuria), bowel serosa (pain and adhesions), **fallopian tube** (scarring -> ectopic pregnancy). Lesions look like yellow-brown "**gunpowder / powder-burn**" or raspberry/blueberry nodules (old blood).
- **Symptoms:** cyclical pelvic pain before, during and after menstruation, **dysmenorrhoea, dyspareunia**, pain on defecation/urination, **infertility**, rectal pain.
- **Complication:** increased risk of **endometrioid (and clear cell) ovarian carcinoma** at the site.
- **SGD answers:** the diagnosis is endometriosis (29-year-old with pain starting 2-3 days before menses, dyspareunia, infertility, dark brown peritoneal lesions on laparoscopy). **Theories:** retrograde menstruation (Sampson), coelomic (peritoneal) metaplasia, lymphatic/haematogenous spread. **Infertility:** adhesions and tubal distortion, an inflammatory peritoneal environment hostile to sperm

and embryo, and ovarian dysfunction. **Cytokines raised:** IL-1, IL-6, IL-8, TNF-alpha (and prostaglandins; high local estrogen), explaining pain and inflammation.



slide 59 *- Endometriosis*

Test yourself (slides 61, 63): brown nodules on the serosa of uterus, tubes and appendix, microscopy -> **endometrial stroma (and glands)** (endometriosis). Uterus 3 times normal size, symmetric, dysmenorrhoea, no masses -> **endometrial glands and stroma in the myometrium (adenomyosis)**.

	Endometriosis	**Adenomyosis**
Where	Outside the uterus	Inside the myometrium
Uterus	Normal size	Diffusely enlarged, boggy, tender
Age	Reproductive (younger)	Older, multiparous (about 40s)
Main features	Pelvic pain, infertility, chocolate cysts	Heavy painful periods

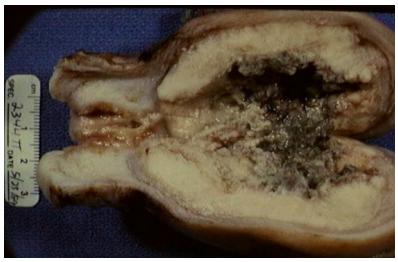
3.6 Uterine tumours

Endometrial carcinoma (most common invasive carcinoma of the female genital tract)

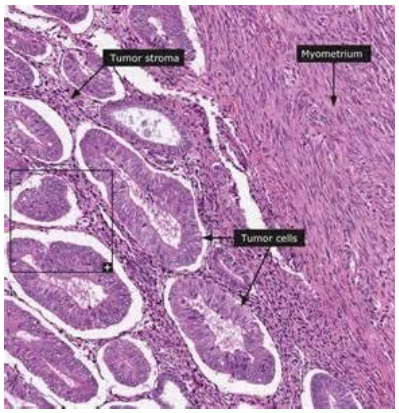
- Typically **postmenopausal bleeding** (always concerning, since cancer must be excluded; endometrial thickness above about 4 mm on ultrasound in a postmenopausal woman is abnormal; slide 76 notes cervical stenosis with **pyometra** or **haematometra** is highly suggestive).
- **Risk factors (estrogen excess):** early menarche/late menopause, nulliparity, obesity, diabetes, hypertension, chronic anovulation/PCOS, estrogen-secreting ovarian tumours (granulosa cell), unopposed estrogen therapy and **tamoxifen**, atypical hyperplasia, **Lynch syndrome/HNPCC** (colon, endometrial and ovarian cancers).

	Type I (hyperplasia pathway)	**Type II (sporadic pathway)**
Share	About 75%	About 25%
Driver	Long-term estrogen exposure	Estrogen-independent, arises in **atrophic endometrium**
Precursor	Atypical (complex) endometrial hyperplasia	No clear precursor (serous intraepithelial carcinoma)
Genes	**PTEN** inactivation (also mismatch-repair genes, MSI, KRAS, PIK3CA)	**p53 (TP53) mutation**
Age	About 55-65 (avg 60)	About 65-75 (avg 70)
Histology	**Endometrioid** adenocarcinoma (resembles normal endometrial glands)	**Serous** adenocarcinoma: papillary tufts with **psammoma bodies** (also clear cell, mixed Mullerian)
Behaviour	Good prognosis	Aggressive

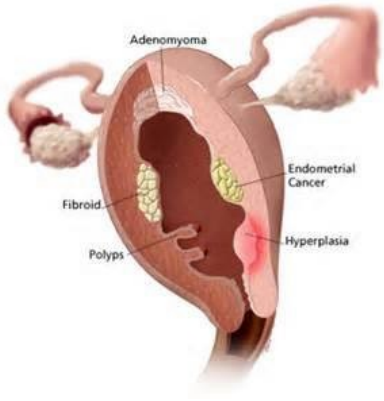
- **Grading/prognosis:** the **depth of myometrial invasion** is prognostically important. Gross: tan polypoid endometrial mass.



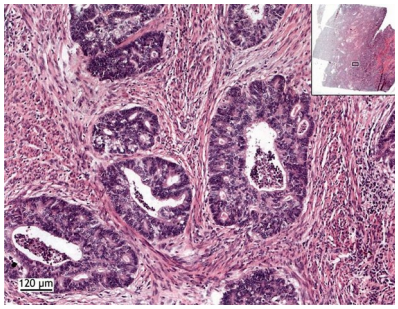
slide 72 *- Endometrial adenocarcinoma*



slide 73 *- Endometrial adenocarcinoma*



slide 73 *- Endometrial adenocarcinoma*



slide 77 *- Endometrial adenocarcinoma | Endometroid endometrial adenocarcinoma | Endometroid endomet*

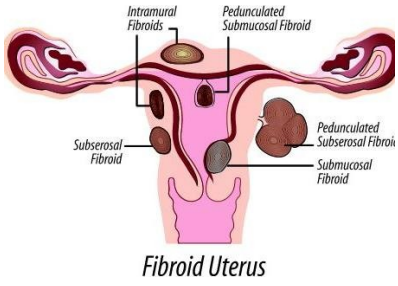
Test yourself (slides 79 and the SGD cases): 72-year-old with bleeding, microcytic anaemia and atypical hyperplasia: risk factor = **unopposed estrogen**. 52-year-old obese woman with chronic anovulation and crowded glands without stromal invasion = **endometrial hyperplasia** (due to unopposed estrogen; obesity makes estrone in fat; **PTEN** mutation; hyperplasia **with atypia carries a roughly 25-30% progression risk**, without atypia about 1-3%). 62-year-old, obese, diabetic, nulliparous with painless bleeding and atypical crowded glands = **endometrial carcinoma, type I (endometrioid)**. Postmenopausal bleeding is always concerning because **endometrial cancer is the commonest cause in a woman with a 14 mm lining and risk factors**; the patient group at highest risk is **obese, postmenopausal women**.

Leiomyoma (fibroid) and leiomyosarcoma

	Leiomyoma	**Leiomyosarcoma**
Nature	**Most common tumour in females**; benign smooth-muscle tumour of the myometrium	Malignant smooth-muscle tumour
Age	Premenopausal, higher in African-American women	Postmenopausal
Number	Often **multiple**	**Single**
Estrogen	Grows with estrogen, **grows in pregnancy, shrinks after menopause**	-
Gross	Well-circumscribed, rubbery, white-tan, **whorled** cut surface; subserosal, intramural or submucosal	Bulky, soft, **necrosis and haemorrhage**

Histology	Bland smooth-muscle cells, no atypia	**Cellular atypia, mitoses (10-20 per hpf), necrosis**
Progress	**Does not become sarcoma**	Arises **de novo** (not from fibroids)
Presentation	Often asymptomatic; **menorrhagia** (heavy bleeding, iron-deficiency anaemia), pelvic mass, pressure symptoms (urinary frequency), pain, **infertility**	Bleeding, mass

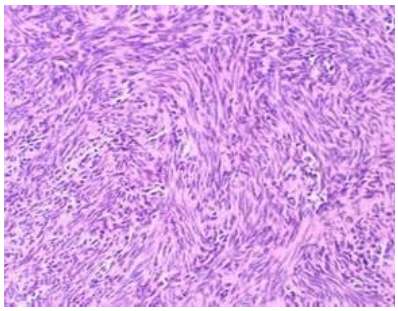
- **SGD answers:** fibroids arise from **myometrial smooth-muscle cells**, are monoclonal and commonly carry **MED12** mutations (an exon 2 mutation in the mediator complex gene found in about 70% of fibroids). They enlarge in pregnancy because estrogen and progesterone are high; they shrink after menopause when estrogen falls. They cause infertility by distorting the endometrial cavity (submucosal fibroids), blocking tubal openings and impairing implantation.



slide 67



slide 68 *- Uterine leiomyomas*



slide 68 *- Uterine leiomyomas*

Test yourself (slides 70, 78): 43-year-old with 6-8 day heavy periods and a twice-normal nodular uterus = **leiomyomata**. 50-year-old with a 9 cm solid mass, cellular spindle cells with 10-20 mitoses/hpf = **leiomyosarcoma**.

3.7 SGD summary: abnormal uterine bleeding (AUB) causes

Use the mnemonic **PALM-COEIN**: **P**olyp, **A**denomyosis, **L**eiomyoma, **M**alignancy and hyperplasia (structural); **C**oagulopathy, **O**vulatory dysfunction, **E**ndometrial, **I**atrogenic (e.g. IUD, hormones, tamoxifen), **N**ot yet classified (non-structural). Age matters: adolescents (anovulation, bleeding disorders), reproductive (pregnancy complications, fibroids, polyps, adenomyosis), postmenopausal (**endometrial cancer until proven otherwise**).

4. Developmental Disorders of the Reproductive Tract (Disorders of Sex Development)

Sources: Teacher deck "Developmental Disorders of Reproductive Tract" (69 slides), Pathoma Ch 17 reproductive slides, First Aid for background.

4.1 Embryology you need first (plain language)

- Every embryo starts **bipotential**: it can become male or female. It has both sets of internal ducts: **Wolffian (mesonephric)** ducts (male plumbing) and **Mullerian (paramesonephric)** ducts (female plumbing).
- The **SRY gene** (on the Y chromosome) switches the gonad into a **testis**. Then the testis makes two hormones:

1. **Anti-Mullerian hormone (AMH)** from **Sertoli cells**: it makes the Mullerian ducts **disappear** (no uterus, tubes, upper vagina). 2. **Testosterone** from **Leydig cells**: it keeps the Wolffian ducts (epididymis, vas deferens, seminal vesicles). In the skin, enzyme **5-alpha-reductase** turns testosterone into **DHT**, which builds the **external genitalia** (penis, scrotum, prostate).

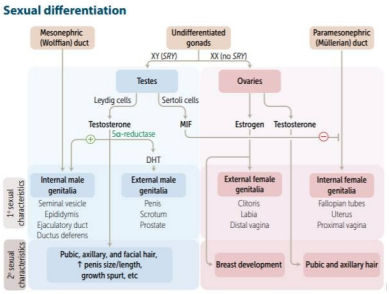
- With **no SRY** the gonad becomes an **ovary**, no AMH is made, the **Mullerian ducts persist** (uterus, tubes, upper two-thirds of the vagina), the Wolffian ducts regress and the external genitalia stay female.
- Rule of thumb: **Sertoli = AMH = removes uterus. Leydig = testosterone = builds male internal organs. DHT = builds male external organs.**
- **DSD (disorder of sex development)** = chromosomal, gonadal or anatomic sex do not line up. The deck groups them by karyotype: **sex chromosome DSD**, **46,XY DSD** and **46,XX DSD**.

4.2 Sex chromosome disorders

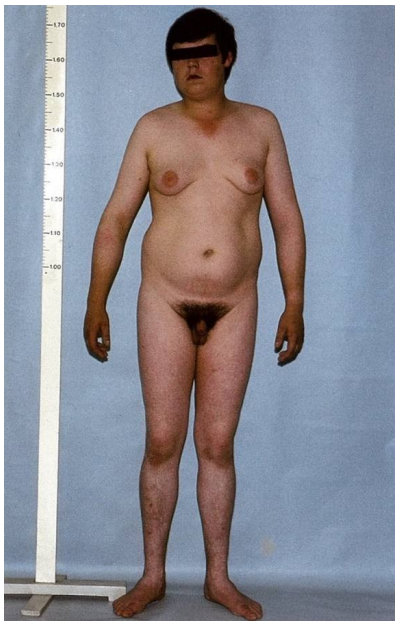
	Klinefelter syndrome	Turner syndrome	Double Y (47,XYY)
Karyotype	**47,XXY** (a Barr body is present, the inactive X)	**45,XO** (no Barr body); mosaics 45,XO/46,XX are milder	**47,XYY** or 46,XY/47,XYY mosaic
Cause	Nondisjunction in meiosis, older maternal age	Usually paternal meiotic nondisjunction (or mitotic error -> mosaic)	Paternal meiotic nondisjunction
Phenotype	**Male**, one of the commonest causes of male hypogonadism	**Female** with streak ovaries	Male, often undiagnosed
Key findings	Tall, slim, **eunuchoid** body (long arms and legs because growth plates close late), **gynaecomastia**, small firm testes, little facial/body hair, reduced fertility, possible learning difficulty. Associated: mitral valve prolapse, **increased risk of breast and testicular cancer**	Short stature (one SHOX gene), **webbed neck** (from cystic hygroma), shield chest with wide-spaced nipples, cubitus valgus, short 4th metacarpals, low hairline, high arched palate, **primary amenorrhoea**, infertility, **lymphoedema of hands and feet** in newborns	**Tall**, severe acne, normal fertility, mild motor/language delay, more ADHD, autism, learning disability
Heart/kidney	-	**Bicuspid aortic valve, coarctation of the aorta**, aortic dissection,	-

		hypertension; **horseshoe kidney**	
Other	-	Hashimoto thyroiditis, type 2 diabetes, osteoporosis, **gonadoblastoma** (if Y material)	-
Hormones	**High FSH and LH**, low testosterone	**Hypergonadotropic hypogonadism**: low estrogen, high FSH and LH	Normal
Treatment	Testosterone replacement	Estrogen + progestin, growth hormone, remove streak gonads if Y material	Usually none

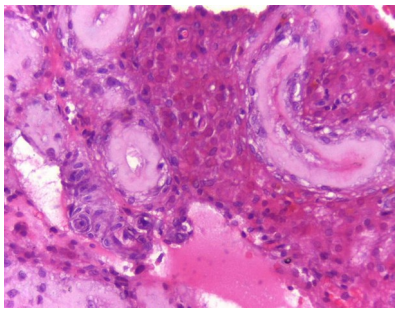
- **Why Klinefelter has high FSH/LH:** seminiferous tubule dysgenesis -> loss of Sertoli cells -> **low inhibin B -> high FSH**; Leydig dysfunction -> **low testosterone -> high LH**. High LH/FSH drives **testosterone to be converted to estrogen** (gynaecomastia). **Histology:** hyalinised, fibrotic seminiferous tubules with prominent hyperplastic Leydig cells.
- **Turner heart rule:** Turner = **left-sided** heart lesions (bicuspid valve, coarctation); Noonan = right-sided. Turner shows brachial-femoral delay (weaker leg pulses) with coarctation.



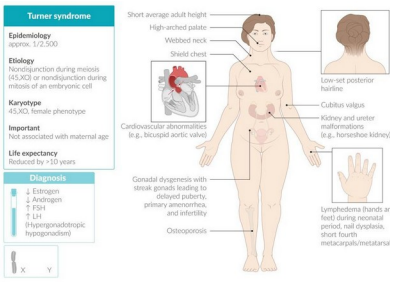
slide 3



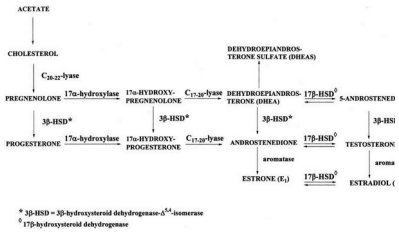
slide 9



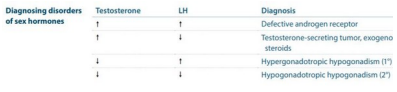
slide 10 *- KLINEFELTER SYNDROME | Prominent, hyperplastic Leydig cells are visible, and the se*



slide 16



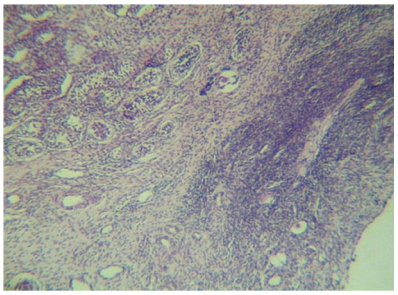
slide 30



slide 43

Hormonal changes in oligospermia	
Ejaculatory duct obstruction	• Normal FSH, LH, testosterone
Primary hypogonadism	• Low testosterone • Elevated FSH, LH
Secondary (hypogonadotropic) hypogonadism	• Low testosterone • Low/normal FSH, LH
Partial androgen resistance	• Elevated LH, testosterone • Normal FSH

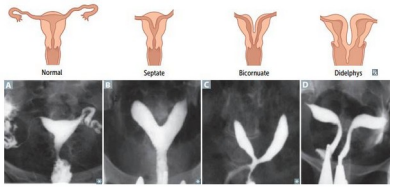
slide 44



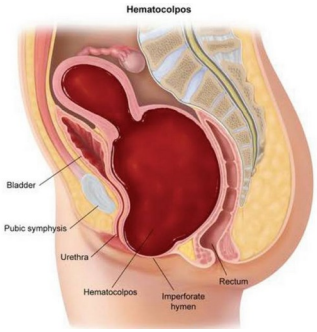
slide 50 *- OVOTESTIS | Seminiferous tubules | Primordial follicles*

UTERUS	BREASTS	DISORDERS
⊕	⊖	Hypergonadotropic hypogonadism (eg, Turner syndrome, genetic mosaicism, pure gonadal dysgenesis) Hypogonadotropic hypogonadism (eg, CNS lesions, Kallmann syndrome)
⊖	⊕	Uterovaginal agenesis in genotypic female or androgen insensitivity in genotypic male
⊖	⊖	Male genotype with insufficient production of testosterone

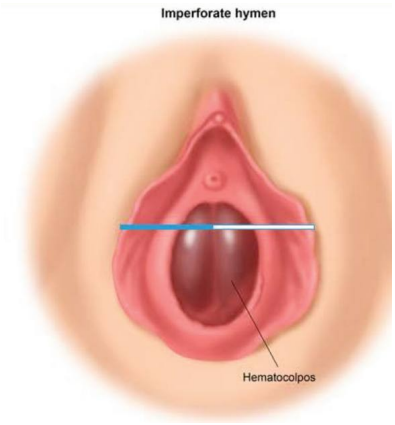
slide 59



slide 65



slide 68



slide 68

4.3 46,XY DSD (a male karyotype with under-masculinised or female-appearing body)

Condition	Defect	What happens	Hormones / key test	Treatment
Androgen insensitivity syndrome (AIS) (most common cause of 46,XY DSD)	**X-linked recessive** mutation of the **androgen receptor**	Complete form: **female external genitalia and breasts** (testosterone is aromatised to estrogen), **blind-ending vagina**, **no uterus or tubes** (testes still make AMH), absent male internal organs, **undescended testes** (in labia, inguinal canal or abdomen), scant pubic hair, **primary amenorrhoea**	Before puberty high testosterone; after puberty **high LH, high estrogen, normal/high testosterone, no virilisation**	Estrogen; remove the gonads after puberty (they can turn malignant, e.g. gonadoblastoma); psychological support. Partial AIS: depends on phenotype and gender identity
5-alpha-reductase deficiency	Autosomal recessive **SRD5A2** loss-of-function	**Cannot make DHT**: female or ambiguous external genitalia at birth (pseudovaginal perineoscrotal hypospadias), but **male internal organs**	Normal/high testosterone, **low DHT, testosterone:DHT ratio >20**	Gender identity decides: gonadectomy + estrogen, or testosterone; correct hypospadias

		and, at puberty, high testosterone causes **masculinisation, phallic growth and testicular descent** ("penis at 12")		
17-beta-hydroxysteroid dehydrogenase 3 deficiency	Autosomal recessive **HSD17B3**; cannot convert androstenedione to testosterone	Ambiguous or female-looking genitalia (blind vaginal pouch, clitoromegaly) with testes; **virilises at puberty** (male hair, voice, gynaecomastia)	**hCG stimulation test**: high androstenedione, low testosterone:androstenedione	Surgery and/or gonadectomy
Kallmann syndrome	>20 gene mutations; **GnRH neurons fail to migrate** from the olfactory placode to the hypothalamus	**Hypogonadotropic hypogonadism + anosmia** (no sense of smell); delayed/absent puberty, infertility, cryptorchidism and low sperm count in males, primary amenorrhoea in females; may have renal agenesis, cleft lip/palate. Can be 46,XY **or 46,XX**	Low GnRH, FSH, LH, sex steroids; otherwise normal pituitary function	Testosterone or estrogen/progesterone; gonadotropins or pulsatile GnRH for fertility
Persistent Mullerian duct syndrome	**AMH or AMH receptor (AMHR2)** mutation	**Normal male** external genitalia, but a **uterus, tubes and upper vagina** are present; cryptorchidism, inguinal hernia, haematospermia; discovered during orchiopexy or hernia repair	Low AMH (ELISA)	Orchiopexy, remove Mullerian remnants
46,XY complete gonadal dysgenesis (Swyer syndrome)	**SRY** mutation in about 20%, other male-development genes	Streak gonads, so **no AMH and no testosterone**: **female phenotype with a uterus and tubes** (Mullerian ducts persist) but no ovaries; **primary amenorrhoea**, absent breast development, infertility	**Hypergonadotropic hypogonadism** (high FSH, LH)	Estrogen/ progesterone; remove streak gonads (high gonadoblastoma risk)

Memory hook: AIS = XY with breasts and no uterus (receptor broken, AMH still works). **Swyer = XY with a uterus and no ovaries** (no testis, so no AMH). **PMDS = normal male with a uterus** (AMH broken).

4.4 46,XX DSD (a female karyotype with masculinised or abnormal development)

Condition	Defect	Key points
Congenital adrenal hyperplasia (CAH) (most common cause of 46,XX DSD)	Usually **21-hydroxylase deficiency** (see the Adrenal chapter)	Excess androgens in utero masculinise the external genitalia of a genetic female (ambiguous genitalia, **clitoromegaly**); normal ovaries, uterus and tubes
Exogenous androgens in pregnancy	Maternal drugs/tumours	Same virilisation
Ovotesticular DSD (formerly true hermaphroditism)	Often **SRY translocation** (from Y to an X or an autosome); karyotype usually 46,XX (more common than 46,XY)	**Ovotestis** (both seminiferous tubules and primordial follicles on biopsy); ambiguous genitalia; often bilateral; infertility
Aromatase deficiency	Autosomal recessive **CYP19A1** (cannot convert testosterone to estrogen)	Ambiguous genitalia at birth, **maternal virilisation during pregnancy** (fetal androgens cross the placenta), then no breast development, primary amenorrhoea, **virilisation**, **tall stature with osteoporosis**, insulin resistance
46,XX gonadal dysgenesis	Unknown gene mutations	Normal female genitalia, then **streak ovaries**, no puberty, primary amenorrhoea, infertility
46,XX testicular DSD	**SRY translocation onto an X**	Male phenotype with small testes, **no Mullerian structures**, gynaecomastia, **azoospermia** (infertile), short stature; **FISH/PCR for SRY**; testosterone treatment

4.5 Female tract anomalies

- **Mullerian agenesis / aplasia (Mayer-Rokitansky-Kuster-Hauser, MRKH)**: **46,XX** with **functioning ovaries and normal secondary sexual characteristics** (breasts and pubic hair are normal) but a **missing or hypoplastic uterus, cervix and upper vagina**; presents as **primary amenorrhoea**. Agenesis = organ absent because the tissue never formed; aplasia = tissue formed but the organ did not develop.
- **Mullerian duct fusion anomalies:**

Anomaly	What happened
Unicornuate	One Mullerian duct failed to develop
Didelphic	Complete lack of fusion: **double uterus, double cervix, double vagina**
Bicornuate	Partial fusion failure; **bicornis unicollis** (two uterine horns, one cervix) or **bicollis** (two cervices, with or without a vaginal septum)
Septate	Ducts fused but the septum persists (subseptate if partial); strongly linked to **recurrent miscarriage**

- **Asherman syndrome:** endometrial adhesions/fibrosis from **D&C** or post-inflammatory scarring (chlamydia); abnormal bleeding, **secondary amenorrhoea**, infertility and recurrent pregnancy loss.
- **Imperforate hymen:** the central cells of the Mullerian eminence in the urogenital sinus fail to disintegrate. **Primary amenorrhoea with periodic lower-abdominal pain** (cryptomenorrhoea) and a

bulging bluish membrane at the vulva (**haematocolpos**, blood trapped in the vagina). In newborns, secretions can accumulate as a swelling at the introitus and may resolve spontaneously.

4.6 Quick decision guide: "Female look, primary amenorrhoea?"

Breasts	Uterus	Likely diagnosis
Present	Absent	**MRKH** (46,XX) or **complete AIS** (46,XY, testes, scant pubic hair)
Absent	Present	**Turner** (45,XO), **Swyer** (46,XY), 46,XX gonadal dysgenesis
Present	Present (blocked)	**Imperforate hymen**, transverse vaginal septum

Karyotype and a hormone profile (FSH/LH) sort the rest: **high FSH/LH = ovaries/testes failing** (Turner, Swyer), **low FSH/LH = hypothalamic/pituitary problem** (Kallmann).

5. Ovary: Functional Anatomy, PCOS and Ovarian Tumours

Sources: Teacher deck "Ovary" (42 slides), SGD handout "Ovarian Tumors", Pathoma Ch 13.5 (Ovary, lecture notes pp. 279-283) and Pathoma slides Ch 17.

5.1 Functional anatomy of the ovary (the logic behind everything else)

The functional unit is the **follicle**: an oocyte surrounded by **granulosa cells** (innermost, right around the oocyte) and **theca cells** (outside the granulosa cells).

Cell	Location	Stimulated by	What it does
Theca cell	Surrounds the granulosa cells	**LH**	Makes **androgens** (androstenedione, testosterone)
Granulosa cell	Right around the oocyte	**FSH**	Uses **aromatase** to convert theca androgens into **estrogen (estradiol)**

- An **estradiol surge** triggers the **LH surge**, which causes **ovulation** (this marks the start of the *secretory* phase of the endometrial cycle).
- After ovulation the follicle becomes the **corpus luteum**, which mainly secretes **progesterone**. Progesterone drives the secretory phase and prepares the endometrium for a possible pregnancy.

Memory hook: "LT, FG": **L**H acts on **T**heca (androgens); **F**SH acts on **G**ranulosa (estrogen). The two cell, two hormone model is the key to PCOS, to granulosa and Sertoli-Leydig tumours, and to why an androgen tumour causes virilisation.

Normal functional cysts (not tumours)

Cyst	Cause	Significance
Follicular cyst	A follicle that did not rupture or that degenerates and fills with fluid	Normal in small numbers; no clinical significance
Hemorrhagic corpus luteum cyst	Bleeding into a corpus luteum (corpus luteum fails to break down and persists)	Normal; commonly found in **early pregnancy**. If large it can rupture and cause pain

5.2 Polycystic ovarian syndrome (PCOS, Stein-Leventhal syndrome)

Who: about 5% of reproductive-age women. The classic patient is an **obese young woman** with **oligomenorrhea or secondary amenorrhea, hirsutism and infertility**, often with acanthosis nigricans.

Cause: unknown, but the teacher's pathophysiology slide gives the best model: **insulin resistance with high insulin** is the driver (more insulin and IGF acting on the ovary), plus a **high LH relative to FSH (LH:FSH greater than 2)**.

The chain of events (teacher slide 5, explained step by step):

16. **Peripheral insulin resistance leads to hyperinsulinemia.**

17. Insulin acts on the skin to cause epidermal hyperplasia and hyperpigmentation: **acanthosis nigricans**.
18. Insulin and high LH push **theca interna cells to make too much androgen**. The granulosa cells cannot convert it all to estrogen, so androgens build up. Result: **hirsutism, acne, male-pattern hair loss, raised testosterone**.
19. High LH relative to FSH disrupts follicle maturation. Follicles start to grow but **never rupture**, so there is **anovulation or oligo-ovulation**, **infertility** and many small **subcapsular follicle cysts** ("string of pearls").
20. **Adipose tissue** converts the extra androgen to **estrone** (more fat means more estrone). Estrone gives negative feedback on FSH, which keeps FSH low and sustains the cycle.
21. Insulin **suppresses SHBG** (sex hormone binding globulin) in the liver, so **free androgens and free estrogens rise**.
22. Without ovulation there is **no corpus luteum and no progesterone**, so the endometrium sees **unopposed estrogen**. This causes **endometrial hyperplasia and a higher risk of endometrial carcinoma** (see chapter 3).
23. Years later (10-15) the insulin resistance can progress to **type 2 diabetes** (see chapter 1).

Why it happens: The vicious circle is: insulin resistance -> androgens up -> anovulation -> no progesterone -> unopposed estrogen. Treat the circle at any point (weight loss and metformin lower insulin, oral contraceptives suppress ovarian steroid production and give progestin to protect the endometrium).

Labs: high LH, **low or normal FSH (LH:FSH greater than 2)**, high testosterone, often high fasting glucose or insulin. **Gross: bilaterally enlarged ovaries** with a thick smooth capsule and multiple subcortical cysts. **Micro:** multiple **follicular cysts** with a hyperplastic, luteinised theca layer and a thickened ovarian capsule. **Treatment:** oral contraceptives or **medroxyprogesterone (Provera)** to suppress ovarian steroid production and protect the endometrium; weight loss, metformin; ovulation induction (clomiphene) if pregnancy is desired; anti-androgens for hirsutism.

How to read this picture: On ultrasound look for **enlarged ovaries with a dozen or more small follicles lined up around the periphery** (the "string of pearls"), with a normal-sized uterus. That is what the teacher MCQ describes.



slide 4 *- Polycystic ovarian disease*

High yield: Test yourself (slide 7). A 33-year-old with BMI 32, hirsutism, infertility, hyperglycaemia and enlarged ovaries with peripheral follicles. At greatest risk for: **A. Endometrial hyperplasia** (unopposed estrogen from anovulation). Adenomyosis, chronic endometritis and malignant mixed Mullerian tumour are not linked, and borderline mucinous tumour is an epithelial tumour unrelated to PCOS.

5.3 Ovarian tumours: classification by cell of origin

The ovary has **three cell lineages**, and each can form a tumour. Knowing the cell of origin predicts the age, the marker and the behaviour.

Lineage	Tumours	Share and key points
Surface (coelomic) epithelium	Serous, mucinous, endometrioid, clear cell, Brenner	**Most common (about 70%).** Mostly older women. Worst prognosis of the gynaecological cancers because they present late
Germ cells	Dysgerminoma, yolk sac tumour, embryonal carcinoma, choriocarcinoma, teratoma	**About 15%.** Children and young women. **All malignant except mature cystic teratoma**
Sex cord-stroma	Granulosa-theca cell tumour, Sertoli-Leydig cell tumour, fibroma/thecoma	Hormone-producing (estrogen or androgen); fibroma is the exception
Metastatic	Krukenberg (gastric), colon, breast, endometrial, appendix (pseudomyxoma)	Often **bilateral**

Memory hook: Epithelial = old, germ cell = young, sex-cord = hormones. Surface epithelial tumours peak in the 60s, germ cell tumours in children and teens, and sex-cord tumours announce themselves with estrogen or androgen effects.

5.4 Surface epithelial tumours

Symptoms: vague abdominal discomfort, bloating, early satiety, or urinary frequency from bladder compression. Often **asymptomatic until far advanced**, so malignant tumours usually present with spread.

Benign, borderline and malignant: how to tell them apart

	Benign (cystadenoma)	Borderline	Malignant (cystadenoCARCINOMA)
Gross	**Single (unilocular) cyst, smooth thin wall, simple flat lining**, clear fluid	In between: some papillary projections	**Complex, multiloculated, thick shaggy lining**, solid areas, haemorrhage, necrosis, usually over 15 cm, adhesions
Micro	Single layer of serous (ciliated) or mucinous (columnar, mucin-filled) cells	Stratification and tufting, **no stromal invasion**	Stratified cells, papillae, **stromal invasion**, nuclear atypia, many mitoses
Age	Premenopausal (30-40)	Middle age	**Postmenopausal (60-70)**
Behaviour	Cured by excision	Low malignant potential; better prognosis than carcinoma	Aggressive

Cystadenoma is the commonest benign ovarian tumour and presents with vague abdominal pain. **Cystadenocarcinoma** is the commonest malignant ovarian tumour, usually **bilateral**, with spread outside the ovary at diagnosis.

Serous tumours (filled with watery fluid)

- Serous cystadenoma: unilocular cyst, watery clear fluid, lined by **ciliated tubal-type epithelium**. CA-125 is typically normal because there is little tumour mass.
- **Serous (high-grade) carcinoma:** most common malignant tumour. Papillary structures with marked nuclear atypia and many mitoses. **Psammoma bodies** (concentric lamellar calcifications) are characteristic. **BRCA1 (and BRCA2) mutations** are strongly linked; many are now thought to arise from the **fimbriated end of the fallopian tube** (serous tubal intraepithelial carcinoma, STIC) and implant on the ovary. Germline carriers are offered **prophylactic bilateral salpingo-oophorectomy** (with or without mastectomy). **TP53** mutations are almost universal.
- **Spread:** by **transcoelomic seeding**. Cells exfoliate from the surface and implant on the peritoneum, omentum ("**omental caking**"), diaphragm and bowel serosa. This is also why **ascites** is common.
- **Marker: CA-125**, used to monitor response to treatment and to detect recurrence. It is **not a screening test** (raised in many benign conditions such as endometriosis, menstruation and PID).

Mucinous tumours (filled with mucin)

- Large, **multiloculated cysts** with thick gelatinous mucus. They can reach 25 cm or more because mucin keeps accumulating inside the cysts without causing early symptoms.
- Lined by tall columnar mucin-producing cells. Benign versus malignant is decided by **architectural complexity, stratification, atypia and invasion**.
- If mucinous tumour is **bilateral** or if mucin is found **throughout the peritoneum** (**pseudomyxoma peritonei**, "jelly belly"), suspect a **metastasis from an appendiceal or GI primary** rather than an ovarian primary. A true primary mucinous ovarian tumour is usually unilateral and large.

Endometrioid tumours

- Made of **endometrial-like glands**, may arise from **endometriosis**. Mostly **malignant**.
- About **15% coexist with an independent endometrial carcinoma** of endometrioid type. Obesity (excess estrogen) raises the risk of both. **Lynch syndrome** is associated. Mutations include PTEN, ARID1A and beta-catenin.
- Differential: metastasis from endometrial cancer versus two synchronous primaries.

Brenner tumour

- Rare. Nests of **bladder-like (transitional) epithelium** in a fibrous stroma. Usually **benign**; can be borderline or malignant. Histology: nests with a **longitudinal "coffee-bean" nuclear groove**.

Table 22-5 WHO Classification of Ovarian Neoplasms

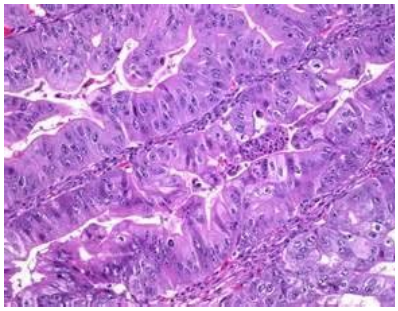
Surface Epithelial-Stromal Tumors
Serous tumors
Benign (cystadenoma, cystadenofibroma)
Borderline (serous borderline tumor)
Malignant (low- and high-grade serous adenocarcinoma)
Mucinous tumors, endocervical-like and intestinal type
Benign (cystadenoma, cystadenofibroma)
Borderline (mucinous borderline tumor)
Malignant (mucinous adenocarcinoma)
Endometrioid tumors
Benign (cystadenoma, cystadenofibroma)
Borderline (endometrioid borderline tumor)
Malignant (endometrioid adenocarcinoma)
Clear cell tumors
Benign
Borderline
Malignant (clear cell adenocarcinoma)
Transitional cell tumors
Benign Brenner tumor
Brenner tumor of borderline malignancy
Malignant Brenner tumor
Epithelial-stromal
Adenosarcoma
Malignant mixed müllerian tumor
Sex Cord-Stromal Tumors
Granulosa tumors
Fibromas
Fibrothecomas
Thecomas
Sertoli-Leydig cell tumors
Steroid (lipid) cell tumors
Germ Cell Tumors
Teratoma
Immature
Mature
Solid
Cystic (dermoid cyst)
Monodermal (e.g., struma ovarii, carcinoid)
Dysgerminoma
Yolk sac tumor
Mixed germ cell tumors
Metastatic Cancer From Non-ovarian Primary
Colonic, appendiceal
Gastric
Pancreaticobiliary
Breast

slide 8

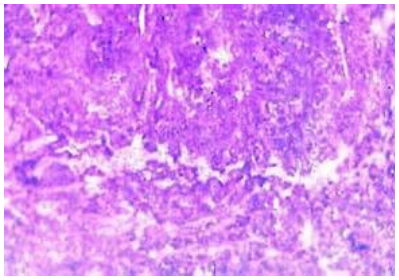


slide 13 *- Ovarian cystadenocarcinoma*

slide 8



slide 14 *- Ovarian cystadenocarcinoma | Ovarian mucinous cystadenocarcinoma | Ovarian serous cystade*



slide 14 *- Ovarian cystadenocarcinoma | Ovarian mucinous cystadenocarcinoma | Ovarian serous cystade*

High yield: Test yourself (slide 12). 43-year-old, 7 cm unilocular, watery, papillary excrescences, no stromal invasion: **D. Borderline serous tumour** (papillae but no invasion).

High yield: Test yourself (slide 17). 47-year-old, 10 cm fluid-filled cyst, ascites, malignant epithelial clusters around psammoma bodies: **B. Serous cystadenocarcinoma**.

High yield: Test yourself (slide 18). 50-year-old, bilateral irregular unilocular cystic masses of 10 and 8 cm: **B. Serous cystadenocarcinoma** (bilateral, malignant, psammoma-type). Mucinous cystadenoma would be multilocular and usually unilateral; teratoma, fibrothecoma and endometrioid carcinoma do not fit.

5.5 Germ cell tumours

Derived from the primordial germ cells. **Mature cystic teratoma is benign; all the others are malignant.** Think "the same tumours as the testis" (chapter 2).

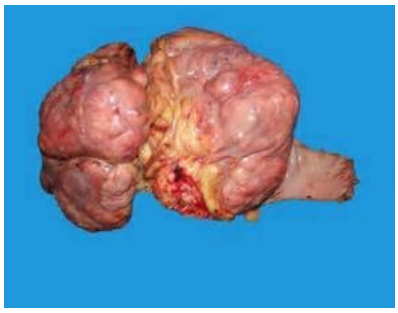
Tumour	Who	Cells / key histology	Marker	Behaviour
Mature cystic teratoma (dermoid cyst)	Reproductive-age women; **most common germ cell tumour**	Tissue from **all three germ layers**: skin, hair, teeth, sebum (ectoderm), bone and cartilage (mesoderm), thyroid, bronchus, gut (endoderm)	None	Benign (over 95%)
Immature teratoma	Children and young women	**Immature (embryonic) tissue**, especially **neuroepithelium**	AFP sometimes	Malignant; grade by the amount of immature neural tissue
Struma ovarii	Rare monodermal teratoma	Mostly **thyroid tissue**	Thyroid hormones	Can cause **hyperthyroidism**
Dysgerminoma	Young women (teens-20s); **most common malignant germ cell tumour** (about 50%)	Sheets of large round cells with **clear cytoplasm (glycogen) and central nuclei**, "**fried egg**" look, fibrous septa with **lymphocytes**	**LDH** (hCG sometimes slightly raised)	**Radiosensitive, good prognosis**. Ovarian counterpart of **testicular seminoma**

Yolk sac (endodermal sinus) tumour	**Most common germ cell tumour in children**, young girls	Resembles the yolk sac; **Schiller-Duval bodies** (glomerulus-like: a central blood vessel surrounded by tumour cells in a space lined by tumour cells)	**AFP**	Aggressive but chemo-responsive
Embryonal carcinoma	Young adults	Large primitive cells, glands, necrosis	hCG, AFP	Aggressive
Choriocarcinoma	Rare in the ovary	Cytotrophoblasts + syncytiotrophoblasts	**High beta-hCG**	Very invasive, spreads **by blood early**, poor response to chemotherapy in the ovarian (non-gestational) form

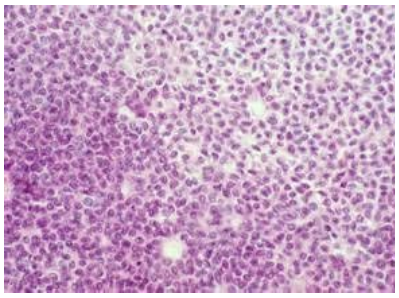
Dysgerminoma risk factors: Turner syndrome and disorders of sex development with a Y chromosome (gonadoblastoma precursor), see chapter 4.

Mature cystic teratoma in depth

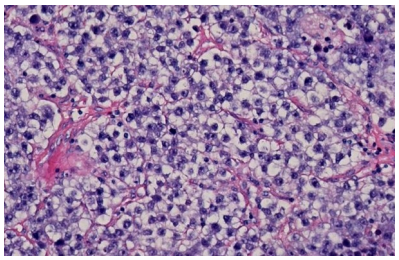
- **Gross:** a cyst filled with **hair, sebum ("greasy material") and sometimes teeth**, with a solid nodule (Rokitansky protuberance).
- **Micro:** mature skin with adnexa, neural tissue, cartilage, thyroid, bronchial and GI epithelium.
- **Imaging:** a cystic mass with **fat and calcification** (tooth or bone) is classic.
- **Complications: torsion** (the most common cause of acute pain, since the heavy mass twists on its pedicle), **rupture** (spill causes **chemical peritonitis**), and **malignant transformation** in about 1%, usually **squamous cell carcinoma** (also sarcoma). **Autoimmune haemolytic anaemia** and **anti-NMDA receptor encephalitis** are rare associations.
- **Struma ovarii** (thyroid-predominant teratoma) can cause **hyperthyroidism** (weight loss, nervousness, sweating, tachycardia).



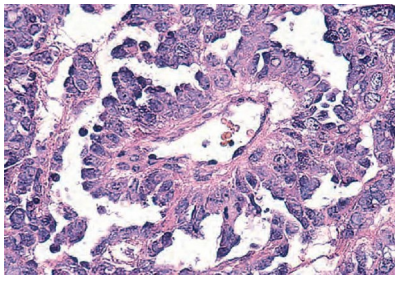
slide 24 *- Ovarian dysgerminoma*



slide 24 *- Ovarian dysgerminoma*



slide 25 *- Dysgerminoma*



***slide 27** - Schiller-Duval body in yolk sac carcinoma*

High yield: Test yourself (slide 22). 40-year-old with weight loss, nervousness, sweating, a 10 cm cystic and solid ovarian mass full of hair and sebum: **struma ovarii, so the lab finding is A. Thyroxine of 11.3 micrograms/dL (elevated)**. HCG would point to choriocarcinoma; low potassium, high estradiol and CA-125 refer to other tumours.

Memory hook: Dysgerminoma = ovarian seminoma (same look, same radiosensitivity, same LDH). **Yolk sac = AFP + Schiller-Duval. Choriocarcinoma = hCG. Teratoma = hair and teeth.**

5.6 Sex cord-stromal tumours

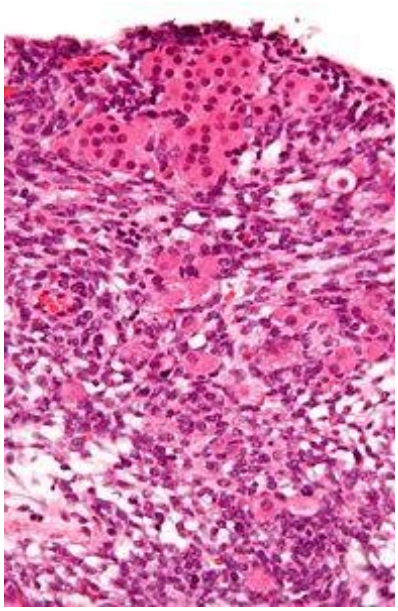
These come from the cells that surround the oocyte (granulosa, theca, Sertoli, Leydig, fibroblasts). They make **hormones**, so they present with endocrine effects.

Tumour	Cell and hormone	Clinical picture	Histology / marker	Behaviour
Granulosa-theca cell tumour	Granulosa cells; **estrogen**	Prepubertal: **precocious puberty**. Reproductive age: irregular menses or dysfunctional uterine bleeding. Postmenopausal: **postmenopausal bleeding** with **endometrial hyperplasia** (and endometrial carcinoma risk)	Yellow-white mass. Polygonal cells with pale **grooved "coffee-bean" nuclei** forming follicle-like structures (**Call-Exner bodies**). Marker: **inhibin**	Potentially malignant (low risk of metastasis); can recur late
Sertoli-Leydig cell tumour (androblastoma)	Sertoli (tubules) and Leydig cells (between tubules); **androgens**	Reproductive age. **Virilisation**: hirsutism, temporal balding, deepened voice, clitoromegaly, amenorrhea. DICER1 mutations	Yellow lobulated solid mass. **Sertoli cells: pale/clear cytoplasm; Leydig cells: abundant eosinophilic cytoplasm; Reinke crystals** (intracytoplasmic). High testosterone with a **high testosterone:androstenedione ratio**	About 75% benign; malignant ones respond poorly
Fibroma / thecoma (fibrothecoma)	Fibroblasts	Often asymptomatic. **Meigs syndrome**: fibroma + ascites + right pleural effusion. Can cause **torsion**	Firm white whorled masses; intersecting bundles of spindle cells producing collagen	Benign. **Most common stromal tumour**. Meigs resolves after removal

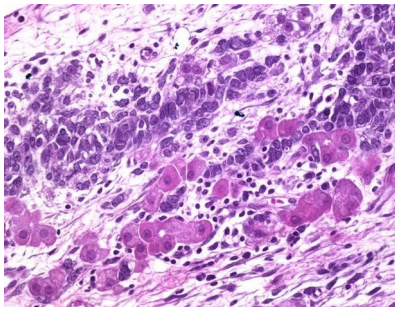
Why it happens: Granulosa cells make estrogen, so the tumour mimics an estrogen-rich state: early puberty in a child, bleeding in an adult, a thick endometrium in an older woman. **Sertoli-Leydig cells make testosterone**, so the patient is virilised.

How to read this picture: Look for **Call-Exner bodies** (small rosettes of cells around an eosinophilic space, like tiny follicles) in granulosa cell tumour. In Sertoli-Leydig look for **tubules lined by Sertoli cells with Leydig cells in the stroma**. In fibroma, whorled **spindle cells with collagen**.

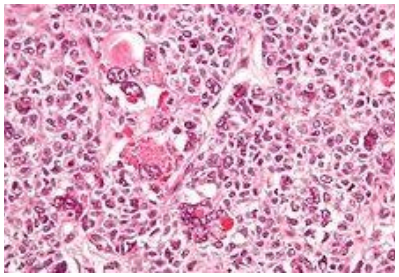
How to tell a Sertoli-Leydig tumour from adrenal causes of virilisation: in an **adrenal** cause (tumour or congenital adrenal hyperplasia) **DHEA-S** and 17-hydroxyprogesterone are high; in an **ovarian** tumour the testosterone is high with normal DHEA-S, and imaging shows a unilateral ovarian mass.



slide 32 *- Sertoli-Leydig cell tumor (ovarian androblastoma) | Leydig cell have abundant eosinophilic or light pink cytopla*



slide 33 *- Sertoli-Leydig Cell Tumor*



slide 35 *- Ovarian granulosa cell tumor | Polygonal granulosa tumor cells w/ pale, grooved, "coffee /*

High yield: Test yourself (slide 38). 48-year-old with irregular bleeding, endometrial hyperplasia on biopsy and a solid ovarian mass: **D. Fibrothecoma** (a thecoma component can make estrogen). The usual teaching answer for a *solid ovarian mass with estrogen effect* is **granulosa-theca cell tumour**; granulosa is not listed here, so choose fibrothecoma. Sertoli-Leydig makes androgens (virilisation) and the other choices do not make estrogen.

High yield: Test yourself (slide 39). 62-year-old, obese, postmenopausal bleeding, well-differentiated endometrial adenocarcinoma. Ovarian tumour most likely: **D. Granulosa cell tumour** (estrogen-secreting tumour causing endometrial carcinoma).

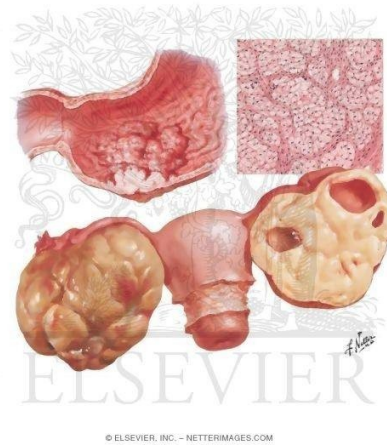
5.7 Metastases to the ovary

Primary sites: **breast, colon, endometrium (also direct extension), stomach**.

- **Krukenberg tumour: bilateral** ovarian metastases from **diffuse-type gastric carcinoma** (also colon, breast). Histology: **mucin-filled signet-ring cells** in an ovarian stroma. Spread is by the

peritoneal/lymphatic route. Prognosis is poor. If a mucinous tumour is **bilateral**, think metastasis first; a primary mucinous carcinoma is usually unilateral.

- **Pseudomyxoma peritonei:** massive mucin in the peritoneum ("**jelly belly**") from a **mucinous tumour of the appendix** (sometimes ovary).



slide 41 *- Metastatic tumors to the ovary- Krugenberg tumor*

5.8 Tumour marker summary

Marker	Tumour
CA-125	Serous epithelial carcinoma (monitoring, **not screening**)
AFP	Yolk sac tumour (some immature teratomas, embryonal)
LDH	Dysgerminoma
Beta-hCG	Choriocarcinoma (and embryonal; also pregnancy, mole)
Inhibin	Granulosa cell tumour (monitor recurrence)
Testosterone	Sertoli-Leydig cell tumour

5.9 SGD answers: Ovarian Tumors (guiding questions)

Serous cystadenoma (32-year-old, 8 cm unilocular cyst, CA-125 normal)

24. Category: **benign surface epithelial (serous) tumour**.
25. Cell of origin: **ovarian surface (coelomic) epithelium**, which takes on a tubal (ciliated, serous) type.
26. Benign features: **unilocular, smooth-walled, no solid areas or papillae, thin wall, clear fluid, unilateral**.
27. Histology resembles **fallopian tube epithelium** (a single layer of ciliated and secretory cells).
28. CA-125 is normal because there is **little tumour mass and no peritoneal spread**; CA-125 rises mainly with tumour bulk and peritoneal involvement.
29. Complications despite benign pathology: **torsion, rupture, pressure symptoms**, and the possibility of coexisting borderline areas (sample the whole tumour).

High-grade serous carcinoma (67-year-old, ascites, CA-125 high)

30. Diagnosis: **high-grade serous carcinoma of the ovary/tube**.
31. Advanced at presentation because the ovary is deep in the pelvis, **symptoms are vague** (bloating, early satiety) and there is no screening test; cells seed the peritoneum early.

32. Precursor: **serous tubal intraepithelial carcinoma (STIC)** in the **fimbriated end of the fallopian tube**.
33. Genes: **TP53 (almost all)**, **BRCA1/BRCA2** (germline in many).
34. CA-125 is a glycoprotein expressed on coelomic-derived epithelium; **bulky peritoneal tumour** releases it.
35. **Transcoelomic spread:** tumour cells exfoliate into the peritoneal fluid, circulate with it and implant on the peritoneum, omentum and diaphragm, forming ascites and omental caking.
36. Poor prognosis because of late stage, widespread peritoneal disease and eventual chemoresistance (PARP inhibitors help BRCA-mutated tumours).

Mucinous tumour (48-year-old, 25 cm multiloculated cyst)

37. **Mucinous cystadenoma** (benign; borderline/carcinoma possible).
38. **Tall columnar mucin-secreting cells** (intestinal or endocervical type).
39. They become huge because **mucin accumulates in the closed cysts** and causes few symptoms until large.
40. Benign vs malignant: **stratification, complex papillae, cytological atypia and stromal invasion** (borderline: atypia without invasion). Sample extensively because areas vary.
41. Mucin throughout the peritoneum: **pseudomyxoma peritonei**.
42. Always exclude **metastasis from the GI tract (appendix, colon, stomach, pancreas)**, particularly if the tumour is bilateral or smaller than about 10 cm.

Endometrioid carcinoma (58-year-old obese woman, endometrial mass + ovarian mass)

43. **Ovarian endometrioid carcinoma**.
44. **Endometriosis** commonly coexists (and a synchronous endometrial carcinoma in about 15%).
45. Mutations: **PTEN, ARID1A, PIK3CA, beta-catenin (CTNNB1), microsatellite instability (Lynch syndrome)**.
46. **Obesity** increases peripheral conversion of androgen to estrone in adipose tissue, giving unopposed estrogen that stimulates the endometrium (and endometriosis).
47. Differential: **metastasis from an endometrial primary** vs **two independent primaries**; also clear cell carcinoma.

Mature cystic teratoma (25-year-old, sudden severe pain)

48. **Mature cystic teratoma (dermoid cyst)**.
49. **All three germ layers** (ectoderm: skin, hair, neural; mesoderm: bone, cartilage, muscle; endoderm: gut, thyroid, bronchus).
50. Calcifications come from **teeth or bone** inside the cyst.
51. Acute pain: **ovarian torsion**.
52. Most common tissue: **ectodermal (skin and adnexa)**.
53. Hyperthyroidism: yes, when thyroid tissue predominates (**struma ovarii**).
54. Rare malignant transformation: **squamous cell carcinoma** (about 1%).

Dysgerminoma (19-year-old, LDH raised, AFP normal)

55. **Dysgerminoma**.
56. **LDH** (and sometimes slightly high hCG).

57. Large uniform cells with clear cytoplasm and central nuclei in sheets and nests, **fibrous septa with lymphocytes**, "fried egg".
58. Testicular counterpart: **seminoma**.
59. Radiosensitive because the tumour is made of rapidly dividing, poorly differentiated germ cells (and also chemosensitive).

Yolk sac tumour (4-year-old, AFP very high)

60. **Yolk sac (endodermal sinus) tumour**.
61. **AFP**.
62. **Schiller-Duval body:** a central capillary (blood vessel) surrounded by tumour cells within a space also lined by tumour cells, looking like a **glomerulus**.
63. Resembles the **extraembryonic yolk sac**.
64. Age: it is the **commonest germ cell tumour in children**; the young age plus AFP is almost diagnostic.
65. AFP falls with treatment and rises with recurrence, so it is used for **follow-up**.

Granulosa cell tumour (52-year-old, bleeding, endometrial hyperplasia, inhibin high)

66. **Granulosa cell tumour**.
67. **Estrogen**.
68. Excess estrogen stimulates endometrial proliferation without progesterone opposition.
69. **Call-Exner bodies** (with coffee-bean grooved nuclei).
70. **Inhibin** (monitors recurrence).
71. Long-term uterine complication: **endometrial hyperplasia and endometrial carcinoma**.

Sertoli-Leydig cell tumour (24-year-old, virilised)

72. **Sertoli-Leydig cell tumour**.
73. **Sex cord-stromal cells** (Sertoli cells forming tubules and Leydig cells in the stroma).
74. Androgens suppress the pituitary (amenorrhea) and cause hirsutism, voice deepening, muscle bulk and clitoromegaly.
75. **Testosterone**.
76. Adrenal causes show high DHEA-S and bilateral adrenal changes; this one is a unilateral ovarian mass with normal DHEA-S.
77. Usually benign (about 75%) and curative with surgery; malignant forms have a poor prognosis.

Krukenberg tumour (60-year-old, bilateral masses, gastric cancer)

78. **Krukenberg tumour** (metastatic signet-ring cell carcinoma).
79. **Stomach** (diffuse-type gastric adenocarcinoma).
80. Bilateral because the cells spread by **transcoelomic and lymphatic/haematogenous** routes and seed both ovaries.
81. Metastatic tumours are usually **bilateral, multinodular, involve the ovarian surface/stroma, with a known primary**; primary tumours are often unilateral and solitary.
82. **Signet-ring cells** with intracellular mucin pushing the nucleus aside.
83. Poor prognosis because the tumour is already metastatic.

5.10 SGD answers: Integrative questions

84. **Cell of origin:** surface epithelium (serous, mucinous, endometrioid, clear cell, Brenner); germ cell (teratoma, dysgerminoma, yolk sac, embryonal, choriocarcinoma); sex cord-stromal (granulosa-theca, Sertoli-Leydig, fibroma); metastatic.
85. **Benign vs malignant epithelial:** see the table in 5.4 (unilocular smooth cyst vs complex multiloculated, thick shaggy, solid areas, invasion, bilateral, ascites; premenopausal vs postmenopausal).
86. **Markers:** CA-125 = serous epithelial carcinoma; AFP = yolk sac; LDH = dysgerminoma; beta-hCG = choriocarcinoma; inhibin = granulosa cell tumour; testosterone = Sertoli-Leydig.
87. **Associations:** BRCA mutations = high-grade serous carcinoma; Lynch syndrome = endometrioid (and clear cell) carcinoma; endometrial hyperplasia = granulosa cell tumour (and fibrothecoma); ovarian torsion = mature teratoma and fibroma; pseudomyxoma peritonei = mucinous tumour of appendix/ovary.
88. **Age:** children = yolk sac tumour (and other germ cell tumours); adolescents/young women = dysgerminoma, teratoma, immature teratoma; reproductive age = mature teratoma, Sertoli-Leydig, borderline/benign cystadenomas, granulosa cell tumour (adult form); postmenopausal = serous carcinoma, fibroma, granulosa cell tumour, Krukenberg.
89. **Histologic hallmarks:** high-grade serous = papillae, marked atypia, psammoma bodies; dysgerminoma = sheets of clear cells with fibrous septa and lymphocytes; yolk sac = Schiller-Duval bodies; granulosa = Call-Exner bodies and coffee-bean nuclei; mature teratoma = mature tissues of all three layers.
90. **Spread:** epithelial carcinoma spreads by transcoelomic seeding of the peritoneum (omental caking), whereas metastases to the ovary arrive by blood, lymph or peritoneum from stomach, colon, breast or appendix.
91. **Bilateral vs unilateral:** bilateral masses raise the possibility of metastasis (Krukenberg), serous carcinoma, or PCOS-type enlargement; a large unilateral mucinous cyst favours a primary ovarian mucinous tumour; unilateral solid hormone-producing masses favour sex cord-stromal tumours.
92. **Age + one lab value:** yes. Child + high AFP = yolk sac tumour; young woman + high LDH = dysgerminoma; older woman + high CA-125 = serous carcinoma; any age + high inhibin = granulosa; high testosterone = Sertoli-Leydig; high hCG = choriocarcinoma.
93. 4-year-old, pelvic mass, high AFP: **yolk sac tumour**.
94. 65-year-old with bloating, weight loss, high CA-125: **serous (surface epithelial) carcinoma**.
95. 5-year-old with precocious puberty and adnexal mass: **estrogen from a (juvenile) granulosa cell tumour**.
96. **Matching** (corrected; the handout table has deliberate errors):

Histological finding	Correct tumour	Handout listed (wrong)
"Fried egg" cells	**Dysgerminoma**	Yolk sac
Call-Exner bodies	**Granulosa cell tumour**	Sertoli-Leydig
Psammoma bodies	**Serous cystadenocarcinoma**	(correct)
Reinke crystals	**Sertoli-Leydig cell tumour** (Leydig cells)	Granulosa
Schiller-Duval bodies	**Yolk sac tumour**	Dysgerminoma

97. **Outside the pelvis:** serous carcinoma gives ascites and omental caking; Meigs syndrome gives a pleural effusion; struma ovarii gives hyperthyroidism; Krukenberg indicates a primary stomach cancer;

granulosa and Sertoli-Leydig give systemic endocrine effects; pseudomyxoma peritonei fills the abdomen with mucin.

98. Tachycardia, tremors and a pelvic mass: **struma ovarii** (monodermal teratoma with functional thyroid tissue).
99. **Meigs syndrome:** ovarian fibroma + ascites + (right) pleural effusion; resolves after resection.
100. Bilateral mass with signet-ring cells: **stomach** (Krukenberg tumour).
101. CA-125: useful for **monitoring** treatment and recurrence, **not for screening** (poor sensitivity early, poor specificity).
102. Mature teratoma: **benign** (malignant change in about 1%, usually squamous cell carcinoma).
103. Neural tissue in a teratoma: if it is mature (well-formed) tissue the teratoma is mature; if **immature neuroepithelium** is present it is an **immature teratoma**, which is potentially malignant.

6. Gestational Disorders: Ectopic Pregnancy, Abortion, Placenta, Hypertension, Molar Disease

Sources: Teacher deck "Gestational Disorders" (55 slides) and Pathoma Ch 13 (Gestational Pathology, lecture notes pp. 284-286) plus Pathoma slides Ch 17.

6.1 Ectopic pregnancy

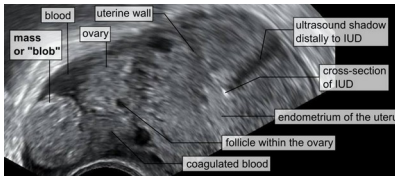
Definition: the fertilised egg implants **outside the uterine endometrium**. The commonest site by far is the **fallopian tube, in the ampulla**. (Rarer sites: isthmus, ovary, cervix, abdominal cavity.)

Why it happens: anything that **scars or slows the tube** so the embryo cannot reach the uterus before it implants.

Category	Risk factors
Anatomic	**PID** (the commonest), previous ectopic pregnancy, previous tubal surgery, endometriosis, ruptured appendix, **Kartagener syndrome** (immotile cilia)
Non-anatomic	Smoking, advanced maternal age, **IUD**, **in vitro fertilisation**

Memory hook: Think **"the pipe is scarred or the cilia are not sweeping."** Pathoma's one-liner: the key risk is **scarring secondary to PID or endometriosis**.

Presentation: a few weeks after a **missed period**: **lower abdominal pain** with guarding, possibly vaginal bleeding, plus ordinary signs of early pregnancy (amenorrhoea, nausea, breast tenderness, urinary frequency). Tenderness over the affected side. **Investigations: beta-hCG is raised but lower than expected for a normal pregnancy** (it rises slowly), and **ultrasound** finds no intrauterine pregnancy. **Danger:** the tube **ruptures** and bleeds: a **surgical emergency** (shock, haemoperitoneum). **Management: salpingectomy** (or salpingostomy), or methotrexate in selected stable cases.



slide 5



slide 6

6.2 Spontaneous abortion (miscarriage)

Definition: loss of pregnancy **before 20 weeks** (after 20 weeks it is called **stillbirth**). Very common: up to about one quarter of pregnancies, mostly in the first trimester. **Presentation:** vaginal bleeding with passage of fetal tissue and cramping pain.

Causes

Group	Causes
Fetoplacental	**Chromosomal abnormalities (up to half of all cases; the single commonest cause, especially trisomy 16)**, congenital anomalies, anembryonic pregnancy ("blighted ovum")
Maternal	Reproductive organ problems (**septate uterus, fibroids, uterine adhesions, cervical incompetence**), systemic disease (**diabetes, hyper- or hypothyroidism**, genetic disorders, infections), **hypercoagulable states such as antiphospholipid syndrome**
Miscellaneous	Trauma, iatrogenic, environmental exposures, unknown

Teratogens: the effect depends on *when* the exposure happens

Time	Effect
0-2 weeks ("all or none")	**Spontaneous abortion** or no effect
3-8 weeks (organogenesis)	**Major organ malformations**
3-9 months	**Organ hypoplasia**, growth restriction

Teratogen	Effect
Alcohol	Fetal alcohol syndrome: most common cause of intellectual disability, facial abnormalities, microcephaly
Cocaine	Growth restriction and **placental abruption**
Thalidomide	Limb defects (phocomelia)
Cigarette smoke	Growth restriction
Isotretinoin	Spontaneous abortion; hearing and visual impairment (also craniofacial and cardiac defects)
Tetracycline	Discoloured teeth
Warfarin	Fetal bleeding
Phenytoin	Digit hypoplasia, cleft lip/palate



Clinical features of spontaneous pregnancy loss [1]

Type	Vaginal bleeding	Fetal activity	Products of conception (POC)	Cervical os	Prognosis
Threatened abortion	• Yes	• Yes	• Intrauterine	• Closed	• Reversible
Inevitable abortion	• Yes	• May be present	• Visible/palpable POC	• Dilated	• Irreversibl
Missed abortion	• No	• No	• No expulsion of the POC	• Closed	• Irreversibl
Incomplete abortion	• Yes	• No	• POC within the cervical canal or uterus	• Dilated	• Irreversibl
Complete abortion	• Yes	• No	• POC completely outside of the uterus	• Closed	• Irreversibl

slide 9

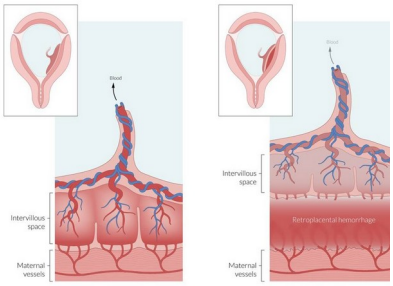
6.3 Antepartum haemorrhage: the three placental bleeds

Antepartum haemorrhage (bleeding in late pregnancy) includes placental abruption, placenta previa, vasa previa, uterine rupture, stillbirth, cervical trauma, and the normal "bloody show".

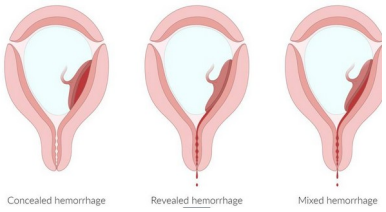
	Placenta previa	Placental abruption	Vasa previa
What	Placenta lies in the **lower uterine segment**, partly or completely covering the **internal os**	**Premature separation** of the placenta from the uterine wall (the decidua) before delivery	**Fetal vessels** run through the membranes over the internal os (**velamentous cord insertion, bilobed or succenturiate placenta**)
Risk	Age over 35, **multiparity**, previous curettage,	**Hypertension, preeclampsia**, trauma,	Velamentous cord, bilobed placenta

	previous caesarean	**cocaine**, smoking, alcohol	
Bleeding	**Painless, bright red**, sudden, third trimester	**Painful**, continuous (may be concealed), uterine tenderness, back pain	Painless, **fetal blood**, sudden after rupture of membranes
Danger	Haemorrhage; usually needs **caesarean**	**Fetal death, DIC and maternal shock**; a **common cause of stillbirth**	**Fetal distress and death** (the baby is bleeding); emergency caesarean

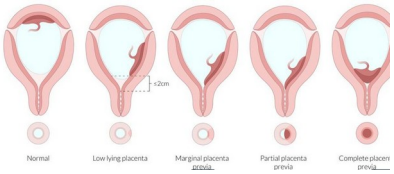
Memory hook: Previa = Painless; aBruption = Bad pain. Previa blocks the exit (never do a digital vaginal exam), abruption rips the placenta off the wall.



slide 14



slide 15



slide 17

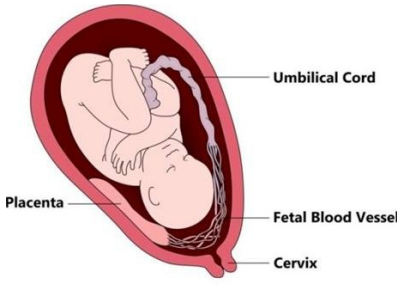
6.4 Postpartum haemorrhage and the placenta accreta spectrum

Causes of postpartum haemorrhage (the **4 T's**): **uterine atony** (commonest), uterine inversion, **retained placenta**, **abnormal placentation**, birth trauma, velamentous cord insertion.

Placenta accreta spectrum: a **defective decidual layer** (Nitabuch layer) lets the placenta attach and fail to separate after delivery.

Type	Meaning
Accreta (up to 75%)	Villi attach to the myometrium but **do not invade** it
Increta	Villi **invade into** the myometrium
Percreta	Villi **penetrate through the myometrium and serosa**; may invade the bladder or bowel

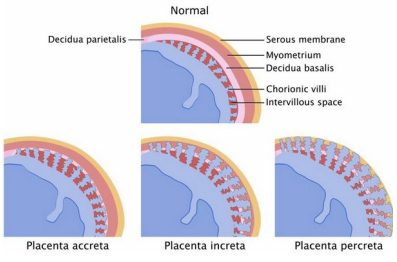
Theories: defective decidua vs excessive trophoblastic invasion. **Risk factors:** previous uterine surgery or **caesarean**, **placenta previa**, multiparity, advanced maternal age. **Presents** with difficult delivery of the placenta and severe **postpartum haemorrhage**; often needs **hysterectomy**.



slide 19



slide 19



slide 22

6.5 Hypertensive disorders of pregnancy

Term	Definition
Chronic hypertension	Diagnosed **before pregnancy or before 20 weeks**
Gestational hypertension	New hypertension **after 20 weeks**, BP of at least 140/90 on two readings at least 4 hours apart, **no proteinuria or end-organ damage**. Mostly asymptomatic (morning headache, fatigue, dizziness). Treat with **hydralazine, labetalol, methyldopa, nifedipine**
Preeclampsia	New hypertension **plus proteinuria or end-organ dysfunction** (Pathoma: hypertension, proteinuria, oedema, usually in the third trimester)
Eclampsia	Preeclampsia **plus seizures** (generalised tonic-clonic) and/or coma
HELLP syndrome	**H**aemolysis, **E**levated **L**iver enzymes, **L**ow **P**latelets; can occur **without** hypertension or proteinuria; about 30% are postpartum; Pathoma: "think of it as **HUS of the liver**"

Risk factors for preeclampsia: thrombophilia (antiphospholipid syndrome), age under 20 or over 35, African-American descent, diabetes, chronic hypertension, chronic renal disease or SLE, **nulliparity, multiple gestation (twins), hydatidiform mole**.

Pathophysiology (explained)

104. The **spiral arteries fail to remodel** properly (trophoblast does not invade the vessel walls), so the placenta stays poorly perfused (**hypoperfusion**).
105. The ischaemic placenta releases **anti-angiogenic and vasoactive factors** (e.g. sFlt-1) into the maternal blood, causing **systemic endothelial dysfunction and vasoconstriction**.
106. Result: **hypertension**, endothelial injury, **microthrombi**, and organ ischaemia.
107. Abnormal maternal-fetal vascular interface in the placenta is the root; it **resolves with delivery** of the placenta.

Organ	Mechanism	Manifestation	Seen in
Kidney	Glomerular endothelial dysfunction, vasoconstriction (glomerular endotheliosis)	**Proteinuria**, reduced function, oedema	Preeclampsia, eclampsia, HELLP
Lung	Increased systemic vascular resistance, left ventricular strain, high permeability, low albumin	**Pulmonary oedema**	Severe preeclampsia, HELLP
Liver	Vasoconstriction and microthrombi in sinusoids (periportal haemorrhage and necrosis)	Liver swelling, high enzymes, **RUQ/epigastric pain**	HELLP, severe preeclampsia
CNS	Vasoconstriction and endothelial damage, microthrombi	**Seizures** (eclampsia), headache, visual disturbance	Eclampsia
Blood	Microthrombi consume platelets and activate coagulation	**Microangiopathic haemolytic anaemia, thrombocytopenia, DIC**	HELLP, severe preeclampsia

Preeclampsia without severe features: headaches, visual disturbance, RUQ/epigastric pain, rapid oedema, hypertension and proteinuria. **With severe features:** BP at least **160/110**, oliguria, headache, visual disturbance, RUQ pain, **pulmonary oedema**, altered mental status. **Complications:** abruption, coagulopathy (DIC), renal failure, pulmonary oedema, uteroplacental insufficiency; may progress to eclampsia and/or HELLP. **Prevention and treatment: magnesium sulfate** infusion prevents seizures (Pathoma: eclampsia and HELLP **warrant immediate delivery**; delivery is the cure). Antihypertensives for severe BP. Management depends on gestational age and severity (expectant if mild and preterm, deliver if severe or at term).

6.6 Sudden infant death syndrome (SIDS)

Pathoma: death of an infant **(1 month to 1 year)** without an obvious cause, often during sleep. **Risk factors:** sleeping prone (on the stomach), exposure to cigarette smoke, prematurity. (Back to sleep reduces risk.)

6.7 Gestational trophoblastic disease (GTD)

Spectrum from benign to malignant: **complete mole, partial mole, invasive mole, choriocarcinoma.** Incidence about 1 per 1,000 pregnancies in the USA, higher in Asia. Risk is higher at **ages under 15 and over 40**.

Complete versus partial mole

	Complete mole	Partial mole
Genetics	An **empty ovum (no maternal DNA)** fertilised by one sperm that duplicates (90% **46,XX**) or by two sperm (10% 46,XY). **All DNA is paternal**	A **normal ovum** fertilised by **two sperm** (or one that duplicates): **triploid, usually 69,XXY** (also 69,XXX; 92,XXXY possible). Maternal and paternal DNA
Fetal parts	**None**	**Present** (a malformed fetus, "two-vessel cord" and syndactyly are typical, like triploidy)
Villi	**Most villi hydropic** (swollen, avascular)	**Only some** villi hydropic
Trophoblast proliferation	**Diffuse, circumferential**	**Focal**, minimal
p57 immunostain	**Negative** (p57 is a maternally expressed imprinted gene; no maternal DNA means no p57)	**Positive**
Risk of invasive mole	15-20%	About 5%
Risk of choriocarcinoma	About **2-3%**	**Rare/minimal**
beta-hCG	**Very high**	Elevated but lower

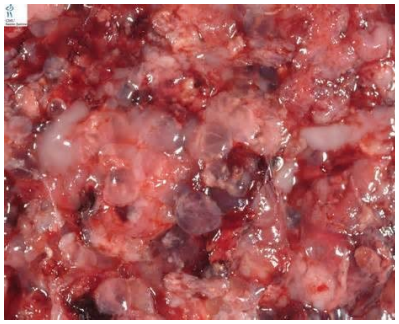
Memory hook: Complete = Paternal only, Pathoma "empty egg"; p57 off. Partial = "Parts of a fetus, Plus a maternal set."

Clinical features (both, but more in complete): uterus larger than dates ("size greater than dates"), first-trimester bleeding, passage of **grape-like** vesicles, **hyperemesis gravidarum**, **early preeclampsia (before 20 weeks)**, **hyperthyroidism** (hCG resembles TSH), and **bilateral theca-lutein ovarian cysts** (hCG over-stimulates the ovaries; large, tender adnexal masses). **Lab: beta-hCG markedly raised** (partial mole is lower than complete, which is lower than choriocarcinoma). **Ultrasound:** complete mole shows an echogenic mass with many hypoechoic cysts ("**snowstorm/honeycomb/bunch of grapes/Swiss cheese**") and **no fetus**;

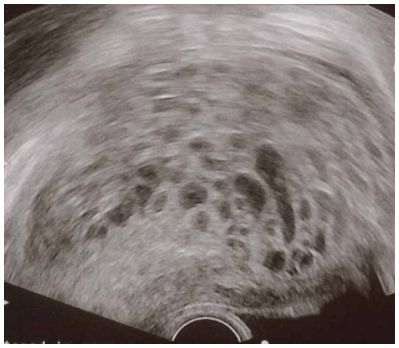
partial mole shows a fetus and may have heart tones. **Treatment: suction evacuation (D&C)**. Follow beta-hCG weekly until undetectable; **contraception for one year** so that a rising hCG from a new pregnancy is not confused with persistent disease. Most patients recover normal reproductive function; some develop gestational trophoblastic neoplasia.



slide 43 *- Hydatidiform mole (Molar pregnancy)*



slide 43 *- Hydatidiform mole (Molar pregnancy)*



slide 46

Invasive mole

A mole that **invades the myometrium or uterine vessels** (hydropic villi extend into the myometrium; the uterus may perforate). Presents with **bleeding and persistently high hCG after evacuation**. Can spread by blood to the **lungs**. Better prognosis than choriocarcinoma; responds to chemotherapy.

Choriocarcinoma

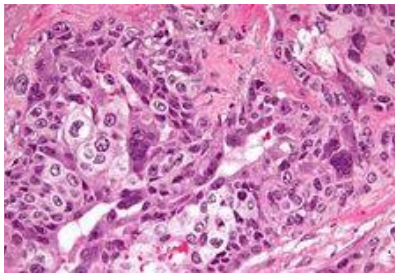
A **malignant tumour of trophoblast**. **Gross:** necrotic haemorrhagic mass. **Micro:** sheets of **cytotrophoblasts (mononuclear, pale) plus syncytiotrophoblasts (multinucleated, dark)**, **no chorionic villi**; both cell types are required for the diagnosis. **Marker: beta-hCG. Spread:** early by **blood** to **lung, brain and liver**. **Treatment: methotrexate**-based chemotherapy; hysterectomy for residual uterine disease. Very responsive. Pathoma: choriocarcinoma may follow a mole, a miscarriage or a normal pregnancy (the **gestational** type, which responds well to chemotherapy), or arise as a germ cell tumour in the ovary or testis (does **not** respond as well).

	Complete mole	Partial mole
KARYOTYPE	46,XX (most common); 46,XY	69,XXX; 69,XXY; 69,XYY
COMPONENTS	Most commonly enucleated egg + single sperm (subsequently duplicates paternal DNA)	2 sperm + 1 egg
HISTOLOGY	Hydropic villi, circumferential and diffuse trophoblastic proliferation	Only some villi are hydropic focal/minimal trophoblasti proliferation
FETAL PARTS	No	Yes (partial = fetal parts)
STAINING FOR P57 PROTEIN	⊖ (paternally imprinted)	⊕ (maternally expressed) Partial mole is P57 positive
UTERINE SIZE	↑	—
hCG	↑↑↑↑	↑
IMAGING	"Honeycombed" uterus or "clusters of grapes" or "snowstorm" on ultrasound	Fetal parts
RISK OF INVASIVE MOLE	15–20%	< 5%
RISK OF CHORIOCARCINOMA	2%	Rare

slide 48



slide 51 *- Choriocarcinoma | Syncytiotrophoblasts are multinucleated and have a dark stai*



slide 51 *- Choriocarcinoma | Syncytiotrophoblasts are multinucleated and have a dark stai*

6.8 Test yourself: teacher MCQs (slides 52-55)

High yield: Slide 52. 15 weeks, malformed stillborn fetus with syndactyly, two-vessel cord, small placenta with scattered 0.5 cm grape-like villi: this is a **partial mole (triploidy)**. Answer: **B. 69,XXY**.

High yield: Slide 53. After D&C for a 14-week mole, bleeding continues and beta-hCG is still 35,000: **E. Invasive mole** (persistent hCG with myometrial invasion). Pulmonary metastases would suggest choriocarcinoma, which is also possible, but invasive mole is the best next step in this scenario.

High yield: Slide 54. After molar evacuation, best follow-up is **B. Serum beta-HCG**.

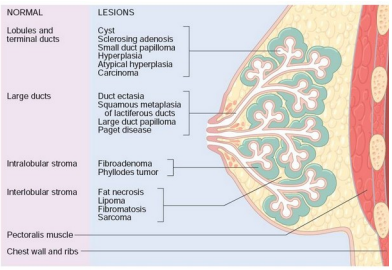
High yield: Slide 55. 12 weeks, large for dates, no fetus, many cystic areas: **E. Complete hydatidiform mole**.

7. Breast Pathology

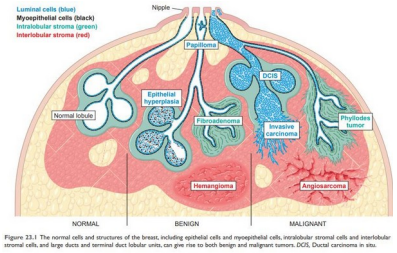
Sources: Teacher deck "Breast carcinoma and disease" (115 slides) and Pathoma Ch 16 (Breast Pathology, lecture notes pp. 320-328) plus Pathoma slides Ch 17.

7.1 Normal anatomy and histology

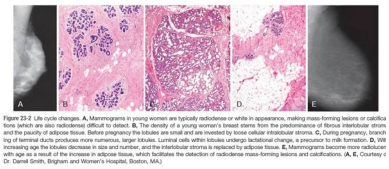
- The breast can develop anywhere along the **milk line** (from the axilla to the vulva); extra nipples or breast tissue (**polythelia/polymastia**) sit along this line.
- In **women** lobules and ducts are densest in the **upper outer (lateral) quadrant**, which is why about **50% of cancers arise in the upper outer quadrant**. In **men** the tissue is densest **under the areola** (subareolar).
- The **terminal duct lobular unit (TDLU)** is the functional unit: the **lobules (acini) make milk** and the **ducts drain it** to the **lactiferous ducts and sinuses** at the nipple.
- Both lobules and ducts have **two cell layers**: an inner **luminal (cuboidal/columnar) epithelial layer** that secretes milk and an outer **myoepithelial layer** that contracts to expel it, sitting on a **basement membrane**.
- Estrogen and progesterone cause lobule and duct proliferation. Prepubertal breast has ducts only; adult breast has lobules with acini and lactiferous ducts in fibrous and fatty stroma.



slide 4



slide 5



slide 8

Changes with the menstrual cycle and pregnancy

- **Proliferative (follicular) phase:** estrogen drives duct proliferation (the "estrogenic phase").
- **Late luteal (secretory) phase:** progesterone plus estrogen cause more **oedema and vacuolisation**, so women feel **pain and fullness** (premenstrual mastalgia).
- **Pregnancy and lactation:** marked proliferation of lobules and acini with **eosinophilic secretions in the lumen**.
- **Galactorrhoea** (milk outside pregnancy and lactation) is caused by **nipple stimulation or a prolactinoma** (see the pituitary chapter). It is **not a sign of cancer**.

7.2 Clinical presentation and imaging

- **Pain (mastalgia)** is the commonest symptom; others are a **palpable mass** and **nipple discharge**. A mass is palpable at about **2 cm**.
- Risk that a mass is malignant: under 40 years about 10%; over 50 years about 60%.
- **Mammography** (screening after about 40 years, and diagnostic) finds non-palpable lesions. It looks for **densities** (invasive carcinoma, fibroadenoma, cysts; the average invasive cancer found by mammography is 1.1 cm versus 2.4 cm for palpation) and **calcifications** (benign: clusters in apocrine

cysts, hyalinised fibroadenoma, sclerosing adenosis; **malignant: small, irregular, numerous, clustered, linear and branching**, as in **DCIS**).

- Breast implant rupture shows prosthetic material surrounded by **foreign-body giant cells** and histiocytes.

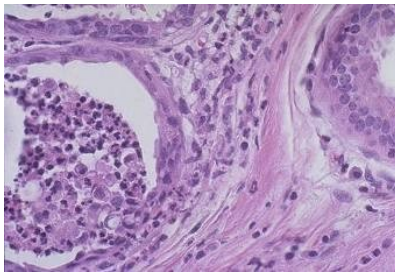
Examination clues

Finding	Meaning
Nipple drawn up or retracted, skin dimpling	Cancer (fibrosis pulling on ligaments); in fibroadenoma the nipple is displaced away
Peau d'orange	Dermal lymphatic obstruction (inflammatory carcinoma, advanced cancer)
Stony hard, fixed mass	Cancer
Firm, rubbery, mobile ("breast mouse")	Fibroadenoma
Fluctuant mass	Cyst

7.3 Inflammatory conditions and fat necrosis

Condition	Cause	Presentation	Biopsy / treatment
Acute mastitis	**Staphylococcus aureus**, during **breastfeeding** (cracks and fissures in the nipple let bacteria in)	Erythematous, painful breast, fever, purulent discharge, possible abscess (**puerperal abscess**)	**Continue draining the breast**, **dicloxacillin**, drainage of abscess
Periductal mastitis (subareolar abscess)	**Smokers**: vitamin A deficiency (relative) causes **squamous metaplasia of lactiferous ducts**; keratin blocks the duct, then ruptures and provokes intense inflammation	Recurrent subareolar abscess and **fistula** to the areola edge, **nipple retraction** (inflammation and fibrosis); staphylococci and anaerobes	Antibiotics, surgery
Mammary duct ectasia	**Not** associated with smoking or squamous metaplasia. **Chronic inflammation with dilation** of subareolar ducts, 5th-6th decade	**Periareolar mass** with thick **green-brown nipple discharge**; can be mistaken for cancer	**Plasma cells and granulomas, foamy macrophages in a dilated duct**
Fat necrosis	**Trauma** (often forgotten)	Painless mass or **mammographic calcifications** (saponification)	**Liquefactive necrosis of adipocytes**, foamy macrophages, giant cells, granulomas, calcification, fibrosis

Memory hook: Smoker with recurrent subareolar abscess = squamous metaplasia + vitamin A. Duct ectasia = plasma cells and a green-brown discharge in an older woman, not a smoker thing.



slide 30 *- Acute Mastitis*

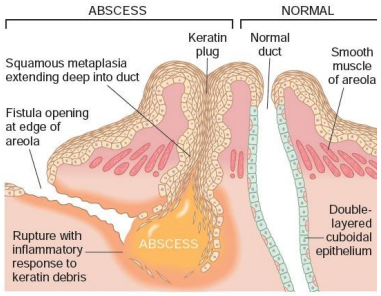


Figure 23-4 Squamous metaplasia of lactiferous ducts. When squamou metaplasia extends deep into a nipple duct, keratin becomes trapped an accumulates. If the duct ruptures, the ensuing intense inflammatory respons to keratin results in an erythematous painful mass. A fistula tract may burrov beneath the smooth muscle of the nipple to open at the edge of the areol

slide 32



slide 34 *- Mammary duct ectasia | Macrophages within lumen*

7.4 Benign epithelial lesions: three groups

These present as a lump or mammographic density/calcification; fibrocystic lumps may **disappear after fine needle aspiration of the cyst**.

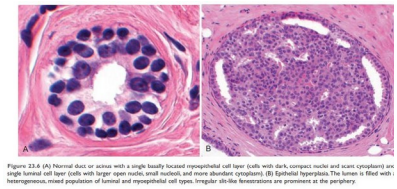
I. Non-proliferative (fibrocystic change)

- **The most common breast change in premenopausal women** (30-60% of women); hormone-mediated, linked to caffeine in older teaching and "subtle ovarian dysfunction". Presents as a **lumpy breast** in the upper outer quadrant, often bilateral and cyclical.
- **Components: cysts** (an unopened cyst looks like a **blue-dome cyst** because of turbid fluid), **fibrosis** (firm nodules), **adenosis** (more acini), and **apocrine metaplasia** (cyst lined by large pink cells with apical snouts, like sweat-gland epithelium) with calcifications.
- **No increased cancer risk.**

II. Proliferative without atypia (about 1.5-2x risk)

Lesion	Key points
Epithelial hyperplasia	More than two cell layers thick lining the ducts
Sclerosing adenosis	**Fibrosis + increased acini (about two-fold) + calcification**; compressed central acini; lobular pattern preserved; it can mimic carcinoma and tubular carcinoma
Complex sclerosing lesion (radial scar)	**Central stellate scar with radiating tear-drop glands**; scarring with no history of trauma or surgery; irregular mass on radiograph that can look like cancer
Intraductal papilloma	**Papillary growth within a dilated duct, with a fibrovascular core lined by both myoepithelial and luminal cells**. Premenopausal women. **Large duct papillomas** sit in the lactiferous sinus: **bloody or serous nipple discharge**, may cause nipple retraction. Small duct papillomas carry a risk of malignancy

Memory hook: Intraductal papilloma vs papillary carcinoma: both bleed from the nipple. The benign one is **premenopausal** and has **myoepithelial cells**; the cancer is **postmenopausal** and lacks **myoepithelial cells**.



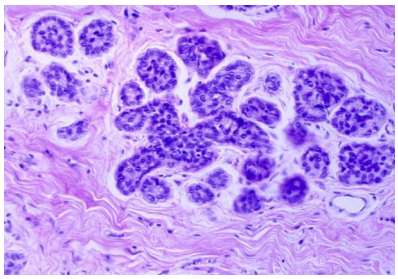
slide 50

III. Proliferative with atypia (about 4-5x risk, both breasts)

- **Atypical ductal hyperplasia (ADH):** monotonous cells with cribriform spaces, often with calcifications; resembles low-grade DCIS (size and extent separate them).
- **Atypical lobular hyperplasia (ALH):** monotonous, loosely adherent cells filling acini; usually an incidental finding without calcifications; resembles LCIS.

Relative risk of cancer	Lesion
No increase	Fibrosis, cysts, apocrine metaplasia, mild adenosis
1.5-2x	Sclerosing adenosis, ductal hyperplasia, papillomas
4-5x	Atypical ductal or lobular hyperplasia
Substantial	Carcinoma in situ

Fibrocystic change versus cancer: fibrocystic change is often **bilateral**, multiple, varies with the **menstrual cycle** (cyclic pain and engorgement) and may regress in pregnancy; cancer is usually **unilateral, single, non-cyclical** and does not regress.



***slide 60** - ALH*

7.5 Stromal and epithelial-stromal tumours

	Fibroadenoma	Phyllodes tumour
Who	**Most common benign breast neoplasm**; usually **under 30**, premenopausal	**Older**, around the 5th decade and **postmenopausal**
Feel	**Well circumscribed, freely mobile, rubbery** ("**breast mouse**")	Fibroadenoma-like but **much larger**, rapidly growing
Hormones	Responds to estrogen: grows in **late menstrual phase and pregnancy**, may be painful	
Histology	**Proliferation of both glands (slit-like ducts lined by epithelium) and fibrous stroma**; the stroma compresses the ducts into slits	**Leaf-like (phyllodes) clefts with a cellular myxoid stroma**; a tumour of the **stromal cells**
Behaviour	Benign (cancer risk 1.5-2x for complex ones)	Usually benign but **recurs**; the stroma can become **malignant (sarcoma)** and metastasise haematogenously

Mammography: fibroadenoma shows uniform densities or coarse calcifications.

Table 23.1 Epithelial Breast Lesions and Risk of Developing Invasive Carcinoma

Pathologic Lesions	Relative Risk (Absolute Lifetime Risk)*
Nonproliferative breast changes (mild hyperplasia, duct ectasia, cysts, apocrine metaplasia, adenosis, fibroadenoma without complex features)	1.0 (~3%)
Proliferative disease without atypia (moderate or florid hyperplasia, sclerosing adenosis, complex sclerosing lesion, fibroadenoma with complex features)	1.5–2 (~5%–7%)
Proliferative disease with atypia (atypical ductal hyperplasia, atypical lobular hyperplasia)	4–5 (~13%–17%)
Carcinoma in situ (lobular carcinoma in situ, ductal carcinoma in situ)	8–10 (~25%–30%)

*Relative risk is the likelihood of developing invasive carcinoma compared to women without any risk factors. Absolute lifetime risk is the percentage of women expected to develop invasive carcinoma in the absence of an intervention.

slide 69

High yield: Pathoma: **only phyllodes tumour is mainly postmenopausal and can be malignant; the others are mainly premenopausal and not usually malignant.**

7.6 Breast carcinoma

Epidemiology: the **commonest non-skin cancer in women** and the **second leading cause of cancer death** (after lung). About **1 in 8** women by age 80.

Risk factors (mostly cumulative estrogen exposure)

Female sex (about 100:1 over men), **age** (mostly postmenopausal, except hereditary), **early menarche and late menopause**, **nulliparity/late first pregnancy**, **obesity** (fat converts androgen to estrogen), **atypical hyperplasia**, **first-degree relative**, African-American race (more aggressive/triple-negative), **BRCA mutations**, radiation exposure. **Sporadic cancers:** about **80% express estrogen receptor (ER)**; estrogen metabolites damage DNA and estrogen drives proliferation. About 20% are ER negative and often overexpress **HER2/neu**.

Table 23.2 Risk Factors for Developing Breast Cancer

Risk Factors	Relative Risk*
Female gender Increasing age Germline mutations of high penetrance Strong family history (>1 first-degree relative, young age, multiple cancers) Personal history of breast cancer High breast density	>4.0
Germline mutations of moderate penetrance High-dose radiation to chest at young age Family history (1 first-degree relative)	2.1–4.0
Early menarche (age <12 years) Late menopause (age >55 years) Late first pregnancy (age >35 years) Nulliparity Absence of breastfeeding Exogenous hormone therapy Postmenopausal obesity Physical inactivity High alcohol consumption	1.1–2.0

*Relative risk is the likelihood of developing invasive carcinoma compared to women without any risk factors.

slide 71

Classification

Group	Types
Non-invasive	**DCIS** (comedo and non-comedo subtypes), **LCIS**

Invasive	**Ductal (no special type)**, **lobular**, medullary, tubular, mucinous (colloid), inflammatory (clinical), others
Special	**Paget's disease** of the nipple

Ductal carcinoma in situ (DCIS)

- Proliferating malignant cells **inside the ducts, confined by the basement membrane**. **10-20% bilateral.** Usually **no mass**; found on mammography as **clusters of branching (linear) microcalcifications**.
- **Comedo type:** high-grade cells with **central necrosis** (comedo necrosis) and dystrophic calcification; ducts are distended up to 10x normal size; may be a 2-3 cm palpable mass.
- **Non-comedo types** (cribriform, solid, papillary, micropapillary): lower grade, monomorphic cells forming **cribriform ("cookie cutter") spaces**; non-palpable.
- **Treatment:** breast conservation by excision plus radiotherapy.

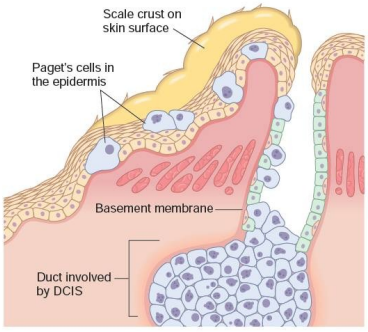


Figure 23-18 Paget disease of the nipple. Ductal carcinoma in situ arising within the ductal system of the breast can extend up the lactiferous ducts and into the skin of the nipple without crossing the basement membrane. The malignant cells disrupt the normally tight squamous epithelial cell barrier allowing extracellular fluid to seep out and form an oozing scaly crust.

slide 85

Paget's disease of the nipple

DCIS (or invasive ductal carcinoma) extends up the lactiferous ducts and **into the nipple skin without crossing the basement membrane**. The malignant cells disrupt the squamous barrier so fluid seeps out as an **erythematous, scaly, weeping crust, sometimes ulcerated**. **Biopsy shows Paget cells (large cells with clear halo) in the squamous epidermis.** It signals an **underlying cancer**.

Lobular carcinoma in situ (LCIS)

- Dyscohesive monomorphic cells distending the acini and lobules, with lobular architecture maintained. **20-40% bilateral**, usually an **incidental biopsy finding** with **no mass or calcifications** (so it is difficult to detect by mammography).
- **Loss of E-cadherin** (an adhesion protein) explains the discohesion. Treatment: close follow-up, **tamoxifen** to reduce risk (it is a marker of high risk for cancer in either breast).

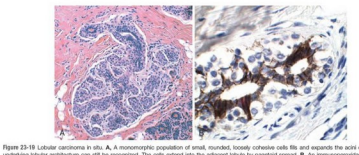


Figure 23-19 Lobular carcinoma in situ. **A**, A monomorphic population of small, rounded, loosely cohesive cells fills and expands the acini of a lobule, underlying lobular architecture can still be recognized. The cells extend into the adjacent lobule by pagetoid spread. **B**, An immunoperoxidase study shows E-cadherin–positive normal luminal cells that have been undermined by E-cadherin–negative LCIS cells spreading along the basement membrane.

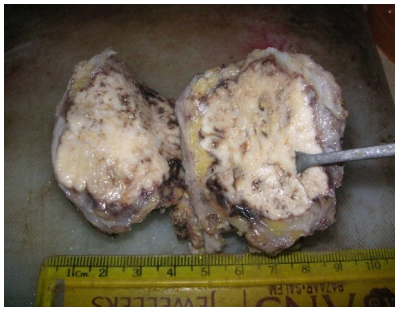
slide 87 *- E – Cadherin*

Invasive ductal carcinoma (no special type)

- **70-80% (Pathoma: over 80%) of all breast cancers.** Presents as a **palpable, rock-hard, immobile mass**; on cut section a **stellate (star-shaped) mass with irregular borders** that has a **grating sound** (calcifications), and a **dense desmoplastic ("scirrhous") stroma**.
- Micro: well-differentiated tumours form tubules/ducts; moderately differentiated show solid nests; poorly differentiated show sheets of pleomorphic cells with mitoses and necrosis. "NOS" is a **diagnosis of exclusion**.

Special invasive types

Type	Key features	Prognosis
Invasive lobular	**Single-file ("Indian file") growth**, often signet-ring cells, **loss of E-cadherin**; postmenopausal; often **bilateral and multifocal**; presents as a vague thickening, not a lump; hard to detect on mammography; associated with LCIS in 90%	Better than ductal; metastasises to unusual sites (peritoneum, GI tract, meninges, ovary)
Medullary	**BRCA1** carriers; soft fleshy well-circumscribed mass (can look like fibroadenoma); **sheets of anaplastic cells with lymphocytes and plasma cells**	Good
Colloid (mucinous)	Older women; soft gelatinous mass; **tumour islands floating in pools of mucin**	Good
Tubular	Late 40s, under 1 cm; well-formed tubules **without myoepithelial cells**, desmoplastic stroma; mistaken for sclerosing adenosis	Excellent
Inflammatory	A **clinical diagnosis** (not a histological type): tumour clogs **dermal lymphatics**, causing a swollen, red, warm breast with **peau d'orange** and no discrete mass; no inflammatory cells (misnomer); can be mistaken for mastitis	**Worst**



slide 93 - *Medullary Carcinoma*

Prognosis and the sentinel node

- **Distant metastasis** is the strongest prognostic factor, but most women present before it. **Axillary lymph node involvement is the most useful prognostic factor.** Others: tumour size, grade (**Nottingham score** = tubule formation + nuclear pleomorphism + mitotic count), inflammatory type, receptor status.
- **Sentinel node biopsy:** dye or tracer injected at the tumour identifies the **first draining node**. If it is negative the axillary dissection is **not performed**, avoiding **lymphoedema** of the arm. After full axillary dissection and mastectomy, **lymphoedema** can occur; chronic lymphoedema can give **angiosarcoma (Stewart-Treves syndrome)**.

Predictive markers: what treatment will work

Marker	Therapy
ER / PR positive (nuclear receptors)	**Antiestrogen therapy** (tamoxifen, aromatase inhibitors)
HER2/neu overexpressed (surface receptor, RAS/MAPK pathway)	**Trastuzumab (Herceptin)**, anti-HER2 antibody (cardiotoxicity warning)
Triple negative (ER-, PR-, HER2-)	Poor targeted options; chemotherapy; more common in African-American women and **BRCA1** carriers

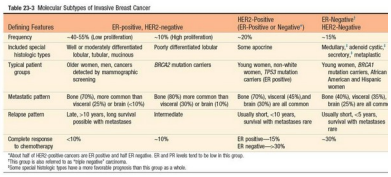


Table 23-3 Molecular Subtypes of Invasive Breast Cancer

Defining Features	ER-positive, HER2-negative		HER2-Positive (ER-Positive or Negative*†)	ER-Negative† HER2-Negative
Frequency	~40–55% (Low proliferation)	~10% (High proliferation)	~20%	~15%
Included special histologic types	Well or moderately differentiated tubular, tubulolar, mucinous	Poorly differentiated lobular	Some apocrine	Medullary,‡ adenoid cystic,‡ secretory,‡ metaplastic
Typical patient groups	Older women, men, cancers detected by mammographic screening	*BRCA2* mutation carriers	Young women, non-white women, TP53 mutation carriers (ER positive)	Young women, *BRCA1* mutation carriers, African American and Hispanic women
Metastatic pattern	Bone (70%), more common than visceral (25%) or brain (<10%)	Bone (80%) more common than visceral (30%) or brain (10%)	Bone (70%), visceral (45%),and brain (30%) are all common	Bone (40%), visceral (35%), brain (25%) are all comm
Relapse pattern	Late, >10 years, long survival possible with metastases	Intermediate	Usually short, ~10 years, survival with metastases rare	Usually short, ~5 years, survival with metastases rare
Complete response to chemotherapy	<10%	~10%	ER positive—15% ER negative—>30%	~30%

*About half of HER2-positive cancers are ER positive and half ER negative. ER and PR levels tend to be low in this group.
†This group is also referred to as "triple negative" carcinoma.
‡Some special histologic types have a more favorable prognosis than this group as a whole.

slide 103

7.7 Hereditary breast cancer

About **10%** of cases: **premenopausal**, multiple tumours, multiple first-degree relatives.

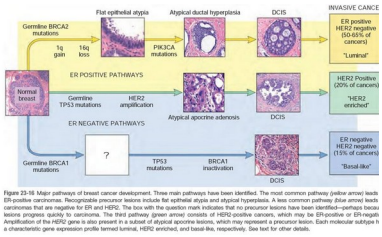
Gene	Chromosome	Cancers
BRCA1	**17**	Breast (about 60%; **triple-negative, medullary**), **ovarian (about 40%, serous)**, male breast 1-2%. Onset 40-50s
BRCA2	**13**	Breast (about 40%), ovarian (about 15%), **male breast (about 6%)**, prostate, pancreas. Onset over 50
TP53 (Li-Fraumeni)	17	**SBLA**: Sarcoma, Breast, Leukaemia, Adrenal cortical carcinoma

Both BRCAs are autosomal dominant tumour-suppressor (DNA repair) genes. **Prophylactic bilateral mastectomy** (and salpingo-oophorectomy) reduces risk but not to zero.

7.8 Male breast

- **Gynecomastia:** proliferation of ducts and stroma from **hyperestrogenaemia or reduced androgen effect**; often idiopathic in adolescents. Older causes: **Klinefelter syndrome, cirrhosis (liver cannot clear estrogen), Leydig cell tumour, drugs** (opiates, marijuana, alcohol, anabolic steroids, spironolactone, digoxin). Histology: ducts surrounded by oedema and chronic inflammation, **no lobules**.

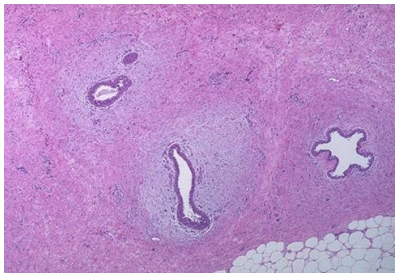
- **Male breast cancer: 1% of all breast cancers**; usually **invasive ductal carcinoma**; subareolar mass in an older man, possibly nipple discharge; presents at high stage. Associations: **BRCA2** and **Klinefelter syndrome**.



slide 110

Gynecomastia	
Clinical findings	• Development of glandular breast tissue in males
Pathophysiology	• Increased estrogen/androgen ratio • Ductal epithelial hyperplasia & stromal fibrosis
Common etiologies	• Increased estrogen levels (eg, cirrhosis, obesity) • Decreased testosterone levels (eg, chronic kidney disease, hypogonadism) • Medication effect ◦ Decreased androgen production (eg, GnRH agonists, ketoconazole) ◦ Decreased conversion to dihydrotestosterone (eg, 5-alpha reductase inhibitors) ◦ Inhibition of androgen receptors (eg, spironolactone, bicalutamide)

slide 112



slide 114 - *Gynecomastia*



slide 115

High yield: Quick breast table to memorise

Clue	Answer
Mobile, rubbery lump in 25-year-old	Fibroadenoma
Large "leaf-like" mass in an older woman	Phyllodes tumour
Bloody nipple discharge, premenopausal	Intraductal papilloma
Blue-dome cyst, lumpy cyclic pain	Fibrocystic change
Green-brown discharge, plasma cells	Duct ectasia
Recurrent subareolar abscess in smoker	Squamous metaplasia / periductal mastitis
Calcifications, no mass, necrosis	DCIS (comedo)
Scaly nipple rash	Paget's disease (underlying DCIS or invasive ductal)
Indian file, E-cadherin lost	Lobular carcinoma
Peau d'orange, no lump	Inflammatory carcinoma
Lymphocytes, BRCA1, soft mass	Medullary carcinoma

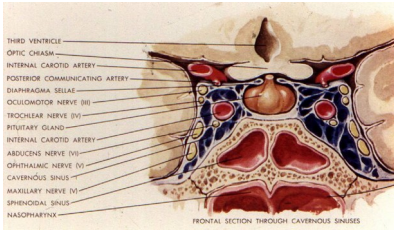
8. Pituitary Gland

Sources: Teacher decks "Pituitary_Updated" (34 slides) and the older "Pituitary" deck (53 slides, content merged here and its pictures included), Pathoma Ch 15.1-15.2 (lecture notes pp. 299-301) and Pathoma slides Ch 18.

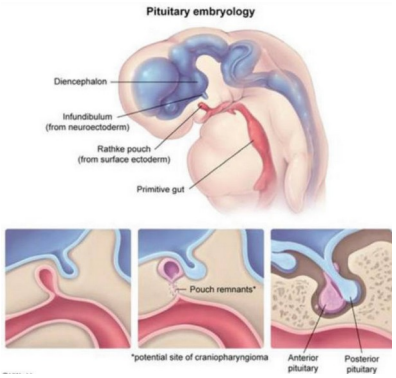
Every pituitary problem is one of three questions: **is a hormone missing, is a hormone in excess, or is a mass pressing on something nearby?**

8.1 Anatomy, embryology and physiology

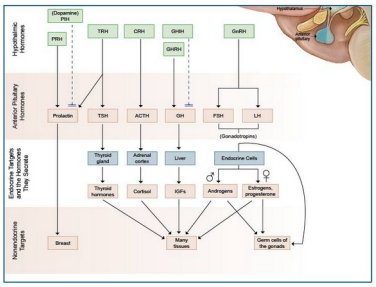
- The pituitary sits in the **sella turcica** (a bony saddle in the sphenoid bone) **directly below the optic chiasm**. That is why **suprasellar masses cause visual field loss**. It is flanked by the **cavernous sinuses** (CN III, IV, VI, V1, V2) and joined to the hypothalamus by the **infundibulum (stalk)**.
- **Anterior lobe (adenohypophysis):** derived from **Rathke's pouch (oral ectoderm)**. Contains acidophils (GH, prolactin), basophils (ACTH, TSH, FSH/LH) and chromophobes (stem cells and degranulated cells).
- **Posterior lobe (neurohypophysis):** derived from the **floor of the third ventricle (neuroectoderm)**; made of **pituicytes (modified glia)** and axons. It **stores and releases ADH and oxytocin made in the hypothalamus.**



slide 3 *- Location of Pituitary : in Sella turcica & Near to optic chiasma.*



slide 5



slide 7 *- Anterior Pituitary Hormones*

Cell type	Hormone(s)	Hypothalamic control
Somatotroph (acidophil)	**Growth hormone (GH)**	GHRH stimulates; somatostatin inhibits
Lactotroph / mammotroph (acidophil)	**Prolactin (PRL)**	**Dopamine inhibits** (PIH); TRH stimulates
Corticotroph (basophil)	**ACTH, MSH, beta-endorphin** (cleaved from POMC)	CRH
Thyrotroph (basophil)	**TSH**	TRH
Gonadotroph (basophil)	**FSH, LH**	GnRH

Posterior hormone	Made in	Function
Oxytocin	**Paraventricular nucleus**	Uterine contraction, milk letdown
ADH (vasopressin)	**Supraoptic nucleus**	Water retention

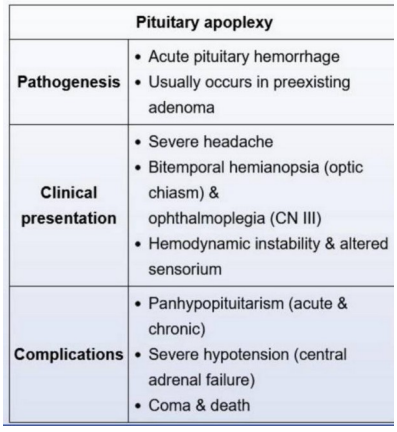
High yield: ADH is made as a pre-propeptide that is cleaved into **vasopressin + neurophysin II + copeptin**. **Neurophysin II** is the carrier protein that travels with ADH down the axon; **copeptin** is a stable marker of ADH release. A **neurophysin gene mutation causes central diabetes insipidus** (teacher MCQ, slide 30).

8.2 Hypopituitarism (loss of function)

Symptoms appear when **over 75% of the pituitary is lost**. Loss usually follows a pattern: **GH, FSH and LH are lost first; TSH and ACTH later.**

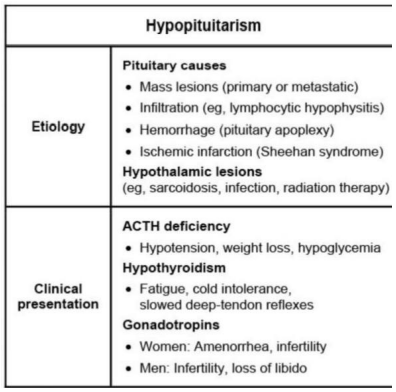
Causes

Cause	Notes
Non-functioning pituitary adenoma	Commonest cause in adults; compresses normal gland
Pituitary apoplexy	Sudden haemorrhage or infarction, usually into a pre-existing adenoma. Risk factors: pregnancy (Sheehan), diabetes, hypertension, sickle cell disease, acute shock
Sheehan syndrome	**Postpartum pituitary infarction.** In pregnancy the anterior pituitary **doubles in size (lactotroph hyperplasia) but its blood supply does not**, so blood loss and hypotension at delivery cause ischaemic necrosis. Presents with **poor lactation (no milk), loss of pubic hair, fatigue** and later panhypopituitarism
Surgery or radiation	Iatrogenic
Ischaemic necrosis, DIC	
Inflammation and infection	Sarcoidosis, TB meningitis, **autoimmune lymphocytic hypophysitis**
Empty sella syndrome	Enlarged sella with **no visible pituitary**. Primary: herniation of arachnoid/CSF into the sella compresses the gland (**obese, hypertensive women**; can be congenital). Secondary: after surgery, radiation, injury. Often asymptomatic (gland is compressed, not destroyed)
Craniopharyngioma	Commonest cause in children (below)
Metastatic tumours	Breast and lung



Pituitary apoplexy	
Pathogenesis	• Acute pituitary hemorrhage • Usually occurs in preexisting adenoma
Clinical presentation	• Severe headache • Bitemporal hemianopsia (optic chiasm) & ophthalmoplegia (CN III) • Hemodynamic instability & altered sensorium
Complications	• Panhypopituitarism (acute & chronic) • Severe hypotension (central adrenal failure) • Coma & death

slide 13



Hypopituitarism	
Etiology	**Pituitary causes** • Mass lesions (primary or metastatic) • Infiltration (eg, lymphocytic hypophysitis) • Hemorrhage (pituitary apoplexy) • Ischemic infarction (Sheehan syndrome) **Hypothalamic lesions** (eg, sarcoidosis, infection, radiation therapy)
Clinical presentation	**ACTH deficiency** • Hypotension, weight loss, hypoglycemia **Hypothyroidism** • Fatigue, cold intolerance, slowed deep-tendon reflexes **Gonadotropins** • Women: Amenorrhea, infertility • Men: Infertility, loss of libido

slide 15

Pituitary apoplexy and Sheehan: presentation

Severe sudden headache and vomiting, sudden visual loss (bitemporal hemianopia; ophthalmoplegia from CN III compression), altered mental status and hypotension. The sudden loss of ACTH produces **an Addisonian crisis** (hypotension, electrolyte abnormalities). **Management:** immediate **glucocorticoid replacement** plus **surgical decompression of the optic nerves** to save vision.

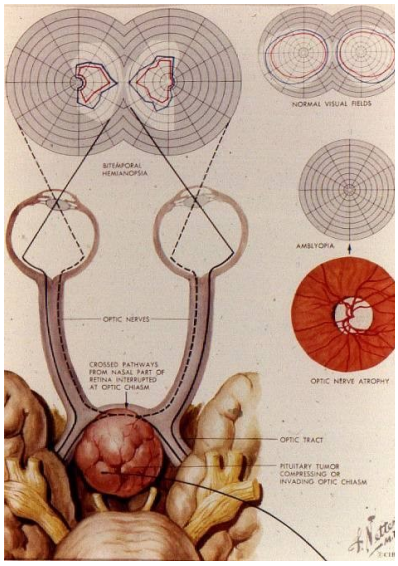
Clinical features by hormone deficiency

Deficient hormone	Consequence
GH	Growth failure/dwarfism (children); abnormal body composition (adults)
TSH	Central (secondary) hypothyroidism
ACTH	Hypocortisolism with **relative preservation of mineralocorticoid** (aldosterone is controlled by the renin-angiotensin system, not ACTH, so there is little hyperkalaemia)
FSH/LH	Loss of libido, infertility, loss of secondary sexual characteristics; oligo/amenorrhoea
ADH (posterior)	Polyuria and polydipsia (central diabetes insipidus)
Mass effect	Non-communicating hydrocephalus, **bitemporal hemianopia**, headache

Diagnostic work-up: basal GH, ACTH, TSH, free T4/T3, LH/FSH (with GnRH stimulation), a combined anterior pituitary stimulation test (GHRH + CRH + GnRH + TRH given together with serial sampling), and **MRI of the sella**.

Bitemporal hemianopia explained

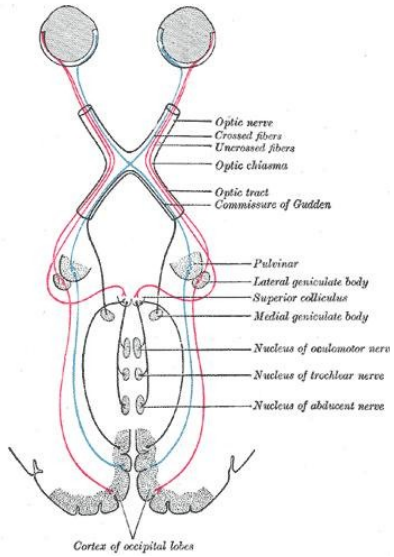
The optic chiasm lies above the gland. A suprasellar mass compresses the **crossing (nasal) retinal fibres**, which carry the **temporal visual fields**. Result: loss of the outer half of the visual field in both eyes. A classic finding in pituitary adenoma, craniopharyngioma and apoplexy.



slide 21 *- Visual Field Problems*



slide 14 *- HYPOPITUITARISM — TUMORS | Craniopharyngioma: Clinical Features | Pituitary Pathology |*



slide 15 *- HYPOPITUITARISM — TUMORS | Mass Effect: Bitemporal Hemianopia | Pituitary Pathology | 1*

Craniopharyngioma

- **Most common childhood suprasellar tumour.** A **benign, slow-growing epithelial tumour from Rathke's pouch remnants** at the infundibulum; may obstruct CSF flow.
- **Gross/micro:** cystic, lined by stratified squamous epithelium; the cyst contains "**motor-oil**" cholesterol-rich fluid; **calcification is common** (visible on CT).

Subtype	Patients	Genetics	Look
Adamantomatous	Children	**CTNNB1 (beta-catenin)**	Wet keratin, calcification, cysts
Papillary	Adults	**BRAF V600E**	No calcification, no cysts

- **Children:** visual loss and **central diabetes insipidus**, adiposity, delayed development, headache and vomiting (raised ICP), hydrocephalus, bitemporal hemianopia, dwarfism, hypothyroidism.
- **Adults:** reduced libido, amenorrhoea, mild spastic leg weakness, headache, failing vision, mental dullness.

Pituitary adenomas

Cell type	Clinical features of tumor
Lactotroph (prolactin)	• Hypogonadism, infertility (men) • Galactorrhea, amenorrhea (women)
Corticotroph (ACTH)	• Cushing disease
Somatotroph (growth hormone)	• Acromegaly (adults) • Gigantism (children)
Thyrotroph (TSH)	• Thyrotoxicosis
Gonadotroph (LH, FSH)	• Nonfunctioning (hypopituitarism)

slide 25

Prolactinoma

Clinical features	• **Premenopausal women**: Oligomenorrhea/amenorrhea, infertility, galactorrhea, hot flashes, decreased bone density • **Postmenopausal women**: Mass-effect symptoms (headache, visual field defects • **Men**: Infertility, decreased libido, impotence, gynecomastia
Laboratory/ imaging	• Serum prolactin (often >200 ng/mL) • Tests to rule out renal insufficiency (creatinine) & hypothyroidism (TSH, thyroxine) • MRI of the brain/pituitary
Treatment	• Dopamine agonist (cabergoline) • Trans-sphenoidal surgery

slide 30

8.3 Hyperpituitarism and pituitary adenomas

Pituitary adenoma: a **benign, well-circumscribed, monoclonal** tumour of one anterior pituitary cell type (polygonal cells). Classified by **function** (endocrine-active vs inactive) and **size** (**microadenoma under 1 cm**, **macroadenoma over 1 cm**). Gross: **reddish-grey, soft, almost gelatinous**, often partly cystic with a calcified rim.

Adenoma	Hormone	Share (Pathoma) / key points
Lactotroph (prolactinoma)	PRL	**Most common**, about 50%
Somatotroph	GH	10-15%
Corticotroph	ACTH	About 10% (Cushing disease; **Nelson syndrome**)
Thyrotroph	TSH	Rare
Gonadotroph	FSH/LH	Rare
Null cell	None	Non-functioning
Mammosomatotroph	PRL + GH	Plurihormonal

Clinical effects of any adenoma: (1) **hormone excess**, (2) **raised intracranial pressure** (headache, vomiting), (3) **visual field defects** (bitemporal hemianopia), and (4) **hypopituitarism** from compression of normal tissue (GH, FSH, LH first; TSH, ACTH later). Earliest symptom in men: loss of libido or infertility; in premenopausal women: oligo/amenorrhoea.

MEN1 (Wermer syndrome): the 3 P's

MEN1 gene (menin). Pituitary adenoma (usually prolactinoma), **P**arathyroid hyperplasia/adenoma (hypercalcaemia), **P**ancreatic neuroendocrine tumour (gastrinoma, insulinoma). Suspect it in a young patient with a pituitary adenoma plus hypercalcaemia.

Prolactinoma

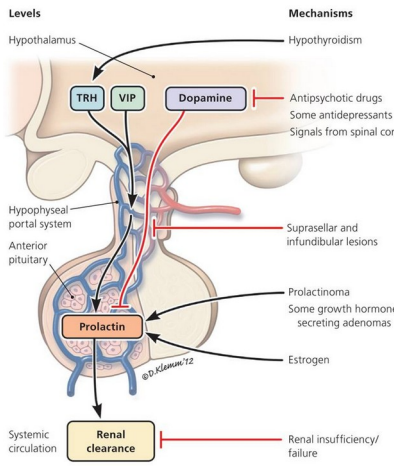
- **Lactotroph (acidophilic) adenoma**, commonest secretory tumour, **young women 20-40**, usually a **microadenoma** in women because menstrual disturbance is noticed early; may calcify.
- **Mechanism:** excess prolactin **suppresses GnRH, which suppresses LH/FSH**, causing hypogonadism.

	Women	Men
Features	**Galactorrhoea, secondary amenorrhoea**, infertility, hot flushes, reduced bone density	**Reduced libido, infertility**, gynecomastia; typically diagnosed late as a **macroadenoma** with headache and visual loss, osteopenia, reduced muscle mass and beard growth. (Pathoma: men usually do not get galactorrhoea as they lack breast lobules)

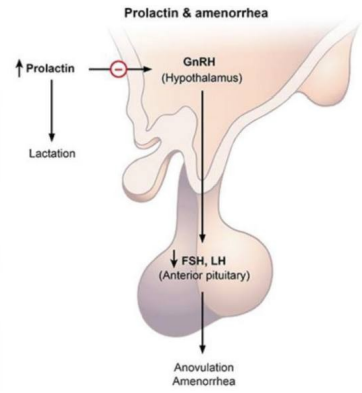
- **Lab:** raised prolactin (**over 200 ng/mL suggests a macroadenoma**); MRI. Rule out other causes of high prolactin: **pregnancy, hypothyroidism (high TRH), renal failure, drugs (dopamine antagonists such as antipsychotics, metoclopramide), stalk compression ("stalk effect": loss of dopamine inhibition)**.
- **Treatment: dopamine agonists** (**cabergoline preferred**, or bromocriptine) shrink the tumour and lower prolactin; trans-sphenoidal surgery if refractory.

High yield: Test yourself (slide 21). 34-year-old man, headaches, bitemporal field loss, prolactin over 20

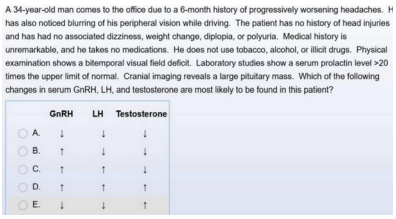
times normal, large mass: GnRH, LH and testosterone are all low. **Answer B (down, down, down)**: prolactin suppresses GnRH, so LH and testosterone fall.



slide 31



slide 32

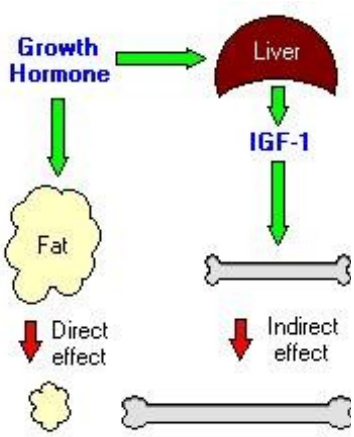


slide 33

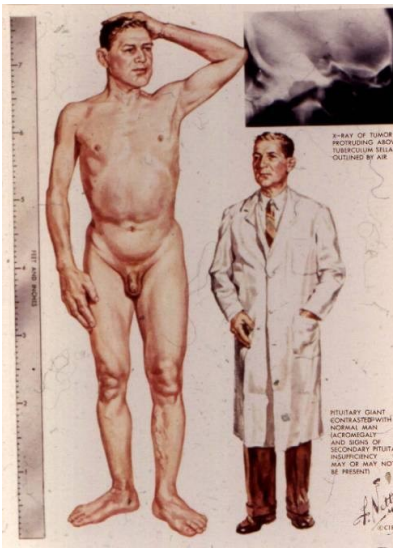
GH excess: gigantism and acromegaly

	Gigantism	Acromegaly
When	**Children, before epiphyseal closure**	**Adults, after epiphyseal closure**
Features	Proportionally increased height, long limbs	Soft tissue and viscera growth (organomegaly), bones of **face, hands and feet**, **prognathism (underbite)**, tooth separation, coarse features, large tongue, **carpal tunnel syndrome**, scoliosis, hyperhidrosis, deepened voice, **obstructive sleep apnoea**, **colon polyps and colorectal cancer**, **hyperglycaemia/diabetes** (GH is anti-insulin)

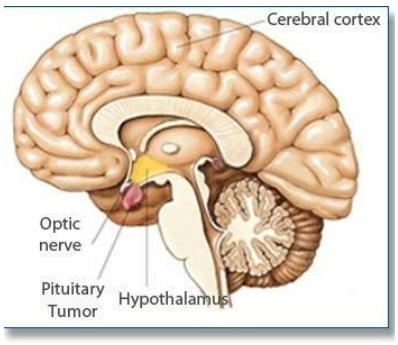
Pathophysiology: a **somatotroph (GH-secreting acidophil) adenoma**. GH stimulates the liver to make **IGF-1 (somatomedin C)**, which drives chondrogenesis and bone and soft-tissue growth. **Diagnosis:** raised **GH and IGF-1**; **failure of an oral glucose load to suppress GH** (the confirmatory test, like the dexamethasone suppression test in Cushing); high phosphate (increased renal reabsorption) and hypercalciuria. **Treatment: trans-sphenoidal surgery** first; **octreotide** (somatostatin analogue); dopamine agonists; **pegvisomant** (GH receptor antagonist); radiation if refractory. **Cause of death: cardiomyopathy/heart failure.**



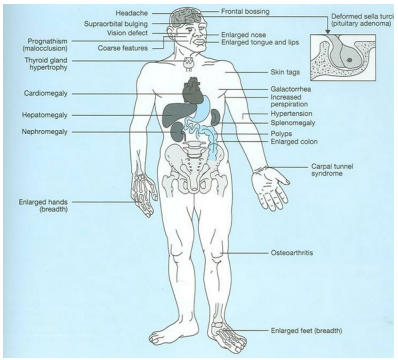
slide 36 *- 36 | Growth hormone*



slide 39 *- Gigantism*



slide 19 *- HYPERPITUITARISM — PROLACTINOMA | Prolactinoma: Overview | Pituitary Pathology | 19*



slide 24 *- HYPERPITUITARISM — GH EXCESS | Acromegaly: Pathophysiology | Pituitary Pathology | 23*

8.4 Posterior pituitary: ADH disorders

Diabetes insipidus (DI)

Polyuria and polydipsia from **ADH deficiency or resistance**.

	Central DI	**Nephrogenic DI**
Problem	**No ADH made/released**	**Kidney does not respond** (V2 receptor/aquaporin-2)
Causes	**Head trauma, sarcoidosis, tumours** (craniopharyngioma, hypothalamic/pituitary), infection, inflammation; also **vasopressinase-induced DI** in the 3rd trimester/puerperium (with oligohydramnios, preeclampsia, hepatic dysfunction)	**Lithium, demeclocycline, severe hypokalaemia, hypercalcaemia**, mutations
Water deprivation + desmopressin	Urine osmolality **rises by more than 50%** after ADH	Rises by **less than 50%**
Treatment	**Desmopressin**	Thiazide diuretic + low-salt diet; stop the drug

Lab: **raised plasma osmolality, low urine osmolality, hypernatraemia, dilute urine in large volume.** Most patients keep their balance if they can drink; dehydration and hypernatraemia occur if they cannot drink or the thirst centre is damaged.

SIADH

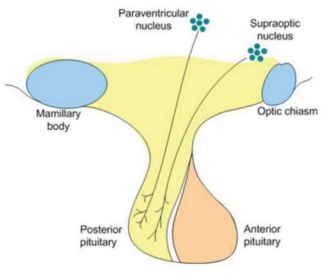
Excess ADH leads to water retention, dilutional euvolaemic hyponatraemia.

Cause category	Examples
Idiopathic	
Paraneoplastic	**Small cell lung carcinoma** (ectopic ADH)
CNS	Head injury, stroke, haemorrhage, infection (TB meningitis)
Pulmonary	Pneumonia, other lung disease
Drugs	Chlorpropamide, **carbamazepine**, morphine, vincristine, cyclophosphamide, SSRIs, NSAIDs

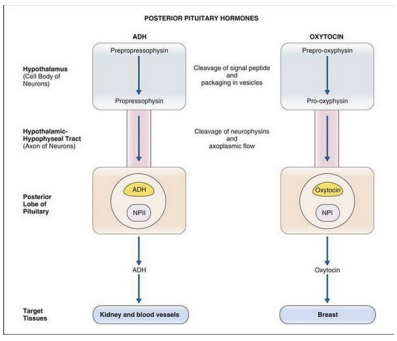
- **Clinical:** often asymptomatic; with sodium below 120 mEq/L: lethargy, weakness, confusion, delirium, seizures (brain-cell swelling). **Euvolaemic** (no oedema or JVD).
- **Labs:** hyponatraemia, low plasma osmolality, inappropriately **concentrated urine**, raised urine sodium.
- **Treatment: free water restriction**, **demeclocycline** (ADH antagonist), **vaptans** (tolvaptan, conivaptan), hypertonic saline only for severe symptoms. **Correct chronic hyponatraemia slowly (no more than 8-10 mEq/L per 24 h)** to avoid **osmotic demyelination (central pontine myelinolysis)**.

High yield: DI vs SIADH (mirror images)

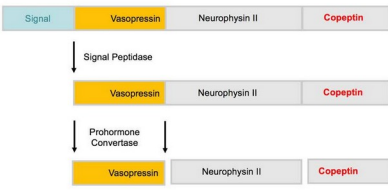
	DI	SIADH
ADH activity	Low/ineffective	High
Serum Na	High	Low
Plasma osmolality	High	Low
Urine osmolality	Low (dilute)	High (concentrated)
Volume	Dehydrated, polyuric	Euvolaemic
Key drug cause	Lithium	Carbamazepine, SSRIs
Treatment	Desmopressin (central); thiazide (nephrogenic)	Fluid restriction, demeclocycline, vaptans



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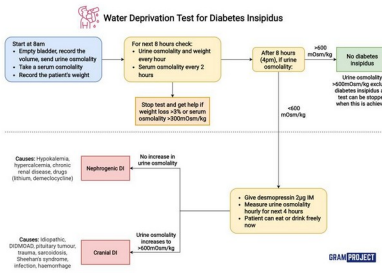
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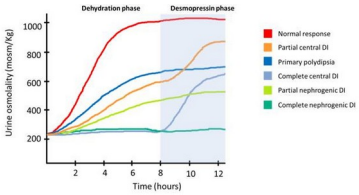
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slide 46



slide 47



slide 48

	Urine Osmolality After Dehydration (mOsm/L)	Urine Osmolality With Vasopressin Injection	Serum Sodium Lev
Normal	>800	<10% increase	Normal
Complete Neurogenic DI	<300	>50% increase	>142 mEq/L
Nephrogenic DI	<300–500	<50% increase	>142 mEq/L
Primary Polydipsia	>500	<10% increase	<137 mEq/L

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Table 1. Laboratory parameters in hypotonic polyuria syndromes

	Serum osmolality (mOsmol/kg H_2O)	Urine osmolality (mOsmol/kg H_2O)	Urine osmolality with water deprivation (mOsmol/kg H_2O)	Serum vasopressin after dehydration (pg/ml)	Change in urine osmolality with desmopressin
Normal	285-295	300-900	> 800	> 2	Little or no increase
Complete CDI	< 295*	< 300, often< 200	< 300	Undetectable	Increase (> 50%)
Partial CDI	< 295*	< 300	300-800	< 1.5	Increase (> 9%)
NDI	< 295*	< 300	< 300	> 5	Increase (< 50%)
PP	< 295	< 300	> 500	< 5	Increase (< 9%)

CDI = central diabetes insipidus; NDI = nephrogenic diabetes insipidus; PP = primary polydipsia
* Serum osmolality is high if thirst/drinking is impaired) [adapted from references 10, 11].

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Syndrome of inappropriate antidiuretic hormone (SIADH)	
Pathophysiology	• Uncontrolled secretion of ADH • Leads to water retention & impaired urinary water excretion
Causes	• CNS disturbances (stroke, hemorrhage, trauma) • Medications (eg, carbamazepine, SSRIs NSAIDs) • Lung disease (eg, pneumonia) • Malignancy (eg, small-cell lung cancer)

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Clinical findings	• Nausea, forgetfulness (mild hyponatremia) • Seizures, coma (severe hyponatremia) • Euvolemia (eg, moist mucous membranes, no edema, no JVD)
Laboratory findings	• Hyponatremia • Low serum osmolality • High urine osmolality • High urine sodium

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8.5 Pituitary key takeaways

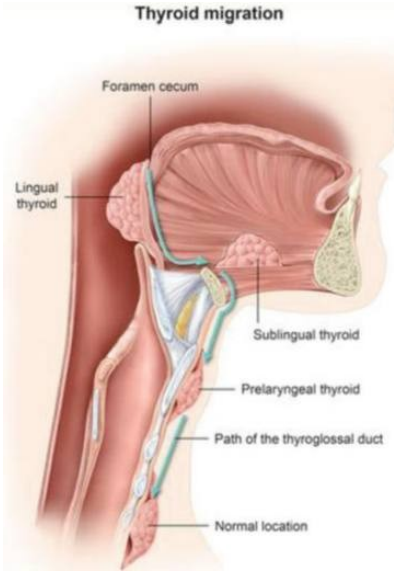
108. The gland lies above the sella and below the chiasm: mass effect gives **bitemporal hemianopia**.
109. **GH, FSH, LH are lost first; TSH and ACTH last.**
110. **Prolactinoma is the commonest secretory adenoma**; think **MEN1** if there is also hypercalcaemia.
111. **Acromegaly = failure of glucose to suppress GH; cardiomyopathy kills.**
112. **DI = dilute urine + high sodium; SIADH = concentrated urine + low sodium.**

9. Thyroid Gland

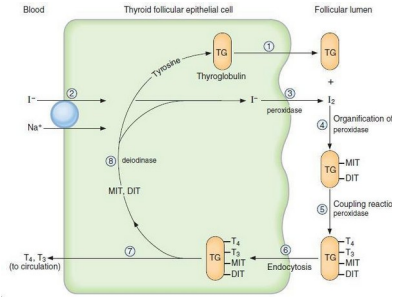
Sources: Teacher deck "Thyroid Disorders" (103 slides) and Pathoma Ch 15.3 (Thyroid, lecture notes pp. 302-310) plus Pathoma slides Ch 18.

9.1 Development, anatomy and hormone physiology

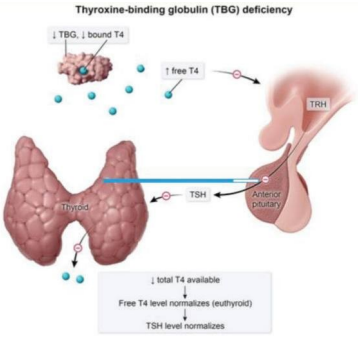
- The thyroid develops as a **down-growth from the floor of the pharynx at the base of the tongue** and migrates caudally in the neck. Its origin persists as the **foramen cecum**. The migration path can leave a **thyroglossal duct cyst** (a midline anterior neck mass that moves upward when the tongue is protruded) or ectopic thyroid tissue (**lingual thyroid**, a mass at the base of the tongue). Blood supply: superior and inferior thyroid arteries.



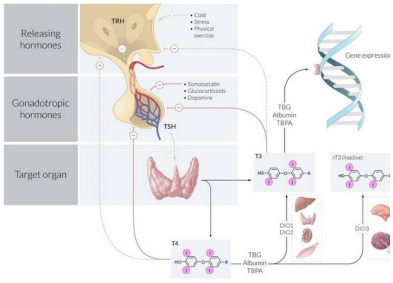
slide 5



slide 6



slide 7



slide 8

Thyroid hormone synthesis

Follicular cells make **thyroglobulin** (a protein rich in tyrosine) and store it as **colloid** in the follicle. The key enzyme is **thyroid peroxidase (TPO)**.

113. **Iodide is trapped** in the follicular cell (Na/I symporter) and oxidised by **TPO**.
114. **Organification:** oxidised iodine attaches to tyrosine in thyroglobulin to give **monoiodotyrosine (MIT)** then **diiodotyrosine (DIT)**.
115. **Coupling:** MIT + DIT = **T3**; DIT + DIT = **T4**.

116. TSH stimulates the cell to take the thyroglobulin back in and cleave off T3/T4 for release.

	T4 (thyroxine)	T3 (triiodothyronine)
Made	Thyroid secretes mostly T4	Mostly made from T4 in tissues (peripheral deiodination); conversion falls in severe illness
Potency	Pro-hormone	**About 10x more potent**
Transport	**Thyroxine-binding globulin 70%, albumin 20%, transthyretin 10%**; free T4 only 0.04%, free T3 0.03%	

Estrogen (pregnancy, oral contraceptives) raises TBG, so **total T4 rises but free T4 and TSH stay normal**; androgen lowers TBG.

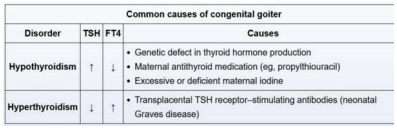
What thyroid hormone does

Binds **nuclear receptors** to change gene transcription. It is essential for **brain and skeletal development** (hence cretinism), **increases BMR** (more Na/K ATPase and oxygen use), **increases LDL receptors** (hyperthyroid = low cholesterol), raises **heart rate and contractility** and **beta-1 receptor sensitivity to catecholamines**, increases **glycogenolysis and gluconeogenesis** (hyperthyroid worsens diabetes), increases **GI motility**, increases **bone resorption** (osteopenia, hypercalcaemia), and maintains **respiratory drive**.

9.2 Goitre (enlarged thyroid)

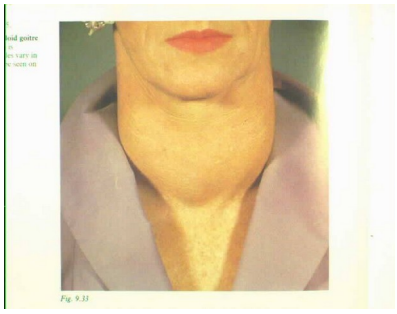
Any enlargement of the thyroid. Causes: **iodine deficiency** (the commonest worldwide), autoimmune disease, **goitrogens** (cassava root with thiocyanate, cruciferous vegetables such as cabbage, cauliflower and Brussels sprouts, milk from regions with goitrogens in grass), and rare inherited **dyshormonogenesis** (e.g. TPO gene defect).

Nontoxic multinodular goitre: low iodine causes low T4, so high TSH, which drives follicular hyperplasia, then nodule formation with fibrosis, haemorrhage and cystic change. Most patients are **euthyroid** (nontoxic). Large goitres compress the **trachea** (dyspnoea), **oesophagus** (dysphagia) and cause **venous congestion**: **Pemberton's sign** (faintness and facial plethora when the arms are raised above the head, which pushes the goitre into the thoracic inlet). Sudden pain and swelling suggest haemorrhage into a nodule. A toxic multinodular goitre (**Plummer syndrome**) makes hormone autonomously.

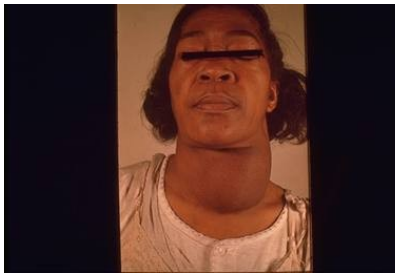


Common causes of congenital goiter			
Disorder	TSH	FT4	Causes
Hypothyroidism	↑	↓	• Genetic defect in thyroid hormone production • Maternal antithyroid medication (eg, propylthiouracil) • Excessive or deficient maternal iodine
Hyperthyroidism	↓	↑	• Transplacental TSH receptor–stimulating antibodies (neonatal Graves disease)

slide 12



slide 14 *- Thyroid Goiter | Definition: any enlargement of the thyroid creating a neck*



slide 14 *- Thyroid Goiter | Definition: any enlargement of the thyroid creating a neck*



slide 15 *- Multinodular Thyroid Goiter | Massive Enlargement May Be Due To Iodine Deficient Diet | Re*



slide 15 *- Multinodular Thyroid Goiter | Massive Enlargement May Be Due To Iodine Deficient Diet | Re*

9.3 Thyroiditis (inflammation of the thyroid)

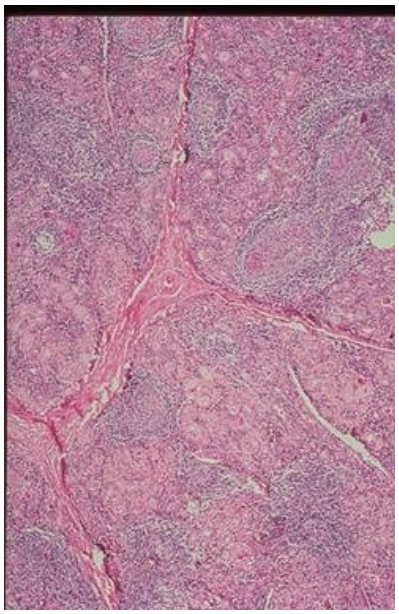
Type	Cause	Clinical	Histology	Lab / course
Acute (infectious) thyroiditis	Bacteria (S. aureus, strep) from pharyngitis, often via a **piriform sinus tract** (4th pharyngeal pouch remnant); fungi in the immunocompromised; also radiation (I-131) and amiodarone	**Painful, tender small goitre**, fever, dysphagia, erythema, lymphadenopathy; pain referred to the throat/ears	Neutrophils, abscess	Normal thyroid function usually
Hashimoto (chronic lymphocytic) thyroiditis	**Autoimmune**, women 45-65; **HLA-DR5** (Pathoma) / DR3; associated with RA, SLE, Sjogren, pernicious anaemia, T1DM, myasthenia; iodine supplementation, amiodarone, interferon-alpha increase frequency	**Painless diffuse goitre**, firm and rubbery, neck tightness; progresses to **permanent hypothyroidism**	**Lymphocytic infiltrate with germinal centres, Hurthle cells (follicular cells with eosinophilic granular cytoplasm), fibrous septa; follicular atrophy**	**Anti-TPO, anti-thyroglobulin, anti-microsomal antibodies**. Initial transient hyperthyroidism ("hashitoxicosis") from gland destruction, then **low T3/T4, high TSH**. Risk of **marginal zone B-cell (non-Hodgkin) lymphoma**
Subacute granulomatous (de Quervain) thyroiditis	After a **viral URI** (post-viral)	**Tender, painful thyroid** in a young or middle-aged woman, fever, malaise; painless variant = silent thyroiditis	**Granulomas with multinucleated giant cells** around colloid, neutrophils	Transient hyperthyroidism (T4 up, TSH down, **I-131 uptake low**) then normal; **self-limited, does not progress to hypothyroidism**
Riedel (invasive fibrous) thyroiditis	Idiopathic; part of **IgG4/systemic fibrosis** (retroperitoneal fibrosis, fibrosing mediastinitis, sclerosing cholangitis)	**Stony-hard, painless, fixed** thyroid that invades neck structures (airway, hoarseness, dysphagia); women 45-65	Dense fibrosis replacing the gland	Surgery to relieve compression; steroids. **Mimics anaplastic carcinoma but patients are younger**

Memory hook: Tender thyroid in a young woman after a cold = de Quervain. Rock-hard painless

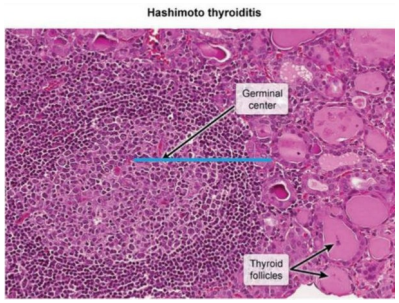
thyroid = Riedel. Painless rubbery goitre + Hurthle cells = Hashimoto.



slide 25 *- Autoimmune Thyroiditis | Firm , Fibrous Or Nodular Consistency | Follicular Atrophy;*



slide 25 *- Autoimmune Thyroiditis | Firm , Fibrous Or Nodular Consistency | Follicular Atrophy;*



slide 26

Postpartum thyroiditis	
Pathophysiology	• Autoimmune destruction of thyroid follicles & release of preformed thyroid hormone • Lymphocytic infiltrates ± germinal centers
Clinical course	• Onset ≤12 months after pregnancy • Transient hyperthyroid phase (↑ T4 & T3, ↓ TSH) • Brief hypothyroid phase (↓ T4 & T3, ↑ TSH) • Return to euthyroid state
Diagnosis	• ↑ Serum thyroglobulin • ↓ Radioiodine uptake • Ultrasound: diffuse thyroid enlargement with ↓ blood flow

slide 27

Differential diagnosis of thyroiditis

Painful thyroiditis Pain & tenderness Systemic symptoms	**Painless thyroiditis** Autoimmune/+TPO antibody Destruction of follicles Release of preformed hormone & thyroglobulin	**Graves** +TSI/TRAb ↑ Thyroid hormone synthesis	**Fibrotic** Replacement of follicles with fibrotic tissue
Subacute (granulomatous, DeQuervain) • Tender goiter • Transient hyperthyroid phase • Postviral syndrome	**Chronic lymphocytic** (Hashimoto) • Common • Usually hypothyroid at diagnosis	**Graves disease** • Overt hyperthyroidism • ↑ RAIU • Ophthalmopathy	**Riedel thyroiditis** • Hard/fixed goiter • Compression of esophagus & trachea
Acute (suppurative) • Fever • Asymmetry • Fluctuance	**Postpartum** • ≤12 months after parturition • Transient hyperthyroid phase		

slide 31

9.4 Hypothyroidism

Primary (low T4, high TSH, the thyroid is the problem): **Hashimoto (the commonest cause in iodine-sufficient countries)**, iodine deficiency, drugs (**lithium**, iodides, radioiodine), thyroidectomy or radiation, congenital enzyme defects, subacute and Riedel thyroiditis. **Secondary:** lack of pituitary TSH (low T4, **low** TSH).

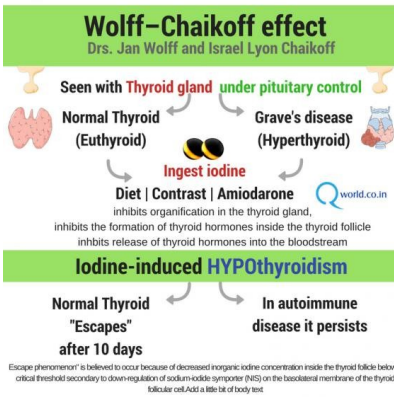
Congenital hypothyroidism (cretinism)

Causes: maternal hypothyroidism early in pregnancy, **thyroid agenesis**, **dyshormonogenic goitre** (e.g. TPO mutation), iodine deficiency. Features: **intellectual disability, dwarfism/short stature, coarse facial features, large protruding tongue (myxoedema), umbilical hernia**. Treat early with levothyroxine.

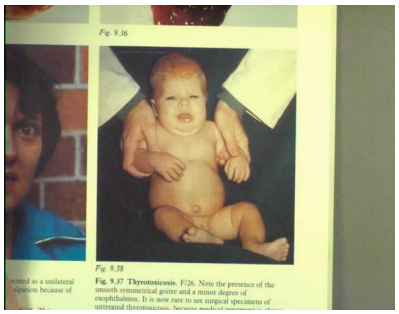
Acquired hypothyroidism (myxoedema in older children and adults)

Myxoedema is non-pitting oedema from **accumulation of glycosaminoglycans (hyaluronic acid) in the interstitium that bind water**, particularly in the **tongue, larynx (deep hoarse voice), face and periorbital area**.

- **Early:** fatigue, lethargy, cold intolerance, constipation, dry skin, brittle nails, **weight gain despite normal appetite**, menorrhagia, **delayed relaxation of deep tendon reflexes**, myalgia.
- **Late:** slow speech, hoarseness, peripheral oedema, pallor, thinning of hair and **lateral eyebrows**, bradycardia, pericardial/pleural effusions, decreased smell/taste/hearing, **hypoventilation** (thyroid maintains the respiratory drive), carpal tunnel syndrome, galactorrhoea (high TRH raises prolactin), amenorrhoea or menorrhagia.
- **Labs: high TSH and low free T4 (primary)**, **high cholesterol** (fewer LDL receptors), raised liver enzymes, **raised creatine kinase**, raised prolactin, **hyponatraemia**, hypoglycaemia, anaemia (often macrocytic). Anti-TPO or anti-thyroglobulin antibodies if Hashimoto.
- **Treatment: levothyroxine (T4)**.
- **Myxoedema coma:** the life-threatening extreme: **hypothermia, hypotension, hypoventilation (hypoxia, hypercapnia), hyponatraemia, hypoglycaemia, bradycardia, altered mentation/convulsions**; triggered by infection, cardiac/respiratory/CNS illness, cold exposure or sedatives in a patient with known hypothyroidism. Treat with **high-dose IV levothyroxine** (plus hydrocortisone and supportive care); high mortality.



slide 34



slide 35 *- Cogenital Hypothyroidism | In iodine deficient endemic areas/ deficiency of thyroid enz*

Congenital hypothyroidism		
Cause	**Free & total T4**	**TSH**
• Primary hypothyroidism ◦ Thyroid dysgenesis ◦ TSH resistance	↓	↑
• Thyroid hormone resistance	↑	↑
• Central hypothyroidism	↓	↓
• Transient due to maternal exposure ◦ Iodine excess or deficiency ◦ TSH receptor–blocking antibodies ◦ Antithyroid medications	↓	↑

slide 36

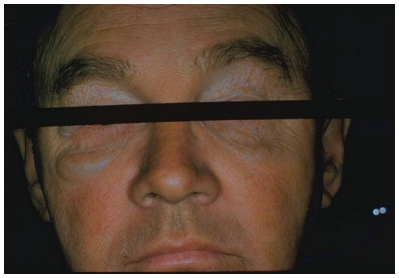
Congenital hypothyroidism	
Clinical manifestations	• Asymptomatic at birth • After maternal thyroxine wanes (weeks to months): ◦ Lethargy, poor feeding ◦ Enlarged fontanelle ◦ Protruding tongue, puffy face, umbilical hernia ◦ Constipation ◦ Prolonged jaundice ◦ Dry skin
Diagnosis	• ↑ TSH & ↓ free thyroxine levels • Newborn screening
Treatment	• Levothyroxine*

*Delayed treatment is associated with neurocognitive dysfunction (eg, ↓ intelligence quotient)

slide 37



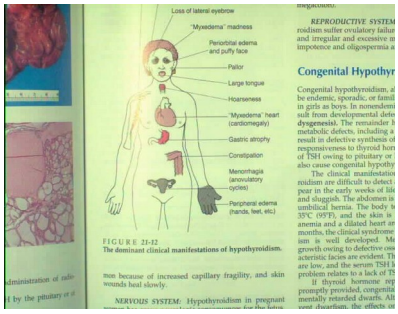
slide 40 *- Acquired Hypothyroidism (Myxedema-special form of edema) (Proteoglycans accumulation in e | Myxedema / Facies | Myxedema / Facies | Pretibial / Myxedem*



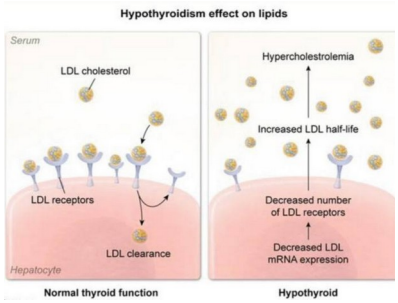
slide 40 *- Acquired Hypothyroidism (Myxedema-special form of edema) (Proteoglycans accumulation in e | Myxedema / Facies | Myxedema / Facies | Pretibial / Myxedem*



slide 40 *- Acquired Hypothyroidism (Myxedema- special form of edema) (Proteoglycans accumulation in e | Myxedema / Facies | Myxedema / Facies | Pretibial / Myxedem*



slide 41 *- Features Of Hypothyroidism*



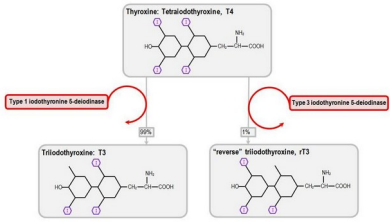
slide 43

Hypothyroid myopathy	• Muscle pain, cramps & weakness involving the proximal muscles • **Delayed tendon reflexes** & myoedema • Features of hypothyroidism

slide 44 *- CK*

Euthyroid sick syndrome (low T3 syndrome)	
Clinical presentation	• Abnormal thyroid function tests in acutely ill patient • Mild central hypothyroid state; asymptomatic
Pathophysiology	• Suppression of 5'-deiodinase by glucocorticoids and inflammatory cytokine (eg, TNF-alpha, IL-1, IFN-beta) • Decreased peripheral conversion of T4 to T3 • Decreased hypothalamic TRH secretion
Diagnostic testing	• Early: Low T3, normal TSH & T4 • Late: Low T3, TSH & T4

slide 45



slide 46

9.5 Hyperthyroidism (thyrotoxicosis)

Thyrotoxicosis is the state of excess thyroid hormone; **hyperthyroidism** is excess thyroid *function*. Not all thyrotoxicosis is hyperthyroidism (e.g. thyroiditis release, factitious).

Causes

Graves disease (commonest), **toxic multinodular goitre (Plummer)**, **toxic adenoma**, **Jod-Basedow** (iodine-induced hyperthyroidism in a goitre), **factitious** (exogenous hormone), TSH-secreting pituitary adenoma, early thyroiditis (Hashimoto initial phase, de Quervain), amiodarone, and **high hCG** (first 4 months of pregnancy, hydatidiform mole, choriocarcinoma, testicular tumours): hCG shares an alpha subunit and cross-reacts with the TSH receptor. Also **struma ovarii** (chapter 5).

Clinical features (due to high BMR and catecholamine sensitivity)

- **General:** heat intolerance, sweating, **weight loss despite increased appetite**, fatigue and weakness (proximal myopathy), anxiety, fine resting tremor, insomnia, hyperreflexia, muscle cramps.
- **Cardiac:** palpitations, **sinus tachycardia, atrial fibrillation**, angina; rarely thyrotoxic cardiomyopathy and heart failure.
- **GI:** frequent stools/diarrhoea (high motility).
- **Metabolic/bone:** hyperglycaemia, **hypocholesterolaemia**, osteopenia, hypercalcaemia.
- **Reproductive:** oligomenorrhoea.
- **Skin/eyes:** warm moist skin, fine hair, onycholysis; **lid retraction, stare and lid lag** (sympathetic overstimulation; seen in any hyperthyroidism).
- **Lab:** sensitive **TSH is the best test** (low in primary hyperthyroidism), raised free T4 and T3. Also anaemia, neutropenia, hypocholesterolaemia.
- **Hypokalaemic periodic paralysis** can complicate it.

Graves disease (diffuse toxic goitre)

- **Most common cause of hyperthyroidism.** Women 20-40. Associated with **HLA-B8 and HLA-DR3** and other autoimmune diseases.
- **Pathogenesis: type II hypersensitivity** (IgG antibodies against the **TSH receptor** that **stimulate** it: thyroid-stimulating immunoglobulins, TSI/LATS, "long-acting thyroid stimulator"). The thyroid grows (hyperplasia and hypertrophy) and makes hormone with no negative feedback, so **TSH is low**.
- **Specific features: diffuse painless goitre with a bruit**, **exophthalmos** (infiltrative ophthalmopathy, proptosis, diplopia), and **pretibial myxoedema** (about 3%).
- **Why exophthalmos and pretibial myxoedema?** Fibroblasts behind the eye and in the shin skin carry **TSH receptors**. The antibodies stimulate them to deposit **glycosaminoglycans (hyaluronic acid, chondroitin sulfate)**, causing inflammation, oedema and fibrosis. The "myxoedema" feels like dough because the swelling is not simple water.
- **Gross/micro:** diffusely enlarged, hyperplastic gland; **tall crowded follicular cells, papillary infoldings, "scalloped" colloid (pale scalloped edges where colloid is resorbed)**, lymphocytic infiltrate.
- **Labs: raised T4 (total and free), T3; low TSH; high I-131 uptake (diffuse)**; TSI positive; low cholesterol; hyperglycaemia.
- **Treatment: beta blockers** (propranolol) for symptoms; **thioamides (methimazole, propylthiouracil)** block TPO (PTU also blocks T4 to T3 conversion); **radioactive iodine (I-131)** (causes permanent hypothyroidism); thyroidectomy.

Thyroid storm

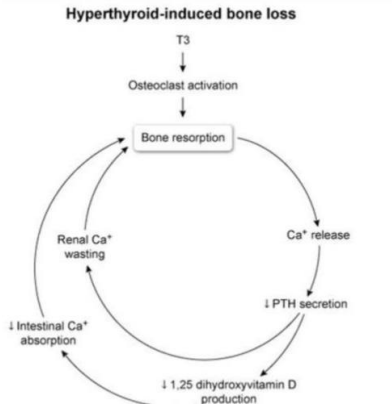
Sudden surge of thyroid hormone, usually triggered by **acute stress** (surgery, childbirth, MI, infection). **Fever above 40 C, sweating, tachycardia or AF, delirium, nausea and vomiting.** Treat with **beta blockers (propranolol)**, **PTU**, **glucocorticoids (hydrocortisone)**, and iodine to block hormone release (the **Wolff-Chaikoff effect**: high iodide transiently shuts down thyroid hormone production). Thyroid storm is the hyperthyroid emergency; myxoedema coma is the hypothyroid emergency.

General manifestations of hyperthyroidism	
Symptoms	• Anxiety & insomnia • Palpitations • Heat intolerance • Increased perspiration • Weight loss without decreased appetite
Physical examination	• Goiter • Hypertension • Tremors involving fingers/hands • Hyperreflexia • Proximal muscle weakness • Lid lag • Atrial fibrillation

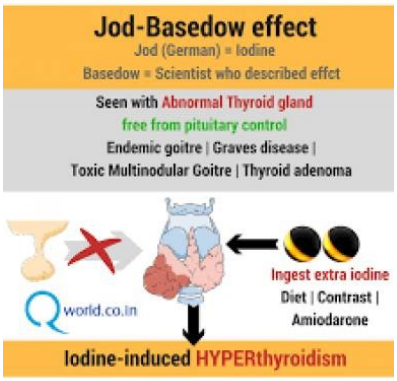
slide 48

Cardiovascular effects of hyperthyroidism	
Increased rate	• Tachycardia/palpitations • Atrial fibrillation
Increased contractility	• ↑ Ejection fraction & cardiac output • ↑ Myocardial oxygen demand & angina • ↑ Pulmonary artery pressure
Decreased afterload	• ↓ Systemic vascular resistance
Additional effects	• ↓ Diastolic pressure • ↑ Systolic pressure • ↑ Pulse pressure • High-output heart failure

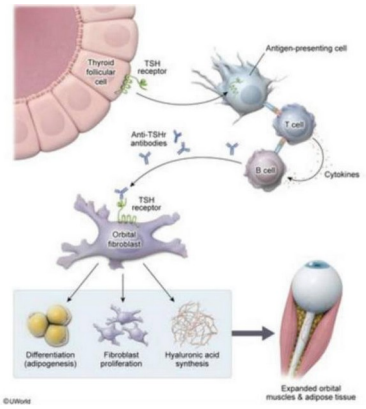
slide 49



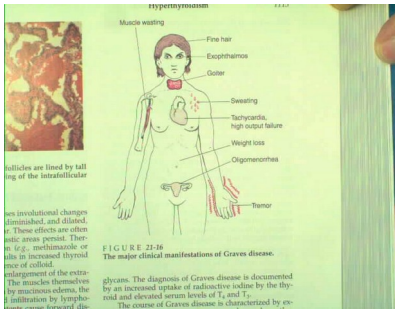
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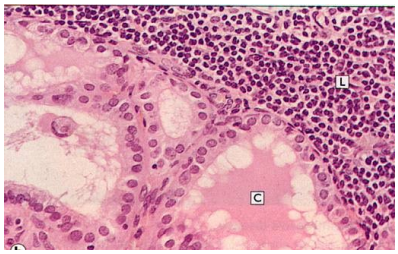
slide 52



slide 55



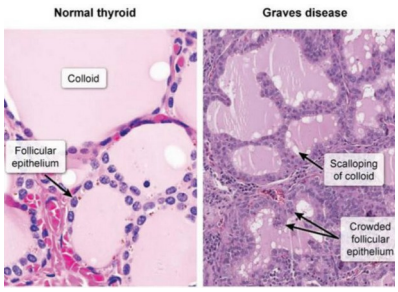
slide 57 *- Features Of Grave's Disease | Long Acting Thyroid | Lab- findings / LATS IgG / Low TSH / H*



slide 59 *- Grave's Disease | Thyroid Enlargement With Exopthalmos | Microscopy: Diffuse T*



slide 59 *- Grave's Disease | Thyroid Enlargement With Exopthalmos | Microscopy: Diffuse T*



slide 60

Thyroid function tests: the pattern table (teacher slide 61, arrows restored)

Disorder	Total T4	TSH	I-131 uptake
Graves disease	High	**Low**	**High (diffuse)**
Factitious thyrotoxicosis	High	**Low**	**Low**
Thyroiditis (infectious, granulomatous)	High early	**Low**	**Low**
Primary hypothyroidism (Hashimoto)	Low	**High**	Low (or variable)
Secondary hypothyroidism	Low	**Low/inappropriately normal**	Low
Estrogen excess	High (raised TBG)	Normal	Normal
Androgen excess	Low (lowered TBG)	Normal	Normal

Memory hook: Hot gland (high uptake) = the thyroid is overproducing (Graves, toxic nodule). Cold/low uptake with thyrotoxicosis = hormone is coming from somewhere else or leaking (factitious, thyroiditis).

Differential diagnosis of thyrotoxicosis			
	Serum TSH	Serum thyroglobulin	Radioiodin uptake
Increased thyroid hormone production (eg, Graves disease)	↓	↑	↑
Destruction of thyroid follicles (eg, subacute thyroiditis)	↓	↑↑	↓
Exogenous thyroid hormone	↓	↓	↓
Central hyperthyroidism (eg, pituitary adenoma)	↑	↑	↑

slide 62

Thyroid autoantibodies	
Antibody	Disease association
Thyrotropin (TSH) receptor antibody	• **Graves disease** • Functional bioassay (thyroid-stimulating immunoglobulin) is also available
Thyroid peroxidase antibody	• **Chronic lymphocytic (Hashimoto) thyroiditis** • Silent thyroiditis • Postpartum thyroiditis • Some patients with Graves disease
Thyroglobulin antibody	• Nonspecific, but more common in hypothyroidism than hyperthyroidism

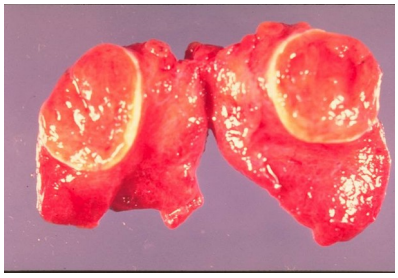
slide 63

9.6 Thyroid tumours

Most thyroid nodules are solitary and **benign**. Work-up: TSH, ultrasound, **radioiodine scan** (**hot** nodules take up iodine and are rarely malignant; **cold** nodules (adenoma and carcinoma) do not and need **fine-needle aspiration (FNA)**). Risk of malignancy is higher in **men and children**, with a history of **neck radiation**, rapid growth, hard fixed nodule, and a cold nodule.

Follicular adenoma

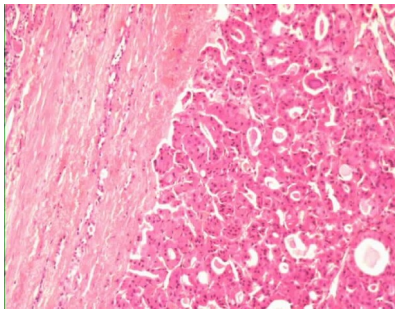
Solitary, **encapsulated** nodule from follicular epithelium (mutations in the **TSH receptor or Gs alpha subunit** in toxic cases). Mostly non-functioning ("cold"), may cause pain from haemorrhage or rarely thyrotoxicosis. Variants: **microfollicular (fetal)** and **Hurthle cell**. FNA cannot tell adenoma from carcinoma because that requires seeing the **capsule** (histology).



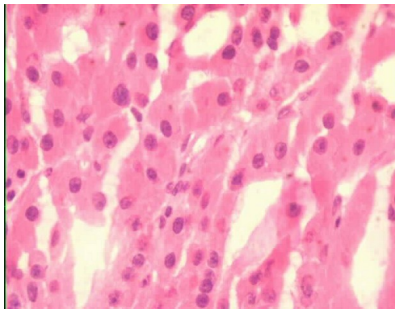
slide 67 *- Follicular Adenoma (most common) (May cause pressure symptoms, pain, and rarely thyrotoxi | Well-demarcated | Cystic Degeneration (hemorhage, fibrosis,*



slide 67 *- Follicular Adenoma (most common) (May cause pressure symptoms, pain, and rarely thyrotoxi | Well-demarcated | Cystic Degeneration (hemorhage, fibrosis,*



slide 68 *- Follicular Adenomas | Microfollicular (Fetal) Type (small follicles containing lit*



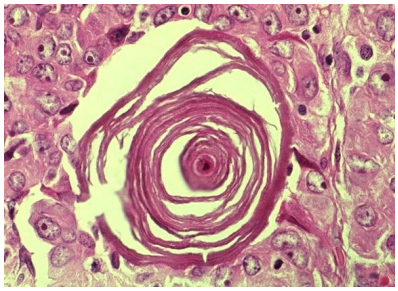
slide 68 *- Follicular Adenomas | Microfollicular (Fetal) Type (small follicles containing lit*

Thyroid carcinomas

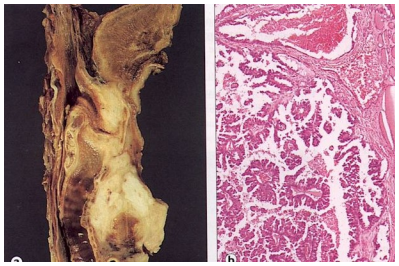
	Papillary (about 80%)	**Follicular (about 15%)**	**Medullary (about 5%)**	**Anaplastic (<5%)**
Who	Young women; **childhood neck**	Middle age, women	Sporadic, or familial with **MEN2**	Elderly

	radiation			
Cell	Follicular	Follicular	**Parafollicular C cells**	Undifferentiated follicular
Histology	**Papillae; diagnosis by nuclear features**: **Orphan Annie eye (ground-glass, clear) nuclei, nuclear grooves ("coffee bean"), nuclear pseudoinclusions; psammoma bodies**	**Follicles resembling the adenoma but with invasion of the capsule and/or blood vessels** (hallmark)	**Sheets of polygonal/spindle cells in an amyloid stroma** (calcitonin deposits; Congo red shows apple-green birefringence)	Giant, spindle, pleomorphic cells; necrosis; keratin positive
Genetics	**BRAF V600E, RET/PTC rearrangements, RAS**	**PAX8-PPAR-gamma fusion, RAS, PI3K**	**RET mutations**	**TP53, RB**
Spread	**Lymphatic**, cervical lymph nodes	**Haematogenous**: bone, lung, liver (follicular is one of the four carcinomas that spread by blood: **renal cell, follicular thyroid, hepatocellular, choriocarcinoma**)	Lymph nodes, then lung, liver, bone	Early, local invasion and distant
Prognosis	**Excellent**	Good	Guarded	**Worst; fatal within about 2 years**
Marker	Thyroglobulin (after thyroidectomy)	Thyroglobulin	**Calcitonin, CEA**	

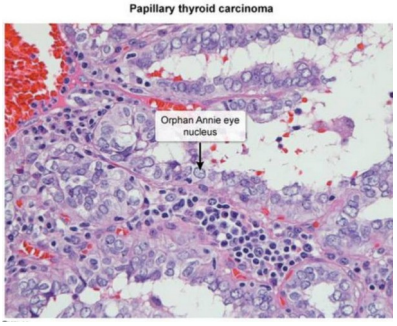
- **Papillary:** firm, painless nodule; FNA then thyroidectomy. Mixed papillary-follicular variants behave as papillary. Larger tumours cause hoarseness and dysphagia.
- **Follicular:** may secrete enough T4 to suppress TSH.
- **Medullary:** calcitonin may cause **flushing and diarrhoea** and lower calcium; about 5% develop Cushing syndrome (ectopic ACTH). **C-cell hyperplasia** is the precursor of the familial form. Familial RET carriers have **prophylactic thyroidectomy**. One third are sporadic, one third familial, one third part of MEN2.
- **Anaplastic:** rapidly growing hard mass with dysphagia and airway compromise. Older than Riedel patients. Small-cell-type tumours are usually lymphoma or medullary carcinoma.
- **Thyroid lymphoma:** mostly **large B-cell (or marginal zone) lymphoma arising in Hashimoto thyroiditis**, women in their 70s, presents with dyspnoea, hoarseness and a rapidly enlarging neck mass; very radiosensitive.



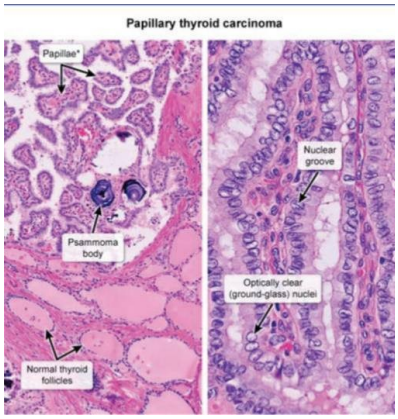
slide 71 *- Papillary Carcinoma: Psammoma Body*



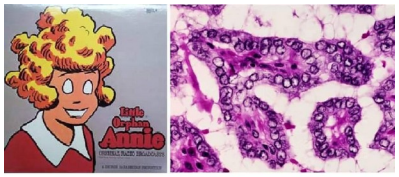
slide 72 *- Papillary Thyroid Cancer | Sclerotic Primary Lesion | Tumor Papillae (lined by neoplast*



slide 73

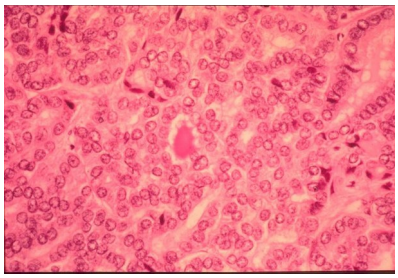


slide 74

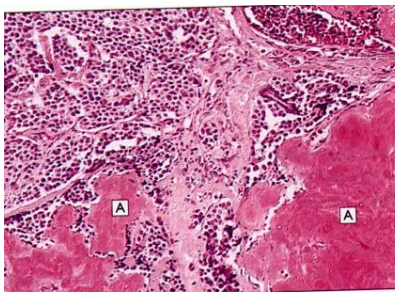


Orphan Annie eye appearance in Papillary Carcinoma thyroid

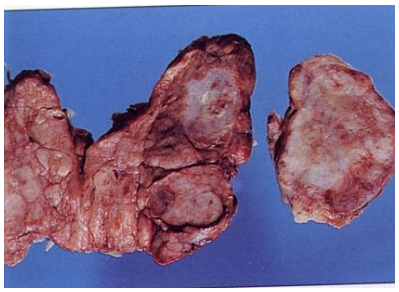
slide 75



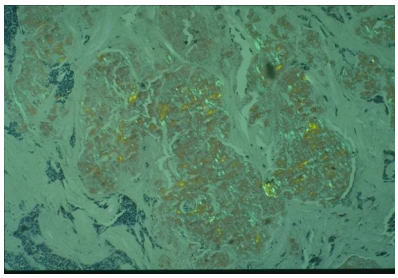
slide 77 *- Follicular Carcinoma (microfollicular, invading vessels in the thyroid parenchyma)*



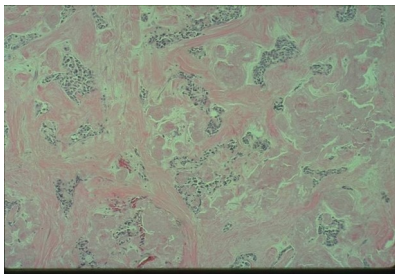
slide 79 *- Medullary Thyroid Cancer | Solid tumor with extension throughout the thyroid | Infiltra*



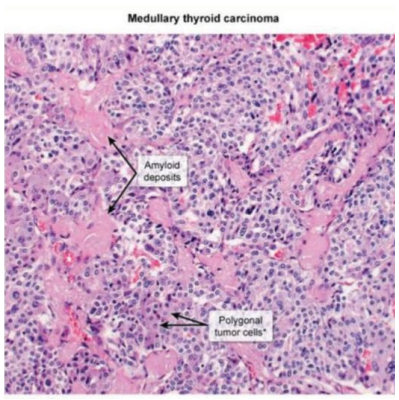
slide 79 *- Medullary Thyroid Cancer | Solid tumor with extension throughout the thyroid | Infiltra*



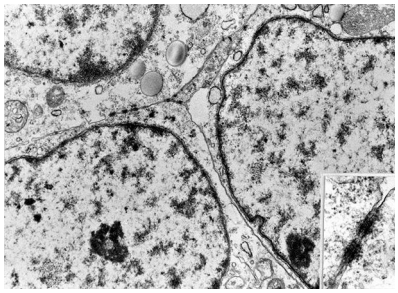
slide 80 *- Medullary Carcinoma | Amyloid Stroma | Congo red stain with apple green birefrin*



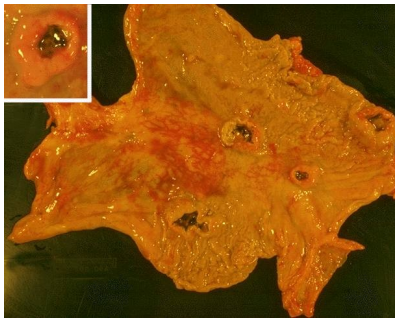
slide 80 *- Medullary Carcinoma | Amyloid Stroma | Congo red stain with apple green birefrin*



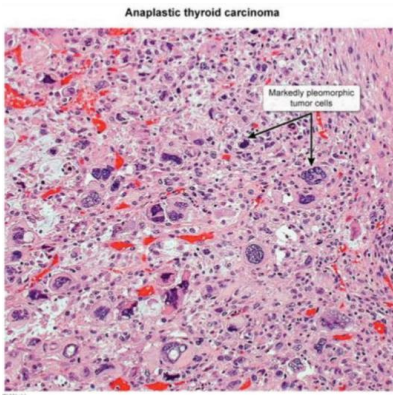
slide 81



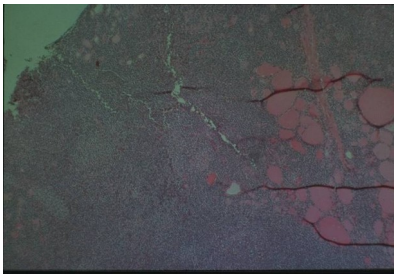
slide 83 *- Anaplastic Thyroid Carcinoma | Metastasis To Stomach | EM Shows Primitive Cells (undifferen*



slide 83 *- Anaplastic Thyroid Carcinoma | Metastasis To Stomach | EM Shows Primitive Cells (undifferen*



slide 84



slide 87 *- Well-differentiated Thyroid Lymphoma*

9.7 Multiple endocrine neoplasia (MEN)

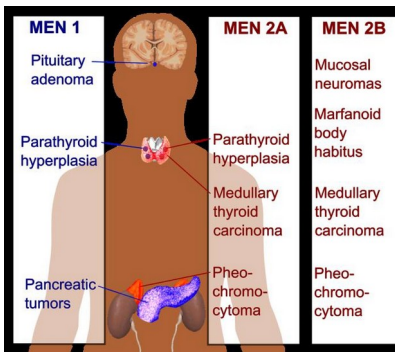
Rare **autosomal dominant** multiglandular syndromes.

Syndrome	Gene	Tumours
MEN1 (Wermer)	**Menin (11q13)**, tumour suppressor	**Pituitary adenoma, parathyroid hyperplasia/adenoma (hyperparathyroidism in over 90%, often first), pancreatic/enteric tumours** (gastrinoma 35% = Zollinger-Ellison; insulinoma; glucagonoma; VIPoma = Verner-Morrison WDHA; somatostatinoma). Non-endocrine: lipomas, facial angiofibromas, collagenomas. "**3 P's**"
MEN2A (Sipple)	**RET proto-oncogene**	**Medullary thyroid carcinoma, pheochromocytoma, parathyroid hyperplasia**. Non-endocrine: Hirschsprung disease
MEN2B (MEN III)	**RET** (different codon, M918T)	**Medullary thyroid carcinoma, pheochromocytoma, mucosal neuromas** (bumpy lips and tongue), intestinal ganglioneuromas, **Marfanoid habitus**; no hyperparathyroidism

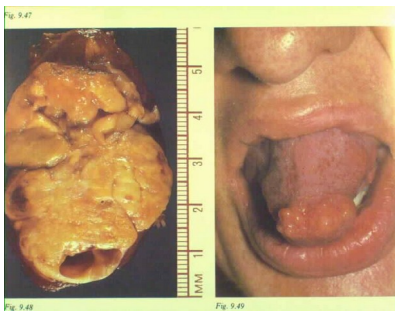
Memory hook: MEN1 = P's (Pituitary, Parathyroid, Pancreas). MEN2A = Medullary thyroid, Pheo, Parathyroid. MEN2B = Medullary, Pheo, Neuromas, Marfanoid body.

Classification of multiple endocrine neoplasia	
Type 1	• **Primary hyperparathyroidism** (parathyroid adenomas or hyperplasia) • Pituitary tumors (prolactin, visual defects) • Pancreatic tumors (especially gastrinomas)
Type 2A	• **Medullary thyroid cancer** (calcitonin) • Pheochromocytoma • Primary hyperparathyroidism (parathyroid hyperplasia)
Type 2B	• **Medullary thyroid cancer** (calcitonin) • Pheochromocytoma • Mucosal neuromas/marfanoid habitus

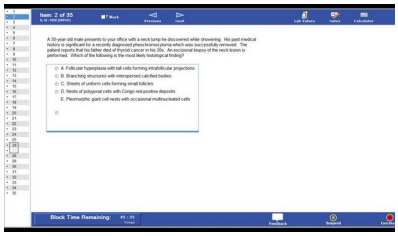
slide 90



slide 91



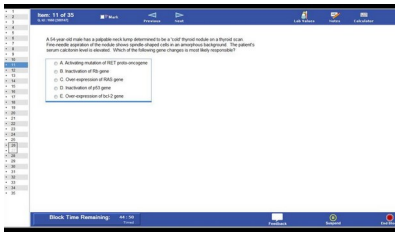
slide 96 *- MEN IIb | Medullary Thyroid Cancer | Mucosal Neuroma of the Tongue*



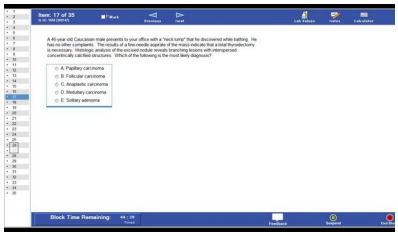
slide 97



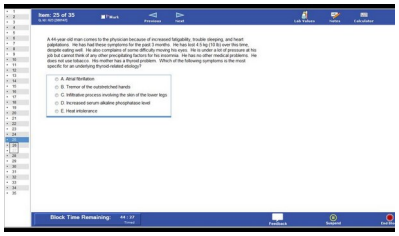
slide 98



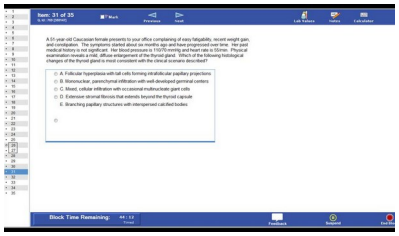
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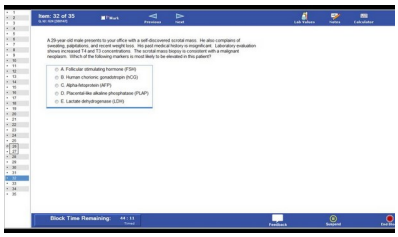
slide 100



slide 101



slide 102



slide 103

10. Parathyroid Glands and Calcium Disorders

Sources: Teacher deck "Parathyroid Disorders" (31 slides) and Pathoma Ch 15.4 (Parathyroid, lecture notes pp. 311-312) plus Pathoma slides Ch 18.

10.1 Calcium physiology: the four players

There are four small glands behind the thyroid. They develop from the **3rd and 4th pharyngeal pouches** (the inferior glands from the 3rd, the superior from the 4th).

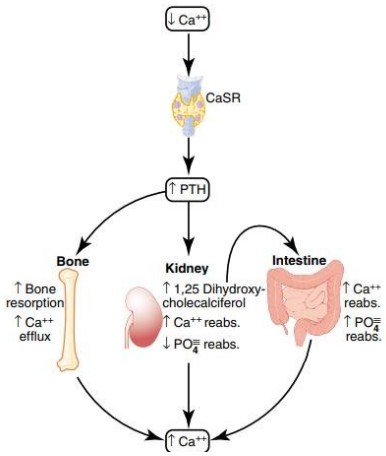
Player	Source	Action
PTH	**Chief cells** of the parathyroid	**Raises free serum calcium**
Vitamin D (calcitriol)	D3 (cholecalciferol) made in skin from 7-dehydrocholesterol by UV; D2 (ergocalciferol) from food. **Liver** makes 25-OH-D (unregulated); **kidney 1-alpha-hydroxylase** makes active **1,25-(OH)2 D** (regulated, stimulated by PTH and low phosphate)	Bone mineralisation; **raises gut calcium and phosphate absorption** and renal reabsorption; **suppresses PTH**
Calcitonin	Thyroid C cells	Lowers serum calcium (minor role in humans)
Phosphatonins (FGF23)	Bone	Increase phosphate excretion

What PTH does:

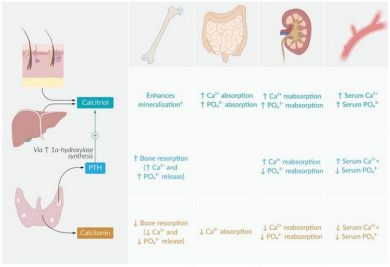
117. **Kidney:** increases **calcium reabsorption** and **phosphate excretion** (losing phosphate matters because phosphate binds free calcium, so less phosphate means more free calcium) and **activates vitamin D**.
118. **Bone:** acts on **osteoblasts**, which release RANKL/M-CSF and so **activate osteoclasts** (more bone resorption). High alkaline phosphatase is a sign of osteoblast activity.
119. **Gut:** more calcium and phosphate absorption, via vitamin D.

PTH signals through **Gs and cAMP**, so some cAMP spills into the **urine**.

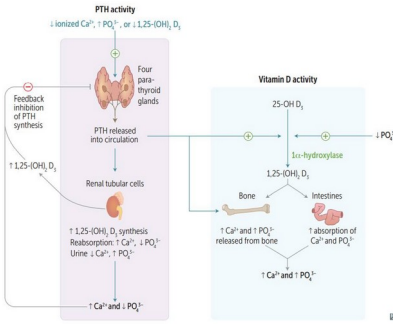
Regulation: the chief cell has a **calcium-sensing receptor (CaSR)**. **High calcium activates the receptor (Gq/Gi), decreases cAMP and decreases PTH release; low calcium raises cAMP and releases PTH.**



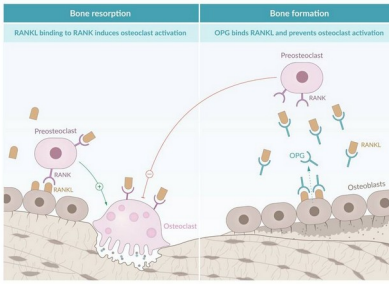
slide 4



slide 5



slide 6



slide 7

10.2 Hyperparathyroidism

Types: **primary** (adenoma, hyperplasia, carcinoma), **secondary** (to low calcium, e.g. chronic renal failure or vitamin D deficiency), **tertiary** (autonomous glands after long-standing secondary), part of **MEN1 and MEN2A**, and **ectopic/paraneoplastic** (non-parathyroid tumour producing PTH; more commonly PTHrP).

Primary hyperparathyroidism

- **Excess PTH from the gland itself.** Causes: **adenoma (about 80%, the commonest cause)**, hyperplasia of all four glands (esp. MEN), very rare carcinoma (palpable in about 50%). Adenomas are rarely palpable.
- **Labs: raised serum calcium (above 10.5 mg/dL) and PTH (high or inappropriately normal), low serum phosphate, raised urinary calcium, raised urine phosphate and cAMP, raised alkaline phosphatase.**
- **Clinical:** mostly **asymptomatic** (found incidentally). Symptoms: "**bones, stones, abdominal groans, psychic moans, and fatigue overtones**".

	Mechanism
Bones	Bone pain, fractures; **osteitis fibrosa cystica (von Recklinghausen disease of bone)**: osteoclastic resorption with fibrous replacement, producing cysts and "**brown tumours**" (haemosiderin-stained giant cell lesions, e.g. jaw), subperiosteal resorption of the phalanges, "salt and pepper" skull
Stones	**Calcium kidney stones** (calcium oxalate), nephrocalcinosis (metastatic calcification); polyuria and polydipsia from **hypercalcaemia-induced nephrogenic DI**
Abdominal groans	**Constipation, peptic ulcers, acute pancreatitis** (calcium activates pancreatic enzymes), nausea, anorexia
Psychic moans	Depression, fatigue, confusion, paraesthesias; psychosis or coma if severe
Other	Muscle weakness, calcium-phosphate deposits in the cornea (band keratopathy) and soft tissues, shortened QT

- **Treatment: surgery** (parathyroidectomy); cinacalcet (calcimimetic); treat hypercalcaemia with fluids, bisphosphonates.

Secondary hyperparathyroidism

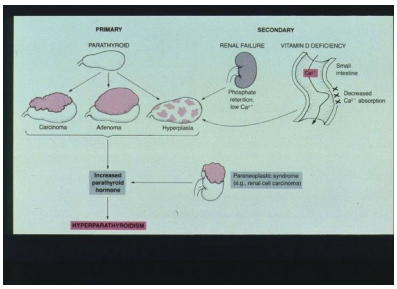
Low calcium drives compensatory PTH release. **Commonest cause: chronic renal failure.** Failing kidneys **retain phosphate** (high phosphate binds free calcium, so calcium falls) and **make less active vitamin D** (so less gut calcium absorption). The glands become hyperplastic. **Labs: high PTH, LOW calcium, HIGH phosphate, high alkaline phosphatase.** (Compare with primary: high calcium, low phosphate.) Renal bone

disease is "**renal osteodystrophy**". Another cause: **vitamin D deficiency** (low calcium, low or normal phosphate).

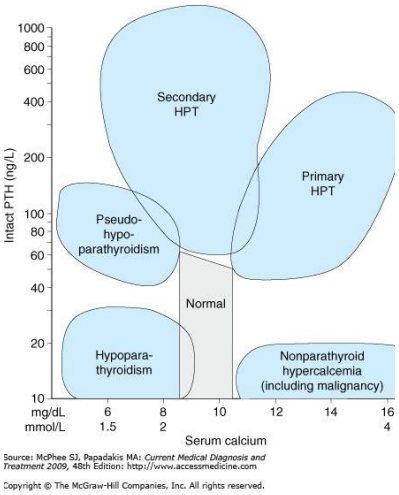
	Calcium	Phosphate	PTH	Alk phos	Urinary Ca
Primary	High	**Low**	High	High	High
Secondary (renal)	**Low**	**High**	High	High	Low
Hypoparathyroidism	Low	High	Low	Normal	Low
Pseudohypoparathyroidism	Low	High	**High**	Normal	Low
FHH	Mildly high	Low/normal	Normal/high	Normal	**Low**

Pathology of the glands

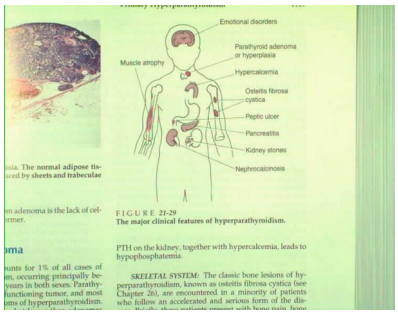
- **Adenoma:** a single enlarged gland, **sheets and trabeculae of chief cells** with loss of normal fat; the other glands are atrophic/suppressed.
- **Hyperplasia: symmetric enlargement of all four glands**; little fat, islands of oncocytes.
- **Bone:** osteitis fibrosa cystica (above).



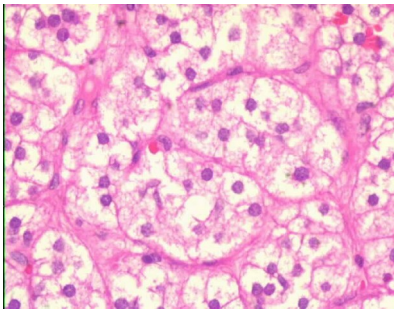
slide 10 *- Causes Of Hyperparathyroidism*



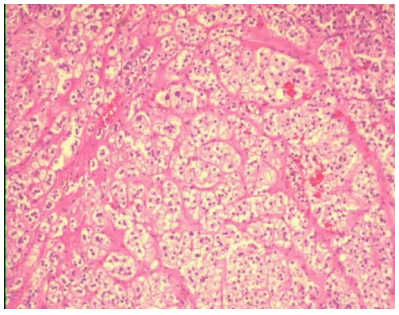
slide 16



slide 17 *- Hyperparathyroidism*



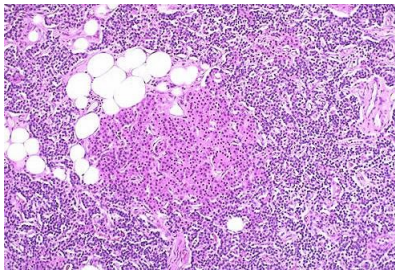
slide 18 *- Parathyroid Adenoma (normal glandular adipose tissue is replaced by hyperplastic plastic*



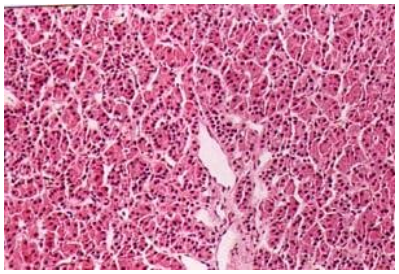
slide 18 *- Parathyroid Adenoma (normal glandular adipose tissue is replaced by hyperplastic plastic*



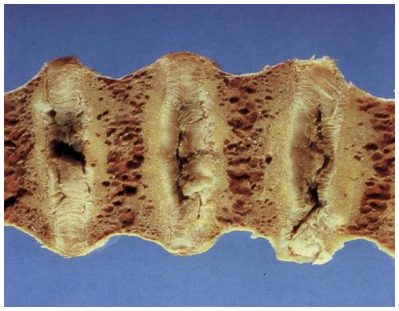
slide 19 *- Parathyroid Hyperplasia | Symmetric enlargement of all four glands | Cellular tissue w*



slide 19 *- Parathyroid Hyperplasia | Symmetric enlargement of all four glands | Cellular tissue w*



slide 20 *- Hyperparathyroid Bone Disease | Tumor with sheets of glandular cells devoid of fat | Osteiti*



slide 20 *- Hyperparathyroid Bone Disease | Tumor with sheets of glandular cells devoid of fat | Osteiti*

10.3 Hypoparathyroidism

Causes: post-thyroidectomy (or parathyroidectomy; the commonest cause, surgical damage), autoimmune (isolated or autoimmune polyglandular syndrome), **DiGeorge syndrome** (22q11 deletion: failure of the **3rd and 4th pharyngeal pouches**, giving **thymic aplasia, hypocalcaemia from absent parathyroids, cardiac defects (outflow tract), cleft palate, abnormal facies**), deposition (iron in haemochromatosis and transfusions, copper in Wilson disease), and **hypomagnesaemia** (needed for PTH secretion and action).
Clinical (low calcium increases Na permeability so nerves fire spontaneously):

- **Acute: perioral numbness and tingling, tetany, carpopedal spasm, muscle cramps, irritability, laryngospasm, seizures**; ECG shows **prolonged QT**.
- **Chronic:** lethargy, anxiety, personality change, **cataracts**, dry scaly skin, brittle nails, hair loss, hyperactive reflexes, defective teeth if onset in childhood.
- **Signs: Chvostek sign** (tapping the facial nerve causes facial twitching); **Trousseau sign** (inflating the BP cuff causes carpopedal spasm).
- **Labs: low PTH, low calcium, high phosphate**, low urine calcium, normal alkaline phosphatase, often low magnesium.

Pseudohypoparathyroidism (PHP) and pseudopseudohypoparathyroidism

- **Pseudohypoparathyroidism (type 1a): end-organ resistance to PTH** because of an inactivating mutation in **GNAS1 (Gs alpha)** on the **maternal** allele (imprinting: the kidney expresses only the maternal allele). So PTH binds but adenylate cyclase/cAMP is not activated. **Labs: high PTH, LOW calcium, HIGH phosphate** (looks like hypoparathyroidism with high PTH).
- **Albright hereditary osteodystrophy: short stature, round face, shortened 4th and 5th metacarpals (and metatarsals)**, sometimes mild intellectual disability, subcutaneous calcifications.
- **Pseudopseudohypoparathyroidism:** same GNAS mutation but inherited from the **father** (maternal allele normal), so the kidney responds to PTH. **Same Albright phenotype but normal PTH, calcium and phosphate.**



slide 25

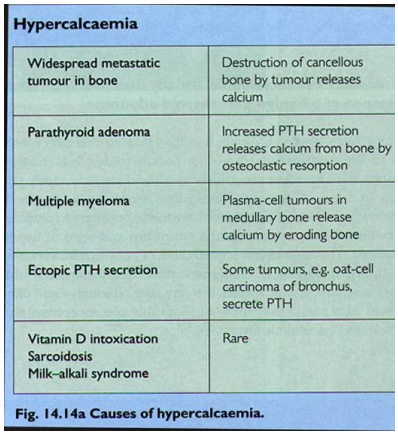
10.4 Familial hypocalciuric hypercalcaemia (FHH)

Inactivating mutation of the **calcium-sensing receptor (CaSR)**: the parathyroid and kidney "think" calcium is low, so **a higher than normal calcium is needed to suppress PTH** and the kidney **reabsorbs too much calcium**. Result: **mild hypercalcaemia, LOW urinary calcium (hypocalciuria), normal or slightly raised PTH**. Benign; do not operate (the key trap versus primary hyperparathyroidism).

10.5 Causes of hypercalcaemia and hypocalcaemia (slides 28-29)

Hypercalcaemia	Hypocalcaemia
Primary hyperparathyroidism (outpatient) and **malignancy** (inpatient): bone metastases/osteolysis, **PTHrP** (squamous cell carcinoma of the lung, etc.), **myeloma**, lymphoma (calcitriol)	**Hypoparathyroidism** (post-surgical, autoimmune, DiGeorge), **pseudohypoparathyroidism**
Granulomatous disease (**sarcoidosis, TB**: activated macrophages make 1-alpha-hydroxylase)	**Vitamin D deficiency** or resistance (rickets/osteomalacia)
Vitamin D or vitamin A intoxication, milk-alkali syndrome	**Chronic renal failure** (high phosphate, low calcitriol)
Thiazide diuretics, lithium	**Acute pancreatitis** (fat saponification), massive transfusion (citrate), rhabdomyolysis
Hyperthyroidism, immobilisation, adrenal insufficiency	**Hypomagnesaemia**, loop diuretics, malabsorption, hungry bone syndrome
FHH	Low albumin (total low, ionised normal)

Symptoms of hypercalcaemia: "bones, stones, groans, moans, psychiatric overtones"; **shortened QT**.
Symptoms of hypocalcaemia: tetany, perioral tingling, seizures; **prolonged QT**.

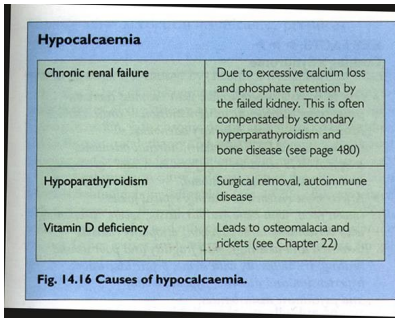


Hypercalcaemia

Widespread metastatic tumour in bone	Destruction of cancellous bone by tumour releases calcium
Parathyroid adenoma	Increased PTH secretion releases calcium from bone by osteoclastic resorption
Multiple myeloma	Plasma-cell tumours in medullary bone release calcium by eroding bone
Ectopic PTH secretion	Some tumours, e.g. oat-cell carcinoma of bronchus, secrete PTH
Vitamin D intoxication Sarcoidosis Milk–alkali syndrome	Rare

Fig. 14.14a Causes of hypercalcaemia.

slide 28 *- Causes Of Hypercalcemia*

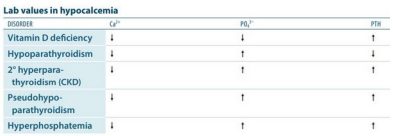


Hypocalcaemia

Chronic renal failure	Due to excessive calcium loss and phosphate retention by the failed kidney. This is often compensated by secondary hyperparathyroidism and bone disease (see page 480)
Hypoparathyroidism	Surgical removal, autoimmune disease
Vitamin D deficiency	Leads to osteomalacia and rickets (see Chapter 22)

Fig. 14.16 Causes of hypocalcaemia.

slide 29 *- Causes Of Hypocalcemia*



Lab values in hypocalcemia

DISORDER	Ca^{2+}	PO_4^{3-}	PTH
Vitamin D deficiency	↓	↓	↑
Hypoparathyroidism	↓	↑	↓
2° hyperpara-thyroidism (CKD)	↓	↑	↑
Pseudohypo-parathyroidism	↓	↑	↑
Hyperphosphatemia	↓	↑	↑

slide 30

11. Adrenal Gland

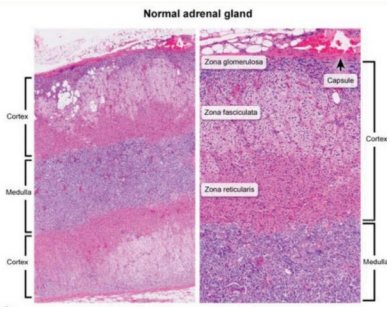
Sources: Teacher deck "Adrenal Disorders" (68 slides), SGD handout "Adrenal Medullary Disorders", Pathoma Ch 15.5 (Adrenal Cortex and Medulla, lecture notes pp. 316-319) plus Pathoma slides Ch 18.

11.1 Anatomy and physiology

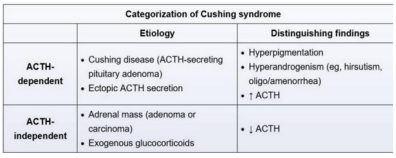
- **Cortex (yellow because of cholesterol)** makes steroid hormones from **cholesterol**. **Medulla** makes catecholamines from **tyrosine**; it is made of **neural crest-derived chromaffin cells**.
- **Three cortical zones: "GFR" from outside in:**

Zone	Hormone	Controlled by
Z. glomerulosa	**Mineralocorticoid (aldosterone)**	**Renin-angiotensin system** (and potassium)
Z. fasciculata	**Glucocorticoid (cortisol)**	**ACTH** (from pituitary, driven by CRH)
Z. reticularis	**Sex steroids (androgens: DHEA, androstenedione)**	ACTH
Medulla	**Epinephrine, norepinephrine**	Sympathetic nerves

Memory hook: Salt (glomerulosa), Sugar (fasciculata), Sex (reticularis): the deeper you go, the sweeter it gets.



slide 4



Categorization of Cushing syndrome

	Etiology	Distinguishing findings
ACTH-dependent	• Cushing disease (ACTH-secreting pituitary adenoma) • Ectopic ACTH secretion	• Hyperpigmentation • Hyperandrogenism (eg, hirsutism, oligo/amenorrhea) • ↑ ACTH
ACTH-independent	• Adrenal mass (adenoma or carcinoma) • Exogenous glucocorticoids	• ↓ ACTH

slide 9

Cortisol: what it does

- Transport: **75% transcortin (CBG)**, 15% albumin, 10% free. Acts on a **nuclear receptor** present in nearly all cells; it can also bind the mineralocorticoid receptor.
- **Licorice** inactivates **11-beta-hydroxysteroid dehydrogenase type 2** (the kidney enzyme that converts cortisol to inactive cortisone), so cortisol acts like aldosterone in the kidney: **sodium retention, hypokalaemia, hypertension**.
- **Glucocorticoid actions:** raise blood glucose (more hepatic glucose output, less uptake by muscle and fat, **gluconeogenesis**), **insulin resistance**, **immune suppression** (inhibits phospholipase A2 so less arachidonic acid, less IL-2, less histamine), **permissive effect on catecholamines** (more alpha-1 receptors; lack of cortisol causes vasodilation and shock), bone resorption, and **negative feedback on ACTH**.
- **Adrenal insufficiency causes hypotension** because low cortisol means vasodilation and low aldosterone means salt and water loss (hypovolaemia).

Steroid synthesis: the enzyme map

Cholesterol -> pregnenolone -> (**17-alpha-hydroxylase**) -> 17-OH-pregnenolone/progesterone -> (**21-alpha-hydroxylase**) -> 11-deoxycortisol/deoxycorticosterone -> (**11-beta-hydroxylase**) -> cortisol/corticosterone. Aldosterone needs 21-hydroxylase and then 11-beta/aldosterone synthase; androgens branch off after 17-hydroxylase. **Blocking one enzyme sends precursors down the other branches.**

11.2 Adrenocortical hyperfunction (hyperadrenalism)

Syndrome	Hormone in excess
Cushing syndrome	Cortisol
Hyperaldosteronism	Aldosterone
Adrenogenital (virilising) syndromes	Androgens

Cushing syndrome (hypercortisolism)

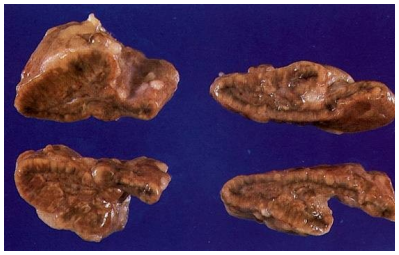
Causes (and what happens to the adrenals):

Cause	Adrenal glands	ACTH
Exogenous glucocorticoids (commonest cause overall)	**Both atrophic** (negative feedback)	Low
Pituitary corticotroph adenoma (Cushing disease; 2nd commonest, about 70% of endogenous)	**Both enlarged (bilateral hyperplasia and nodularity)**	**High**
Adrenal adenoma, carcinoma, or hyperplasia	**One large, the other atrophic**	**Low**
Ectopic ACTH (paraneoplastic: **small cell lung carcinoma**, medullary thyroid carcinoma, pancreatic islet tumours, carcinoids)	**Both large**	**Very high**

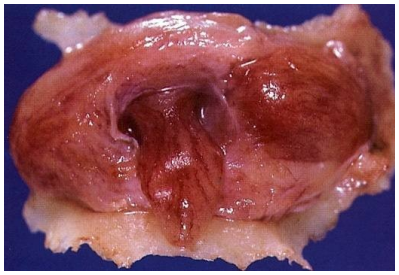
Clinical features: hypertension (more alpha-1 receptors and mineralocorticoid effect), **weight gain with truncal obesity, moon facies, buffalo hump** (cortisol raises glucose, so insulin causes central fat deposition), **proximal muscle wasting and weakness with thin limbs** (amino acids used for gluconeogenesis), **hyperglycaemia/diabetes**, **thin fragile skin, easy bruising and purple abdominal striae** (cortisol impairs collagen synthesis), **osteoporosis**, mood changes (depression, psychosis), hirsutism and menstrual irregularity, immune suppression and **hyperpigmentation only with high ACTH** (pituitary and ectopic).



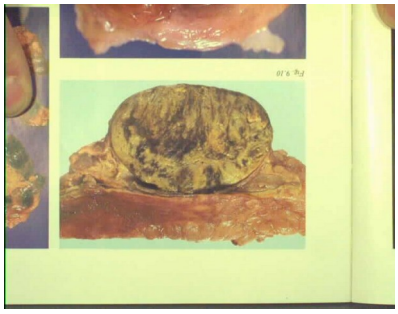
slide 11 *- Adrenal gland diseases | Addison's Disease | Normal Adrenals | Cushing's Disease*



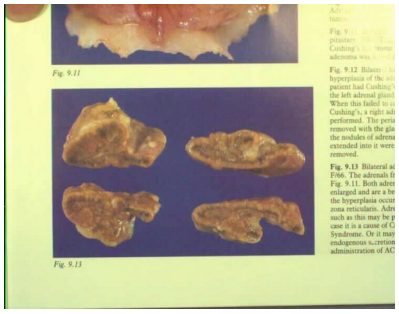
slide 12 *- Cushing's Disease | Pituitary Adenoma Producing ACTH | Bilateral Hyperplasia Of*



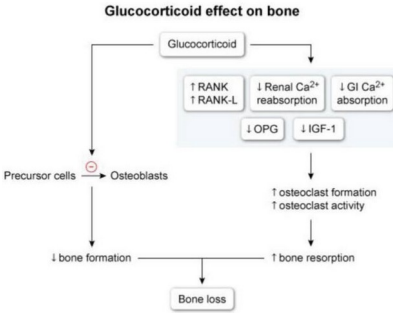
slide 12 *- Cushing's Disease | Pituitary Adenoma Producing ACTH | Bilateral Hyperplasia Of*



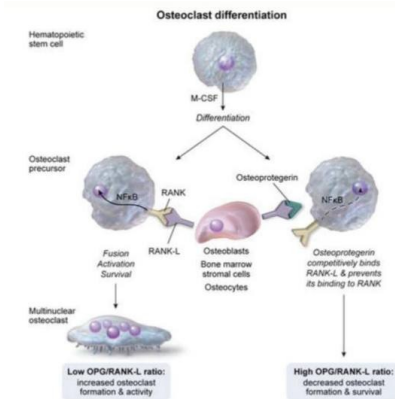
slide 13 *- Causes Of Cushing's Syndrome | Functioning Cortical / Adenoma | Bilateral Diffuse / Cortic*



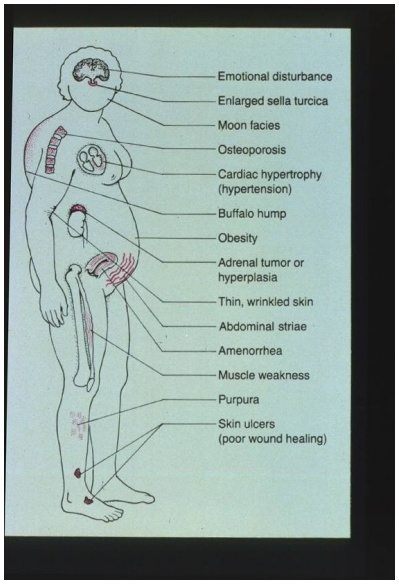
slide 13 *- Causes Of Cushing's Syndrome | Functioning Cortical / Adenoma | Bilateral Diffuse / Cortic*



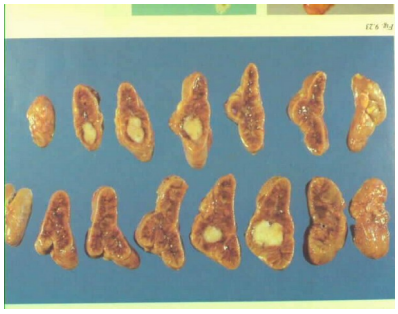
slide 15



slide 16



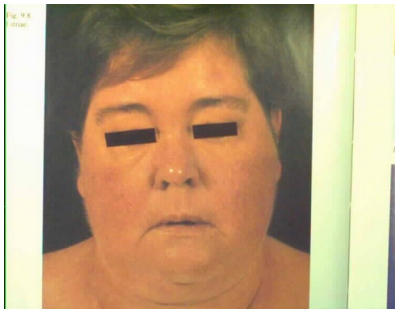
slide 17 *- Cushing's Syndrome*



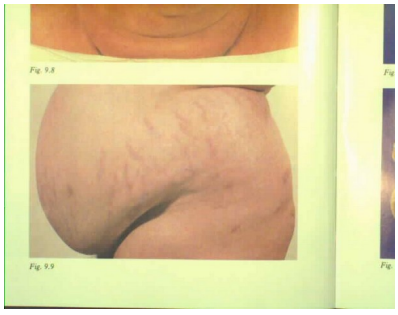
slide 18 *- Ectopic Cushing's Syndrome | Small Cell Lung / Cancer Producing ACTH | Before / Treatmen*



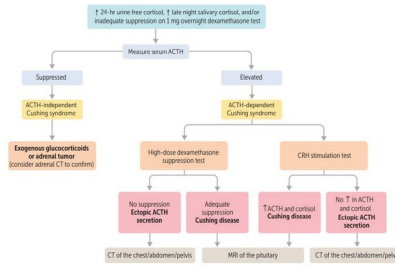
slide 18 *- Ectopic Cushing's Syndrome | Small Cell Lung / Cancer Producing ACTH | Before / Treatmen*



slide 19 *- Cushing's Syndrome | Moon Facies | Central Obesity And Striae*



slide 19 *- Cushing's Syndrome | Moon Facies | Central Obesity And Striae*



slide 20

Diagnosis (teacher slide 21 table, arrows restored):

Test	Pituitary (Cushing disease)	Adrenal	Ectopic ACTH
Serum cortisol	High	High	High
Urine free cortisol	High	High	High
Plasma ACTH	**High (or normal)**	**Low**	**Very high**
Low-dose dexamethasone	Not suppressed	Not suppressed	Not suppressed
High-dose dexamethasone	**Cortisol suppressed**	Not suppressed	Not suppressed

Steps: confirm hypercortisolism (24-hour urine free cortisol, late-night salivary cortisol, low-dose dexamethasone suppression), then check ACTH (low = adrenal source, imaging of the adrenal; high = pituitary MRI vs ectopic source; high-dose dexamethasone separates them: the pituitary adenoma is still partly sensitive to negative feedback, ectopic tumours are not).

Hyperaldosteronism

- **Primary (Conn syndrome): adrenal cortical adenoma** (commonest) or bilateral hyperplasia (carcinoma rare). **High aldosterone, LOW renin** (negative feedback of volume expansion and hypertension).
- **Secondary:** activation of the renin-angiotensin system: **renal artery stenosis (atherosclerosis; fibromuscular dysplasia in young women)**, arteriolar nephrosclerosis, hypovolaemia and oedema states (**cirrhosis, nephrotic syndrome, heart failure, blood loss**), and pregnancy (estrogen raises angiotensinogen). **High aldosterone, HIGH renin.**
- **Presentation:** aldosterone acts on the **principal cells of the collecting duct: reabsorbs Na and excretes K and H**, so **hypertension, hypernatraemia (or high-normal), hypokalaemia (weakness, cramps, arrhythmias) and metabolic alkalosis.** Oedema is typically absent ("aldosterone escape").

	Aldosterone	Na	K	Renin
Primary	High	High/normal	**Low**	**Low**
Secondary	High	Normal	Low	**High**

- **Treatment: spironolactone** (aldosterone receptor blocker); surgery for an adenoma. Secondary: treat the cause (ACE inhibitors, correct stenosis).



slide 22



slide 23

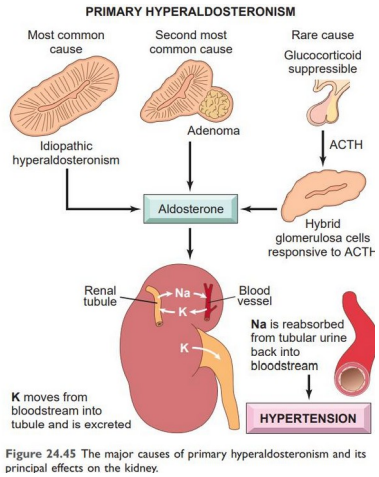
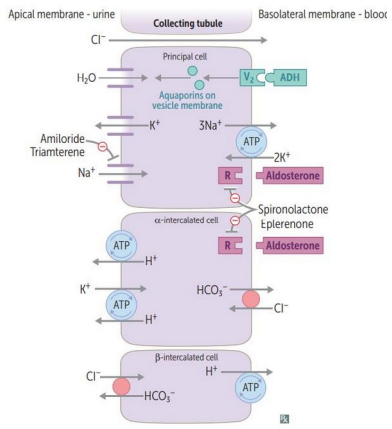
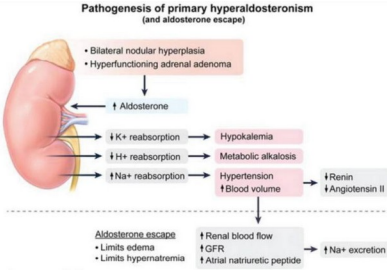


Figure 24.45 The major causes of primary hyperaldosteronism and its principal effects on the kidney.

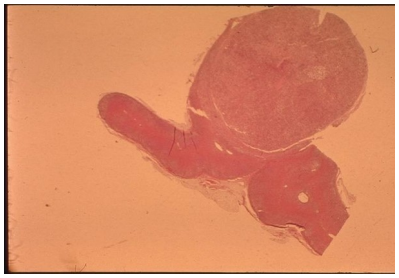
slide 25



slide 26



slide 27



***slide 29** - Conn's Syndrome (Primary hyperaldosteronism) | Cortical Adenoma | Benign Cortical Adenoma / / Aldosterone*

Adrenogenital (virilising) syndromes and congenital adrenal hyperplasia (CAH)

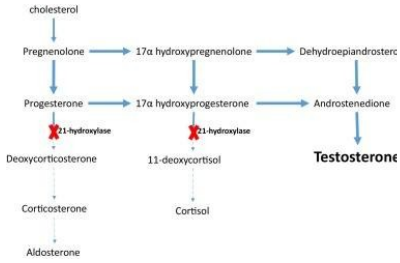
Excess **adrenal androgens** (DHEA and androstenedione) from adrenal tumours, **CAH**, or Cushing disease. Urinary **17-ketosteroids** reflect adrenal androgens (in women and children, nearly all of them come from the adrenal).

CAH mechanism: an enzyme deficiency means **low cortisol, so no negative feedback and high ACTH, which causes adrenal hyperplasia**, and the blocked precursors are shunted to **androgens**.

	21-hydroxylase deficiency (about 90%, commonest)	**11-beta-hydroxylase deficiency**	**17-alpha-hydroxylase deficiency**
Cortisol	Low	Low	Low
Aldosterone	**Low (salt wasting in the classic form)**	**Low, but deoxycorticosterone (DOC), a mineralocorticoid, is HIGH**	**Low aldosterone, but DOC and corticosterone HIGH**
Androgens	**High**	**High**	**Low**
Blood pressure	**Low** (hypovolaemia)	**High** (DOC acts as a mineralocorticoid)	**High**
Potassium	**High**	Low	**Low**
Genitalia	**XX: ambiguous genitalia (clitoral enlargement, virilisation); XY: precocious puberty**	XX: virilisation; precocious puberty in boys	**XY: ambiguous/female genitalia, undescended testes; XX: no secondary sexual development**
Other	Hyponatraemia, hyperkalaemia, acidosis, hypoglycaemia, seizures, **hyperpigmentation** (high ACTH/POMC). Raised **17-hydroxyprogesterone**	Presents in infancy with severe hypertension or childhood with precocious puberty	

Memory hook: "21 = low Pressure, 11 = high Pressure, 17 = no Sex" (21-hydroxylase deficiency: salt-wasting, low BP; 11 and 17: high BP; 17: no androgens or estrogens).

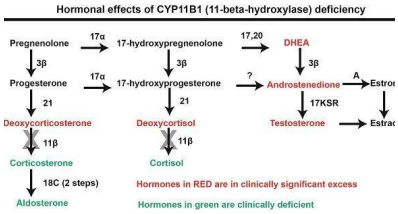
Pathoma: in 11-hydroxylase deficiency you do not see hypomineralocorticoid effects, because 21-hydroxylase makes a weak mineralocorticoid (DOC) and 11-hydroxylase is needed to turn it into a strong one.



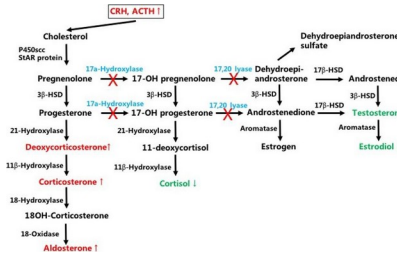
slide 33

Clinical phenotypes of 21-hydroxylase deficiency			
	Classic, salt-wasting	Classic, non-salt-wasting	Non-classic, delayed
Degree of 21-hydroxylase deficiency	Severe	Moderate	Mild
Clinical presentation	Females present at birth with ambiguous genitalia Males present at 1 to 2 weeks with failure to thrive, dehydration, hyperkalemia, and hyponatremia	Females present at birth with ambiguous genitalia Males present at 2 to 4 years with signs of early virilization	Premature pubarche or sexual precocity in school age children Young women can present with acne, hirsutism, and menstrua irregularity

slide 35



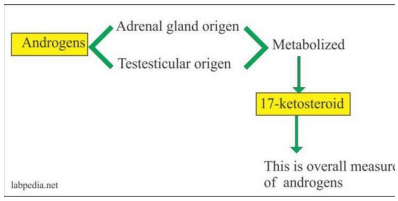
slide 36



slide 38

Congenital adrenal hyperplasia				
Enzyme deficiency	Mineralocorticoids	Glucocorticoids	Androgens	Symptoms
21-hydroxylase	↓	↓	↑	• Salt wasting (vomiting, hypotension, ↓ Na+, ↑ K+) • Ambiguous genitalia in girls
11β-hydroxylase	↑*	↓	↑	• Hypertension, ↓ K+ • Ambiguous genitalia in girls
17α-hydroxylase	↑	↑**	↓	• Hypertension, ↓ K+ • Ambiguous genitalia in boys • Absent puberty
*Low aldosterone but excessive weak mineralocorticoid (11-deoxycorticosterone). **Low cortisol but excessive weak glucocorticoid (corticosterone).				

slide 40



***slide 41** - Urinary 17-ketosteroids | In men: 1/3 is produced from the testosterone produced in th*

11.3 Adrenocortical insufficiency

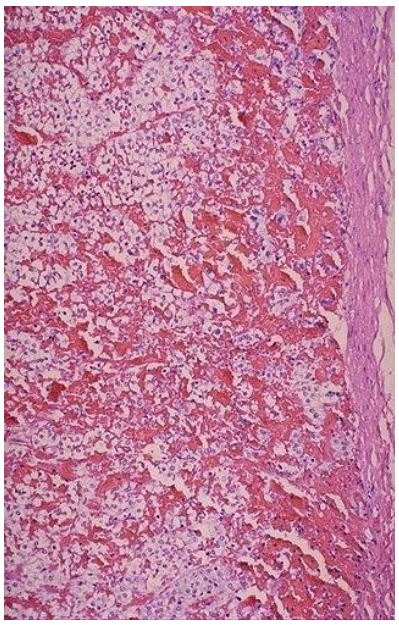
Primary (adrenal disease): acute (adrenal crisis) or chronic (**Addison disease**). **Secondary:** pituitary or hypothalamic disease (low ACTH): **no hyperpigmentation, no hyperkalaemia, aldosterone largely preserved**.

Acute adrenal insufficiency (crisis)

- **Waterhouse-Friderichsen syndrome:** bilateral **adrenal haemorrhage** in **meningococcaemia (Neisseria meningitidis)** from **DIC and vascular injury**; "**sepsis with purpura**", massive hypotension and shock. The adrenals appear as "sacs of blood".
- Sudden **withdrawal of glucocorticoid therapy** in a patient with suppressed adrenals, or stress (surgery, infection) in a patient with chronic insufficiency; pituitary apoplexy (secondary).



***slide 44** - Waterhouse-Friedrischen Syndrome |*

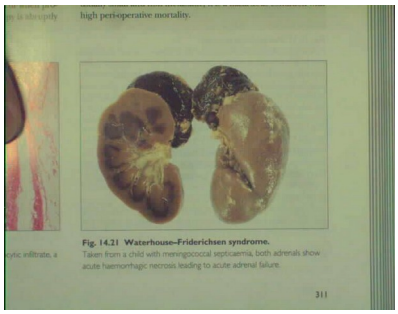


***slide 44** - Waterhouse-Friedrischen Syndrome |*



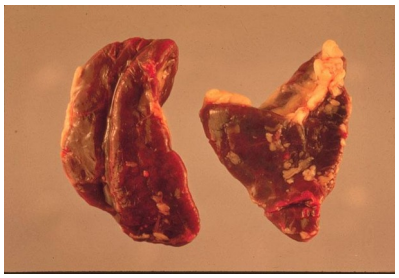
***slide 44** - Waterhouse-Friedrischen Syndrome |*

Sepsis With Purpura From DIC | Bilateral Medullary Hemorrhag



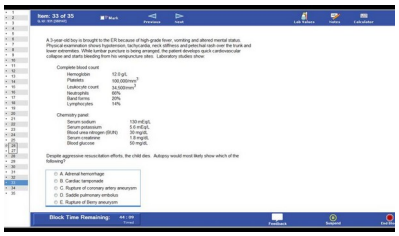
slide 45 *- Waterhouse-Friderichsen Syndrome*

Sepsis With Purpura From DIC | Bilateral Medullary Hemorrhag



slide 45 *- Waterhouse-Friderichsen Syndrome*

Sepsis With Purpura From DIC | Bilateral Medullary Hemorrhag

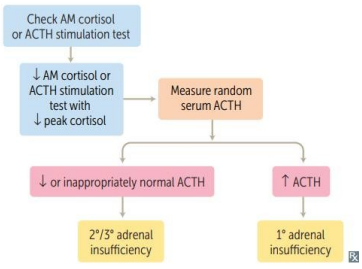


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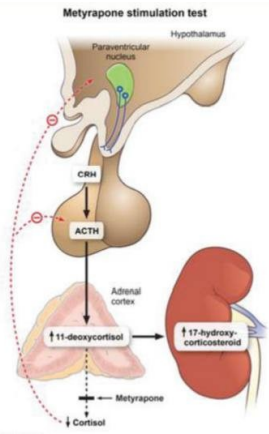
Chronic adrenal insufficiency: Addison disease

Causes: autoimmune adrenalitis (commonest in developed countries), **tuberculosis (commonest in developing countries)**, sarcoidosis, AIDS (CMV, MAC), **metastatic cancer (lung, breast)**, fungal infections (histoplasma, coccidioides), amyloidosis, haemochromatosis. **Pathology:** both adrenals **atrophic**. **Clinical:** weakness, fatigue, weight loss, anorexia, nausea, vomiting and diarrhoea (low cortisol), **hypotension and orthostasis** (90% systolic below 110), **hypoglycaemia**, **hyponatraemia, hyperkalaemia, volume depletion and non-anion-gap metabolic acidosis** (low aldosterone), **hyperpigmentation** (ACTH and MSH rise because both are cleaved from POMC): knuckles, elbows, knees, palmar creases, nail beds, mucosa, new scars. Vitiligo (about 10%), small heart, scant axillary and pubic hair in women (loss of adrenal androgens). **Labs:** low cortisol, **ACTH markedly raised (generally above 200 pg/mL)** in primary disease, eosinophilia, lymphocytosis, neutropenia; anti-adrenal antibodies. **Treatment:** glucocorticoid plus fludrocortisone replacement; stress dosing.

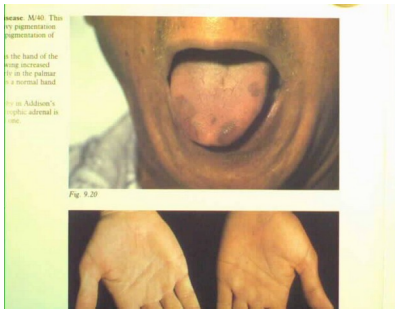
	Primary (Addison)	**Secondary**
ACTH	High	Low
Hyperpigmentation	**Yes**	No
Aldosterone / K	Low / **high K**	Preserved / normal K



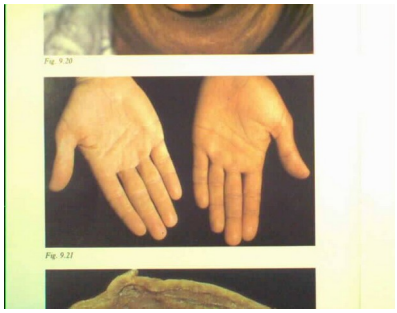
slide 51



slide 52



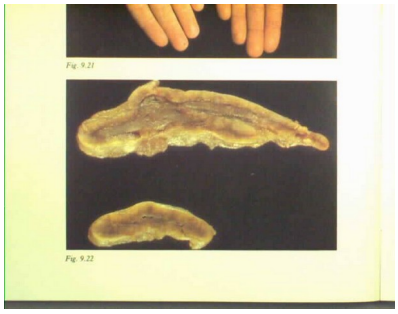
slide 53 *- Addison's Disease | Hyperpigmentation*



slide 53 *- Addison's Disease | Hyperpigmentation*



slide 54 *- Addison's Disease | Amyloidosis | Idiopathic Atrophy*



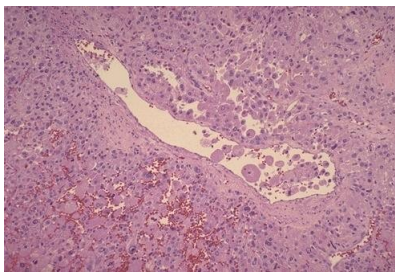
slide 54 *- Addison's Disease | Amyloidosis | Idiopathic Atrophy*

11.4 Adrenal cortical tumours

- **Cortical adenoma:** usually non-functioning; may cause Cushing or Conn syndrome. Yellow (lipid-rich), well-circumscribed.
- **Adrenocortical carcinoma:** rare; more common in children (especially boys); a **bulky mass with haemorrhage, necrosis and cysts**; **haematogenous** spread; often secretes androgens (precocious puberty/virilisation); polygonal eosinophilic cells with pleomorphism. Associated with **Li-Fraumeni (TP53)**.



slide 56 *- Adrenal Cortical Carcinoma | More Common In Children / Boys More Than Girls / Hematogenou*



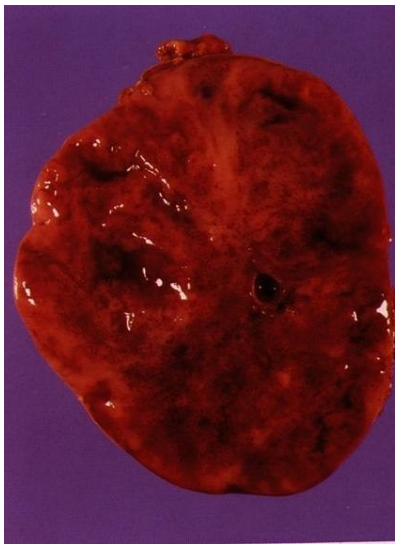
slide 57 *- Adrenocortical Carcinoma | Eosinophilic polygonal cells; Variable nuclear pleomorphism;*

11.5 Adrenal medulla: pheochromocytoma

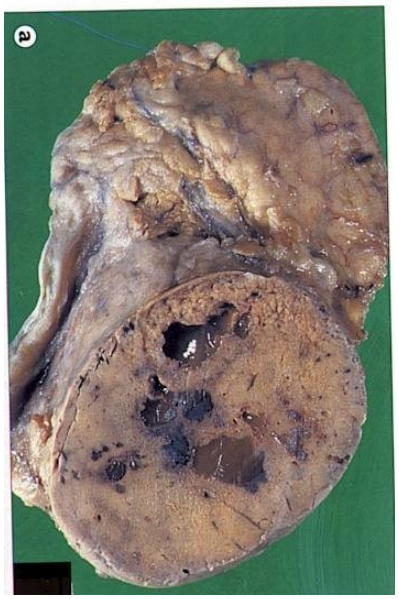
- A tumour of the **chromaffin cells** (neural crest derived), secreting **norepinephrine and epinephrine**. **Gross: mahogany-brown (chromaffin reaction), with fibrosis, central cysts and haemorrhage. Micro: nests of polygonal cells ("Zellballen", balls of cells) with eosinophilic granular cytoplasm separated by sustentacular cells and capillaries.** Electron micrograph: **dense-core neurosecretory granules.**
- **Presentation: episodic hypertension** (or sustained), the classic triad **headache, palpitations, sweating**, plus anxiety, pallor, tremor. Orthostatic hypotension may occur (receptor down-regulation and volume depletion). Unilateral mass on CT.
- **Diagnosis: plasma free metanephrines** (best, most sensitive) and 24-hour urinary metanephrines, VMA and catecholamines; then CT/MRI and MIBG scan.
- **Why metanephrines rather than catecholamines?** Epinephrine and norepinephrine are released episodically and have a very short half-life; **metanephrines are made continuously inside the tumour cells (COMT) and leak out steadily**, so they are far more sensitive. Epi and NE are metabolised to metanephrine and normetanephrine, then (MAO) to VMA.

- **Treatment:** surgery. **Start an alpha blocker first (phenoxybenzamine, an irreversible alpha-1 blocker; or phentolamine/doxazosin), then add a beta blocker. Beta blocker alone is dangerous:** beta-2 receptors normally cause vasodilation, so blocking only beta leaves **unopposed alpha vasoconstriction with a hypertensive crisis**. Mechanical handling of the tumour during surgery can dump catecholamines.
- **Associations: MEN2A and 2B (RET)**, **neurofibromatosis type 1**, **von Hippel-Lindau** (VHL: cerebellar/retinal haemangioblastomas, renal cell carcinoma, pheo), and paraganglioma syndromes (SDH mutations).

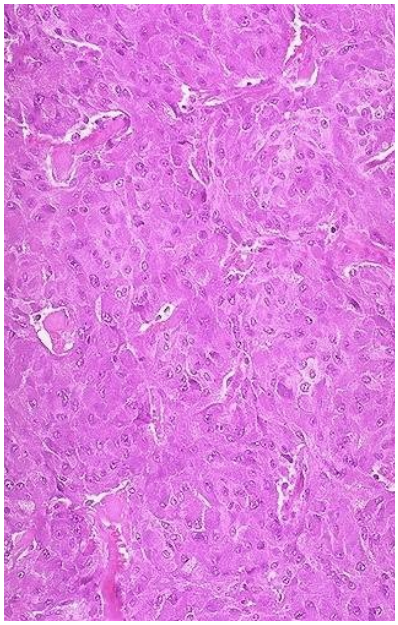
Rule of 10's (the old rule: bilateral, familial, malignant, extra-adrenal, children, normotensive, extra-abdominal, metastatic at diagnosis, each about 10%): 10% bilateral (more in familial), 10% familial, 10% malignant, 10% extra-adrenal (**organ of Zuckerkandl, carotid body, urinary bladder** (headache and palpitations while urinating)), 10% in children, 10% without hypertension.



slide 60 *- Pheochromocytoma | Mahogany-brown gross color; fibrosis | Central cysts and hem*



slide 60 *- Pheochromocytoma | Mahogany-brown gross color; fibrosis | Central cysts and hem*



slide 60 *- Pheochromocytoma | Mahogany-brown gross color; fibrosis | Central cysts and hem*

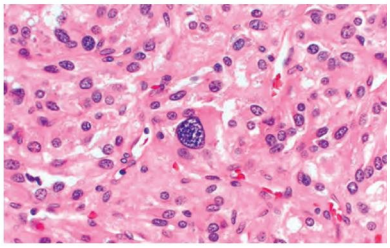
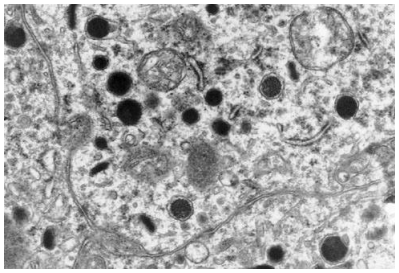


Figure 24.55 Pheochromocytoma demonstrating characteristic nests of cells ("zellballen") with abundant cytoplasm. Granules containing catecholamine are not visible in this preparation. It is not uncommon to find bizarre cells, even in pheochromocytomas that are biologically benign (Courtesy Dr. Jerrold R. Turner, Department of Pathology, Brigham and Women's Hospital, Boston, Mass.)

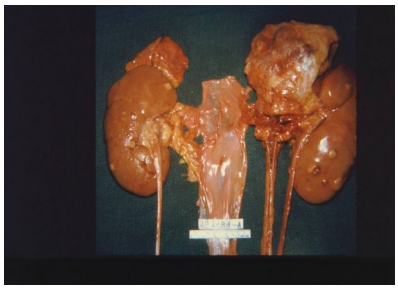
slide 61



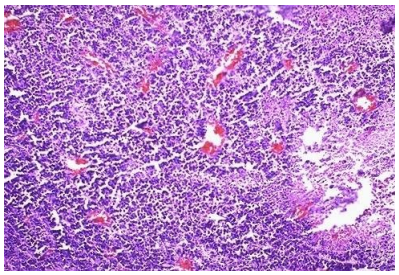
slide 62 *- Pheochromocytoma | Electron micrograph with dense core membrane bound neurosecr*

Neuroblastoma

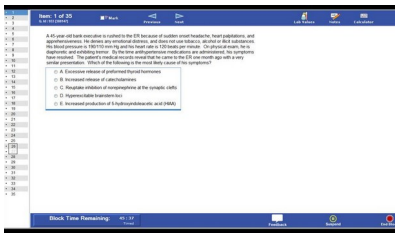
- A tumour of **neural crest cells (sympathetic ganglia and adrenal medulla)**, the commonest extracranial solid tumour in children; **most are under 2 years** (a failure of normal neuroblast maturation). Sites: adrenal medulla, retroperitoneal sympathetic chain, **posterior mediastinum**.
- **Genetics: N-myc (MYCN) amplification** is the key determinant of aggressiveness; poor prognosis with amplification, older age and advanced stage. Infants under 1 year have a better prognosis, with possible spontaneous regression.
- **Clinical:** a **firm abdominal mass that crosses the midline**, abdominal pain and distension, weight loss, bone pain (metastases), fever, anorexia, periorbital ecchymoses ("raccoon eyes") and proptosis (orbital metastases), skull masses, blueberry-muffin skin nodules, hepatomegaly, spinal cord compression, **opsoclonus-myoclonus** (paraneoplastic). Catecholamine excess gives hypertension (less than pheochromocytoma), sweating, palpitations.
- **Labs:** raised urinary **HVA and VMA** (and metanephrines).
- **Micro: small round blue cell tumour** with **Homer-Wright rosettes** (tumour cells around a central pale fibrillar neuropil core), necrosis, calcification.
- **Differential:** Wilms tumour (arises in the kidney, mass does not usually cross the midline, no HVA/VMA).



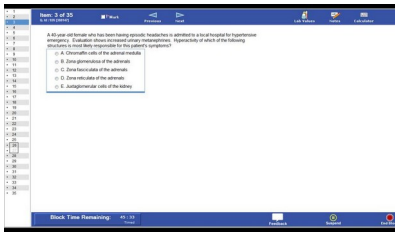
slide 64 *- Neuroblastoma | Large tumor of left adrenal- lobulated, hemorrhagic and cyst*



slide 64 *- Neuroblastoma | Large tumor of left adrenal- lobulated, hemorrhagic and cyst*



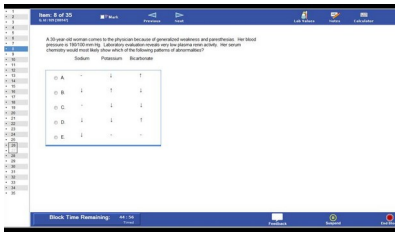
slide 65



slide 66



slide 67



slide 68

11.6 SGD answers: Adrenal Medullary Disorders

Case 1: 38-year-old woman, episodic headache, palpitations, sweating; BP 168/100, HR 110; raised plasma free metanephrines; 4 cm adrenal mass

120. **Chromaffin cells** of the adrenal medulla produce catecholamines.
121. The medulla arises from **neural crest** (ectoderm); the cortex arises from mesoderm.
122. **Pheochromocytoma.**

123. Catecholamine bursts: **alpha-1 vasoconstriction (hypertension)** and **beta-1 stimulation (tachycardia, palpitations)**; sweating and anxiety from sympathetic activation. The episodes follow the intermittent release.
124. Metanephrines (made continuously inside the tumour cell by COMT) are more stable and constant than the short-lived, intermittently released catecholamines, so they are the more sensitive test.
125. **Zellballen** pattern: nests (cell balls) of chromaffin cells surrounded by sustentacular (S100-positive) cells and a rich capillary network.
126. Zellballen = German for "balls of cells".
127. Alpha blockers first control the hypertension caused by alpha-1 vasoconstriction.
128. Beta blockade first leaves alpha vasoconstriction unopposed (blocked beta-2 vasodilation), causing a hypertensive crisis (and, if cardiomyopathy, heart failure).
129. Differential: **essential hypertension, anxiety/panic attacks, hyperthyroidism, carcinoid, cocaine or amphetamine use, MAO inhibitors with tyramine, withdrawal syndromes, primary hyperaldosteronism, other paragangliomas.**
130. **Rule of 10's** (see above).

Case 2: 3-year-old boy, abdominal mass crossing the midline, raised urinary HVA and VMA, small round blue cells in rosettes

131. **Neuroblastoma.**
132. **Neural crest neuroblasts** (sympathetic nervous system precursors).
133. **Homer-Wright rosettes:** a ring of tumour cells around a pink fibrillar (neuropil) centre.
134. The tumour makes catecholamines, which are metabolised to **HVA (from dopamine) and VMA (from epinephrine and norepinephrine)** and excreted in urine.
135. It arises from **immature neural crest cells (neuroblasts)** that fail to differentiate during early development, so it appears in infants and toddlers.
136. It originates from the adrenal or paraspinal ganglia, grows around vessels and **extends across the midline** (Wilms tumour is more intrarenal and does not usually cross).
137. **N-myc (MYCN) amplification.**

11.7 Quick adrenal table

Clue	Answer
Moon face, striae, proximal weakness, high ACTH, suppressed by high-dose dexamethasone	Cushing disease (pituitary)
Hypertension, low K, alkalosis, low renin	Primary hyperaldosteronism (Conn)
Ambiguous genitalia in a girl, hypotension, hyperkalaemia	21-hydroxylase deficiency
Hypertension + virilisation + low K	11-hydroxylase deficiency
Purpura, shock, bilateral adrenal haemorrhage	Waterhouse-Friderichsen (meningococcus)
Tanned skin, hypotension, high K, low Na	Addison disease
Episodic headache, sweating, palpitations	Pheochromocytoma (alpha blocker first)
Abdominal mass crossing the midline, child under 2, N-myc	Neuroblastoma

12. ERS II Pathology Review: Picture Quiz

Source: Teacher deck "ERS II Pathology Review" (50 slides). The deck is mostly pictures with a one-word label in the speaker notes, plus the learning-objective slides for each organ. Use this chapter as a **self-test**: cover the label, name the lesion, then say two things you would expect to see in the clinical history.

12.1 What your teacher says you must be able to do (learning objectives by organ)

Organ	Objectives
Penis	Differentiate congenital malformations; **Peyronie disease vs penile fracture**; **phimosis vs paraphimosis**; precursor lesions: **bowenoid papulosis, BXO (balanitis xerotica obliterans), leukoplakia, erythroplasia** (chapter 2)
Testis	**Epididymitis vs torsion** (gross morphology); **germ cell tumours** (morphology, clinical features, markers); **hydrocele, hernia, spermatocele vs tumour**; lymphatic drainage of the external genitalia (chapter 2)
Ovary	Surface epithelial tumours (gross and micro); **pseudomyxoma peritonei**; germ cell tumours; **estrogen-producing** and **androgen-producing** tumours (chapter 5)
Prostate	**Prostatitis** differential, **BPH**, prostate cancer, **PIN**, **Gleason grading**, staging (chapter 2)
Cervix	**PID** (etiology, pathogenesis, clinical features, complications); **cervical carcinoma** (risk factors, pathogenesis, dysplasia/CIN/LSIL-HSIL/carcinoma in situ, clinical features); identify **koilocytes** (chapter 3)
Breast	Abscess, mastitis, fat necrosis, nipple discharge; cysts, fibrocystic change, fibroadenoma, phyllodes, intraductal papilloma; DCIS, Paget disease, breast cancer and screening (chapter 7)
Thyroid	Cyst/nodule, euthyroid sick syndrome, goitre, hypo- and hyperthyroidism including **thyrotoxicosis and thyroid storm**, thyroiditis (Hashimoto), Graves disease, neoplasms (papillary, follicular, medullary, anaplastic), secondary hypo/hyperthyroidism, infertility due to thyroid disease (chapter 9)
Adrenal	Addison disease, secondary insufficiency, hypocortisolism, Cushing syndrome, hyperaldosteronism, neoplasms (neuroblastoma, pheochromocytoma, carcinoma, adenoma, aldosteronoma, incidentaloma), delayed and precocious puberty, hypertensive endocrine disease (chapter 11)

High yield: Euthyroid sick syndrome (low T3 syndrome): in severe non-thyroidal illness, the conversion of T4 to T3 falls (and reverse T3 rises) while TSH is normal or low; **no treatment**, repeat the tests after recovery. It is not hypothyroidism. (Pathoma also notes that T4 to T3 conversion is reduced in severe illness.)

12.2 Germ cell tumours (testis and ovary): the pictures

Match each picture with the tumour, the marker and one distinguishing feature.

Slide	Tumour	What to look for	Marker
5	**Seminoma** (and ovarian dysgerminoma)	Sheets of uniform large cells with clear cytoplasm and square-cut nuclei, fibrous septa with lymphocytes, "fried egg"	PLAP, LDH, c-KIT; hCG slightly raised if syncytiotrophoblasts
6	**Embryonal carcinoma**	Primitive large cells, glands/papillae, necrosis, haemorrhage	hCG (some), AFP (some), CD30
7	**Yolk sac tumour**	Reticular/lace pattern, **Schiller-Duval bodies**, hyaline globules	**AFP**
8	**Choriocarcinoma**	Cytotrophoblasts and syncytiotrophoblasts, haemorrhage, no villi	**beta-hCG**
9, 10	**Mature teratoma / dermoid cyst**	Cystic with skin, hair, teeth; tissue from all three germ layers	None
11	**Struma ovarii**	Thyroid tissue in the teratoma; may cause hyperthyroidism	Thyroid hormone

12.3 Ovarian tumours: the pictures

Slide	Tumour	What to look for
14	**Serous cystadenoma/cystadenocarcinoma**	Unilocular clear-fluid cyst; papillae and **psammoma bodies** if malignant
15	**Mucinous cystadenoma**	Large multiloculated cyst with gelatinous mucin; tall columnar mucin cells
16	**Granulosa cell tumour**	**Call-Exner bodies**, coffee-bean nuclei; estrogen effects

12.4 Prostate and cervix: the pictures

Slide	Lesion	What to look for
18	**BPH**	Nodular hyperplasia of glands and stroma; glands with **papillary infolding and a double layer (basal + luminal)**, corpora amylacea
19	**Prostatic intraepithelial neoplasia (PIN)**	Crowded glands lined by atypical cells with **prominent nucleoli**, **basal cell layer still present**
20	**Prostate adenocarcinoma**	Small glands **without a basal layer**, prominent nucleoli, back to back; **Gleason pattern** by gland architecture
23, 24	**Cervix** (CIN / koilocytes / carcinoma)	Koilocytes (perinuclear halo, wrinkled raisin nucleus) in LSIL; full-thickness atypia in HSIL; see chapter 3

12.5 Breast: the pictures

Slide	Lesion	What to look for
27	**Fibroadenoma**	Slit-like compressed ducts in a loose myxoid stroma; well-circumscribed
28	**DCIS (cribriform)**	Ducts filled with monomorphic cells forming punched-out round spaces; basement membrane intact
29	Breast lesion (see slide)	Compare to chapter 7
30	**Invasive ductal carcinoma**	Irregular glands and nests in dense desmoplastic stroma
31	**Lobular carcinoma**	Small dyscohesive cells in **single file (Indian file)**, signet-ring cells, E-cadherin negative

12.6 Thyroid: the pictures

Slide	Lesion	What to look for
34	**Hashimoto thyroiditis**	Lymphoid follicles with germinal centres, **Hurthle cells**, atrophic follicles
35	**Riedel thyroiditis**	Dense fibrosis replacing the gland, extending into surrounding tissue
36	**Subacute (de Quervain) thyroiditis**	**Granulomas with giant cells** around colloid
37	**Graves disease**	Tall crowded follicular cells, **scalloped colloid**, papillary infoldings
38	**Follicular adenoma**	Encapsulated nodule of uniform follicles, no capsular or vascular invasion
39	**Follicular carcinoma**	Capsular and/or vascular invasion
40	**Papillary carcinoma**	**Orphan Annie nuclei, grooves, psammoma bodies, papillae**
41	**Medullary carcinoma**	Nests of polygonal cells in **amyloid stroma** (Congo red)

12.7 Adrenal: the pictures

Slide	Lesion	What to look for
44-47	Adrenal pathology images	Match each to Cushing/Addison/Conn/adenoma/carcinoma/haemorrhage using chapter 11
48	**Pheochromocytoma**	**Zellballen** nests of eosinophilic granular cells; brown gross specimen
49	**Neuroblastoma**	**Homer-Wright pseudorosettes**: small round blue cells around a fibrillar core
50	Closing image	

13. Master Quick-Recall Tables

13.1 Tumour markers

Marker	Tumour / use
AFP	Yolk sac tumour (testis, ovary), hepatocellular carcinoma; high in neural tube defects
beta-hCG	Choriocarcinoma, hydatidiform mole, embryonal carcinoma, seminoma (mild), pregnancy
LDH / PLAP	Seminoma, dysgerminoma
CA-125	Serous ovarian carcinoma (monitoring recurrence, not screening)
Inhibin	Granulosa cell tumour
PSA	Prostate cancer (screening/monitoring), BPH, prostatitis
Calcitonin, CEA	Medullary thyroid carcinoma
Thyroglobulin	Follow-up of papillary/follicular thyroid carcinoma after thyroidectomy
Metanephrines, VMA	Pheochromocytoma; VMA/HVA for neuroblastoma
C-peptide	Insulinoma (raised) vs exogenous insulin (low)

13.2 Genes and syndromes

Gene / feature	Condition
BRCA1 (17)	Breast (triple negative, medullary) and serous ovarian cancer
BRCA2 (13)	Breast (including male), ovarian, prostate, pancreas
TP53 (Li-Fraumeni)	SBLA: sarcoma, breast, leukaemia, adrenal cortical carcinoma
RET	MEN2A, MEN2B, medullary thyroid carcinoma, Hirschsprung (loss of function)
MEN1 (menin)	Pituitary, parathyroid, pancreas
BRAF V600E	Papillary thyroid carcinoma, papillary craniopharyngioma
CTNNB1	Adamantomatous craniopharyngioma; endometrioid ovarian tumours
PAX8-PPARG	Follicular thyroid carcinoma
N-myc amplification	Neuroblastoma (poor prognosis)
GNAS1 (Gs alpha)	Pseudohypoparathyroidism (maternal), McCune-Albright
VHL	Hemangioblastoma, renal cell carcinoma, pheochromocytoma
HLA-DR3/DR4	Type 1 diabetes; **DR3/B8** Graves; **DR3/DR5** Hashimoto

13.3 Classic histology signs

Sign	Lesion
Psammoma bodies	Papillary thyroid carcinoma, serous ovarian carcinoma, meningioma, mesothelioma
Orphan Annie eyes and nuclear grooves	Papillary thyroid carcinoma
Amyloid stroma	Medullary thyroid carcinoma; islet amyloid (amylin) in type 2 diabetes
Hurthle cells, germinal centres	Hashimoto thyroiditis

Scalloped colloid	Graves disease
Granulomas with giant cells around colloid	De Quervain thyroiditis
Schiller-Duval bodies	Yolk sac tumour
Call-Exner bodies	Granulosa cell tumour
Reinke crystals	Leydig cell tumour; Sertoli-Leydig tumour
Koilocytes	HPV (condyloma, CIN)
Fried-egg cells	Seminoma/dysgerminoma
Zellballen	Pheochromocytoma
Homer-Wright rosettes	Neuroblastoma
Signet-ring cells	Gastric cancer, Krukenberg tumour, lobular breast carcinoma
Indian-file growth	Lobular breast carcinoma
Comedo necrosis	High-grade DCIS
Paget cells	Paget disease of nipple/vulva
Hydropic villi, absent p57	Complete mole
Cytotrophoblast + syncytiotrophoblast, no villi	Choriocarcinoma

13.4 Hormone-secreting tumour clues

Clinical picture	Tumour
Galactorrhoea, amenorrhoea, bitemporal field loss	Prolactinoma
Acral growth, glucose not suppressing GH	GH adenoma
Moon face, striae, ACTH high, dexamethasone-suppressible	Cushing disease
Hypertension, hypokalaemia, low renin	Conn
Episodic headaches, sweating, palpitations	Pheochromocytoma
Precocious puberty in girl with ovarian mass	Juvenile granulosa cell tumour
Virilisation with ovarian mass	Sertoli-Leydig cell tumour
Hyperthyroidism with ovarian teratoma	Struma ovarii
Postmenopausal bleeding + ovarian mass	Granulosa-theca cell tumour
Large for dates uterus, very high hCG, hyperemesis, theca-lutein cysts	Complete mole
Hypercalcaemia + pituitary tumour	MEN1
Medullary thyroid carcinoma + pheo	MEN2

13.5 Eponyms and signs

Name	What it is
Sheehan syndrome	Postpartum pituitary necrosis
Waterhouse-Friderichsen	Meningococcal adrenal haemorrhage
Addison disease	Primary chronic adrenal insufficiency
Conn syndrome	Primary hyperaldosteronism
Nelson syndrome	Pituitary tumour growth after bilateral adrenalectomy for Cushing
Zollinger-Ellison	Gastrinoma
Verner-Morrison	VIPoma
Stein-Leventhal	PCOS
Meigs	Ovarian fibroma + ascites + pleural effusion
Krukenberg	Bilateral signet-ring metastases to ovary (stomach)
Stewart-Treves	Angiosarcoma in chronic lymphoedema
Peyronie	Penile fibromatosis

Chvostek / Trousseau	Signs of hypocalcaemia
Pemberton	Faintness on raising arms (retrosternal goitre)
Albright hereditary osteodystrophy	Pseudohypoparathyroidism phenotype
Kallmann	Anosmia + hypogonadotropic hypogonadism
Klinefelter / Turner	47,XXY / 45,X